

basis of the computation result output by the heat balance computation unit, corrects one or both of the fuel flow rate and the air flow rate, and gives the corrected flow rate to the turbine inlet temperature calculation unit.

12 Claims, 16 Drawing Sheets

(52) U.S. Cl.

CPC .. *F05D 2260/821* (2013.01); *F05D 2270/303*
(2013.01); *F05D 2270/306* (2013.01); *F05D*
2270/71 (2013.01)

(56)

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FIG. 1

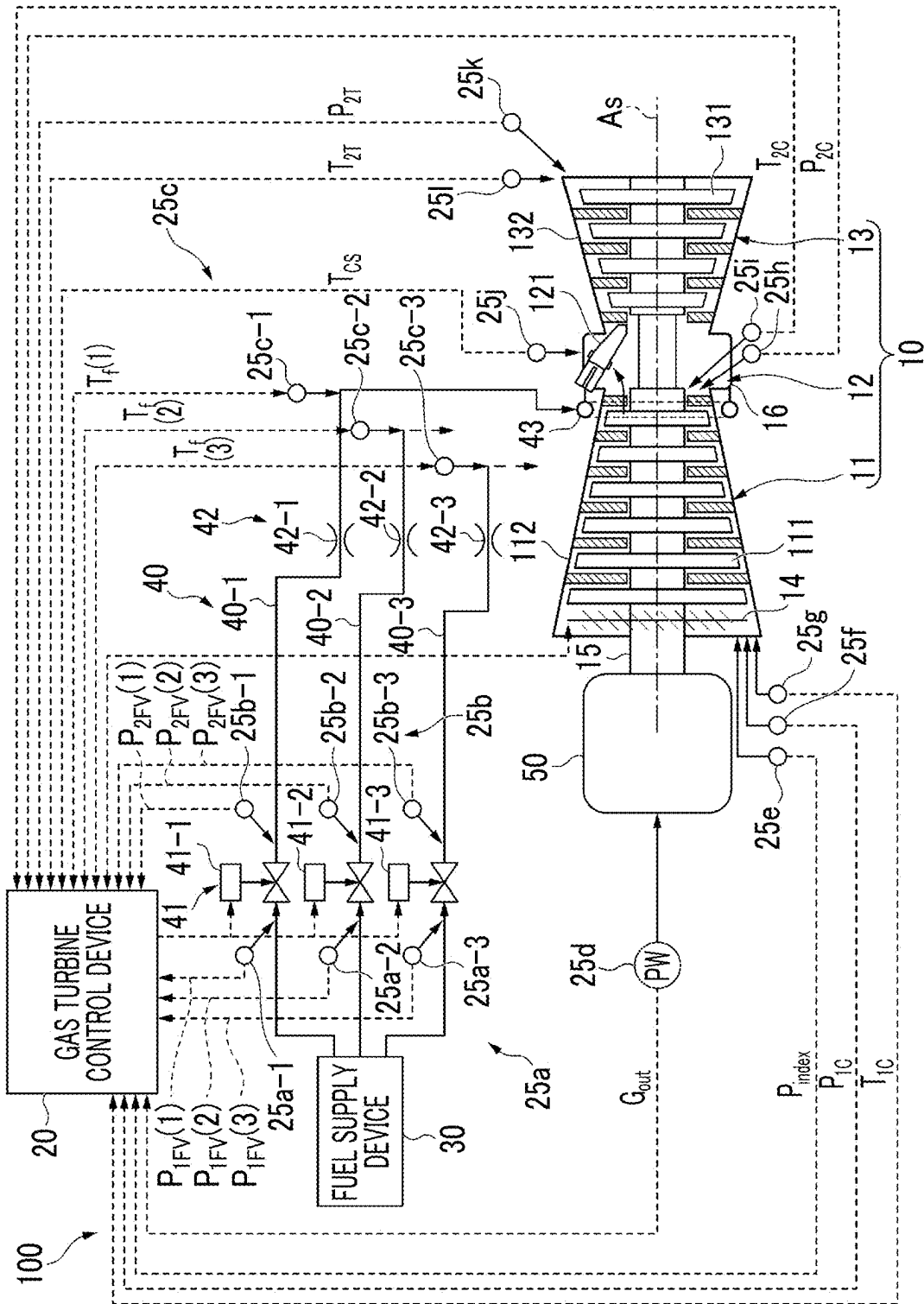


FIG. 2

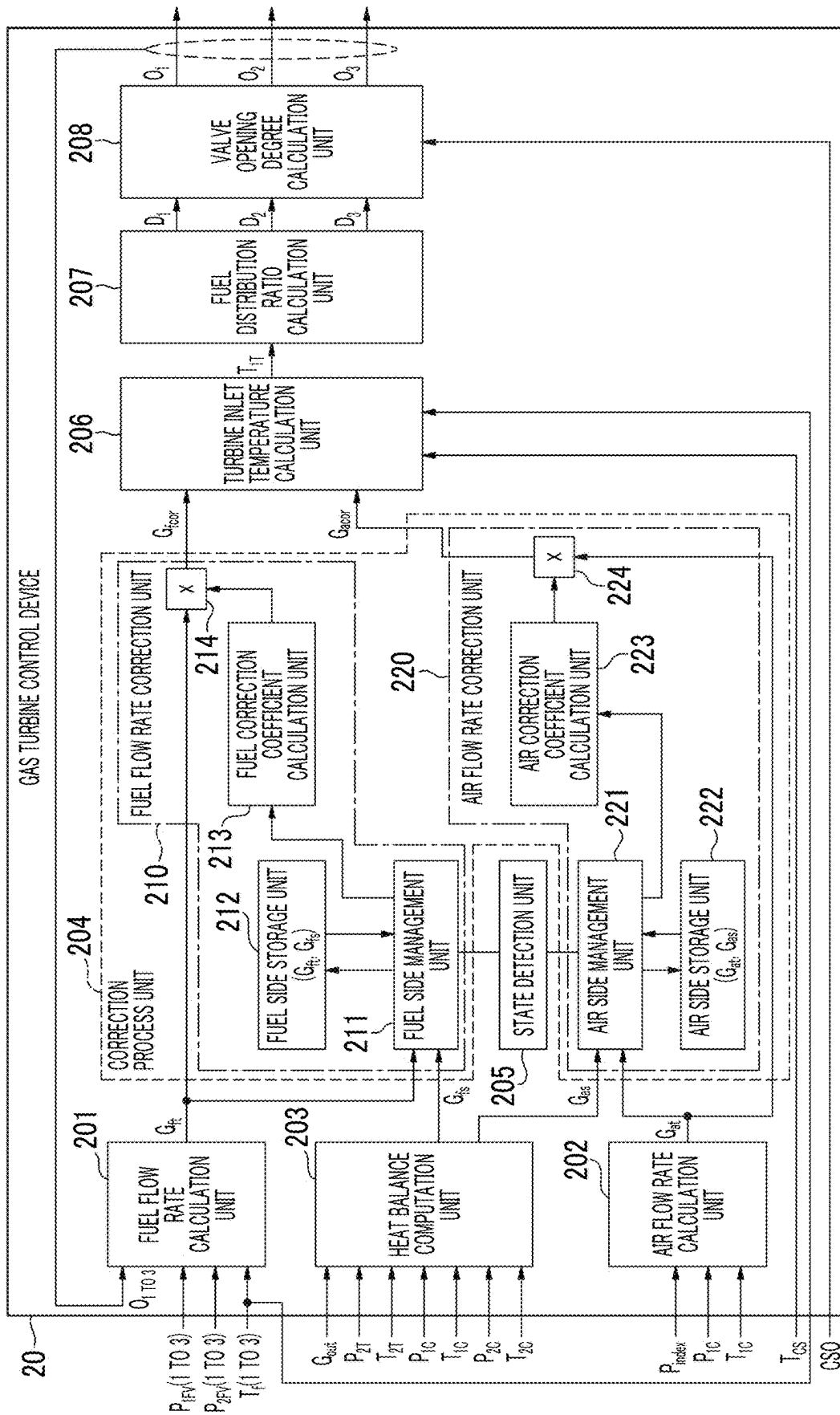


FIG. 3

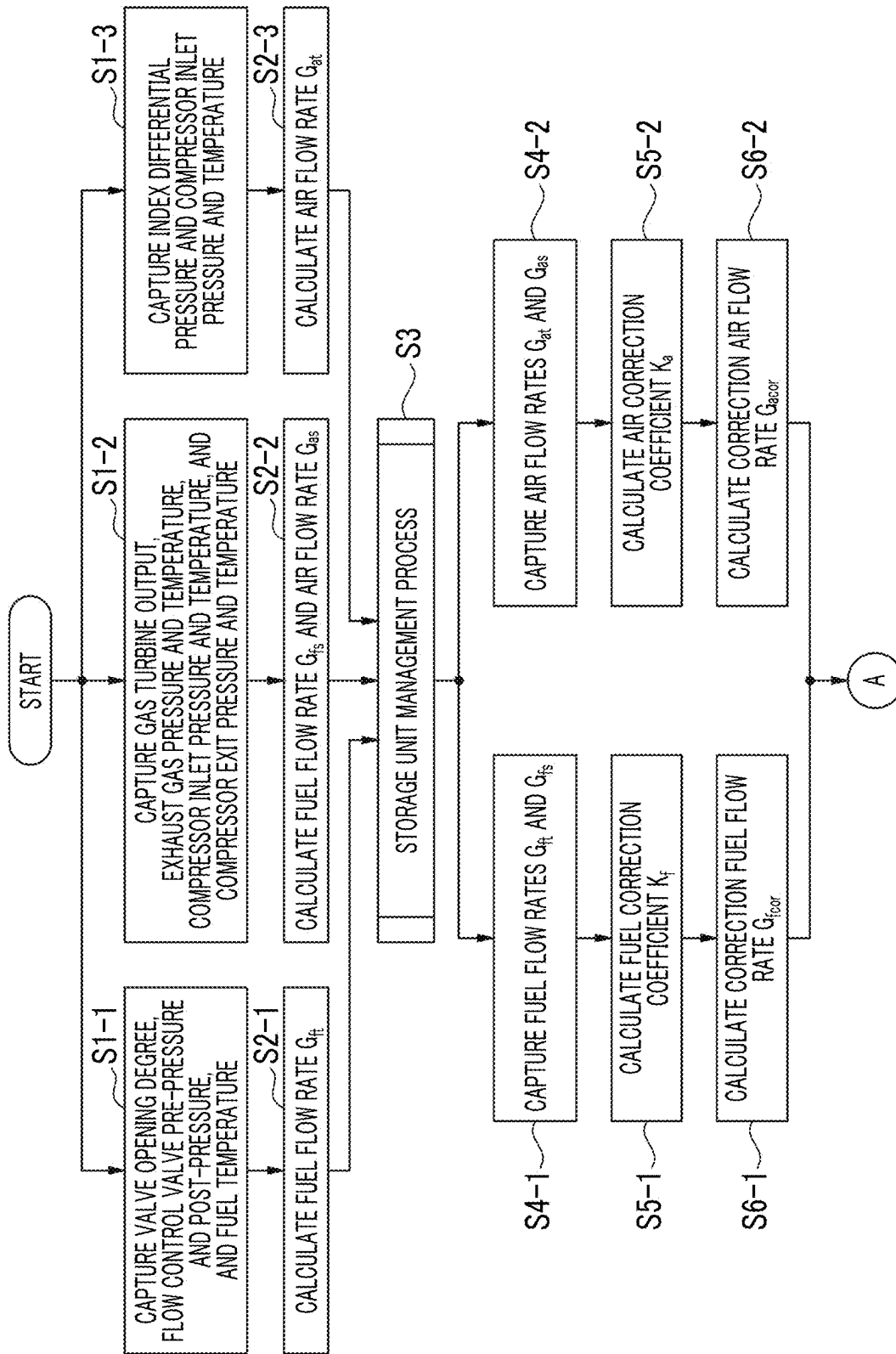


FIG. 4

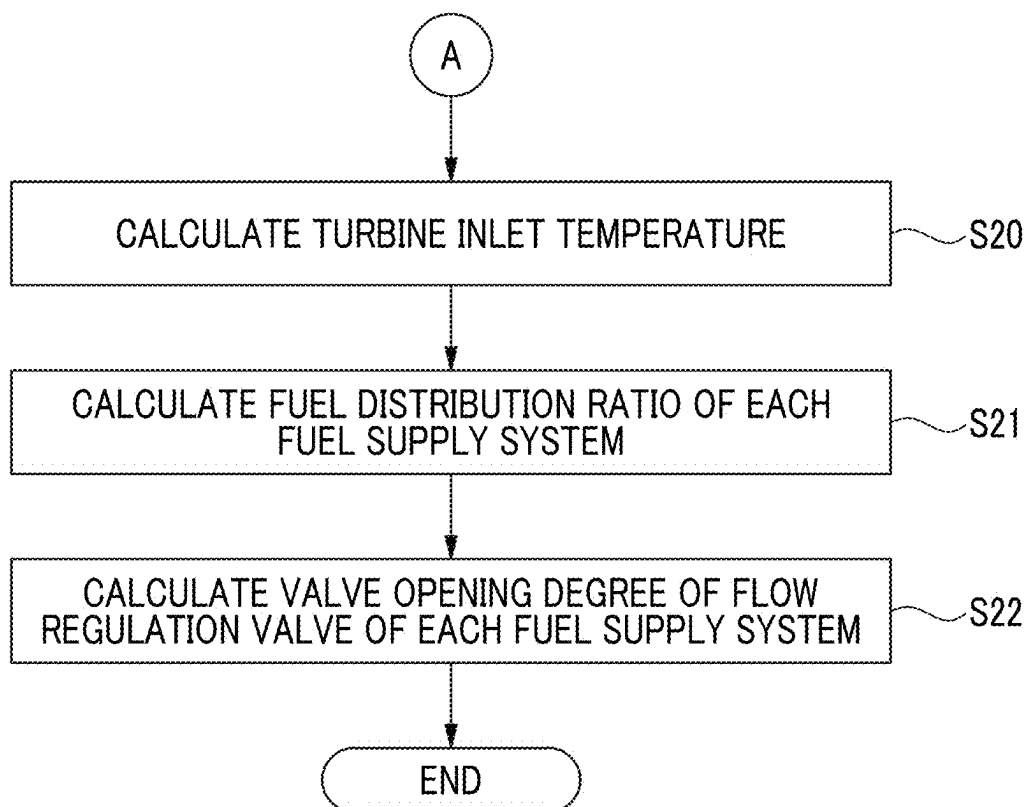


FIG. 5

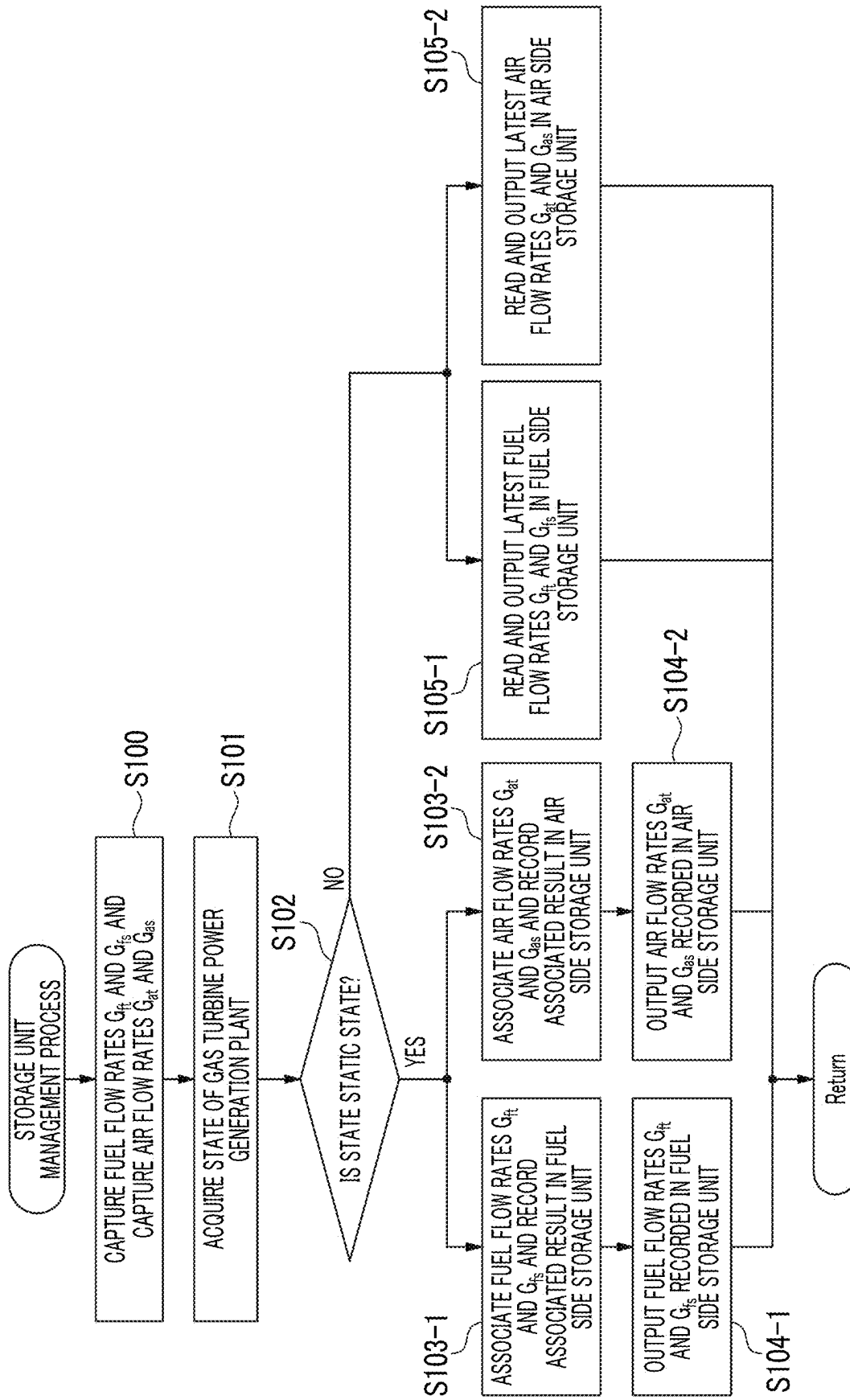


FIG. 6

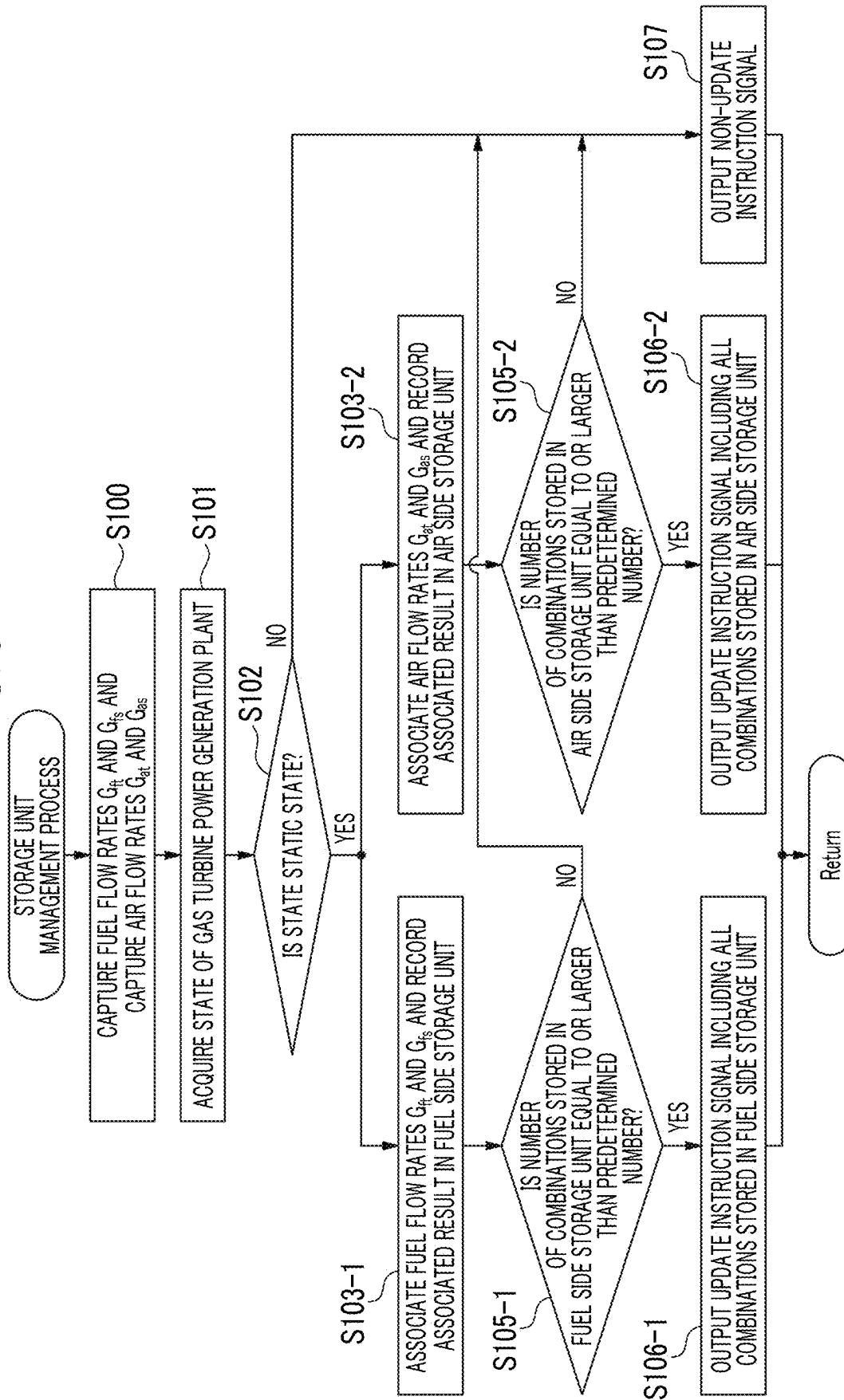


FIG. 7

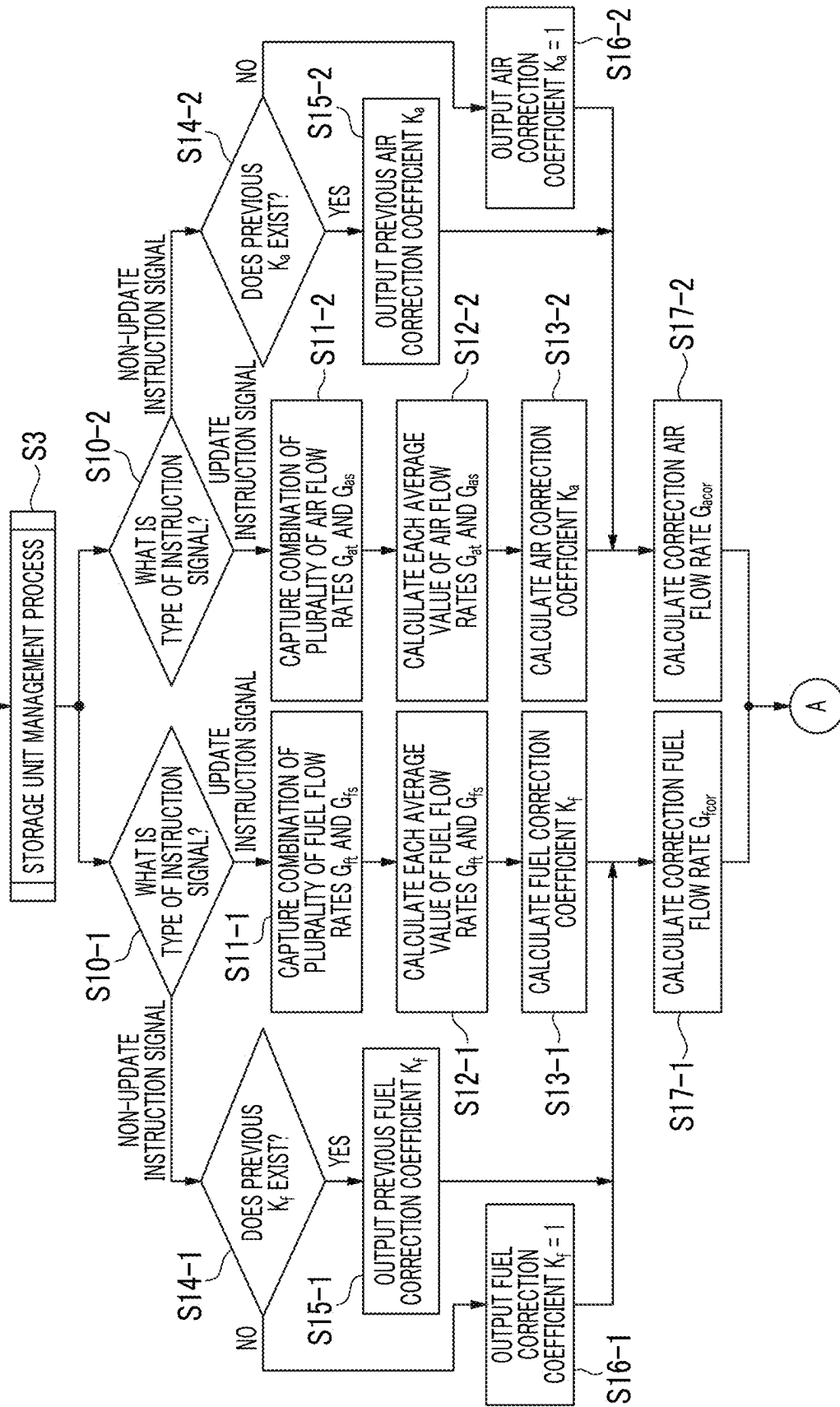


FIG. 8

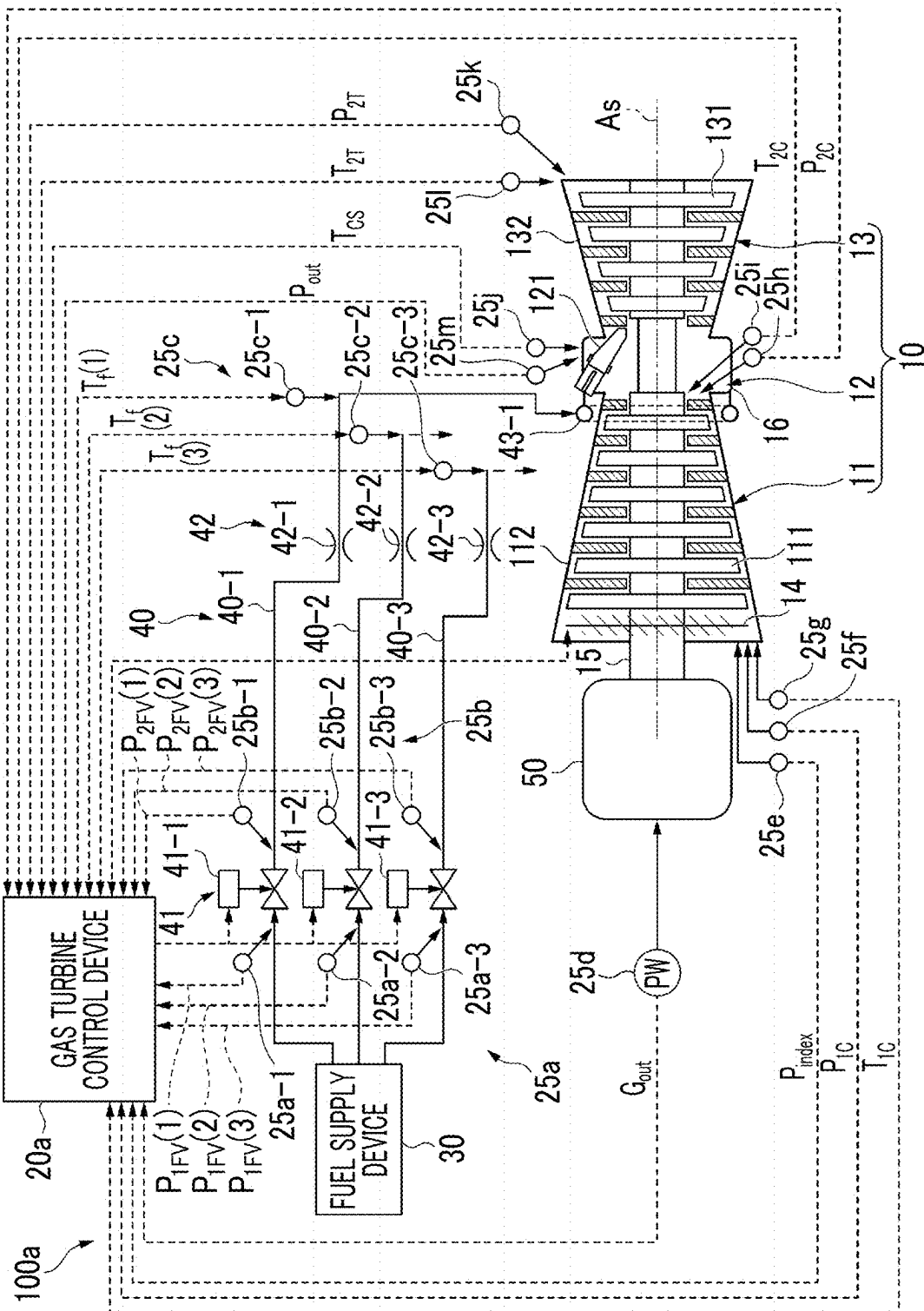


FIG. 9

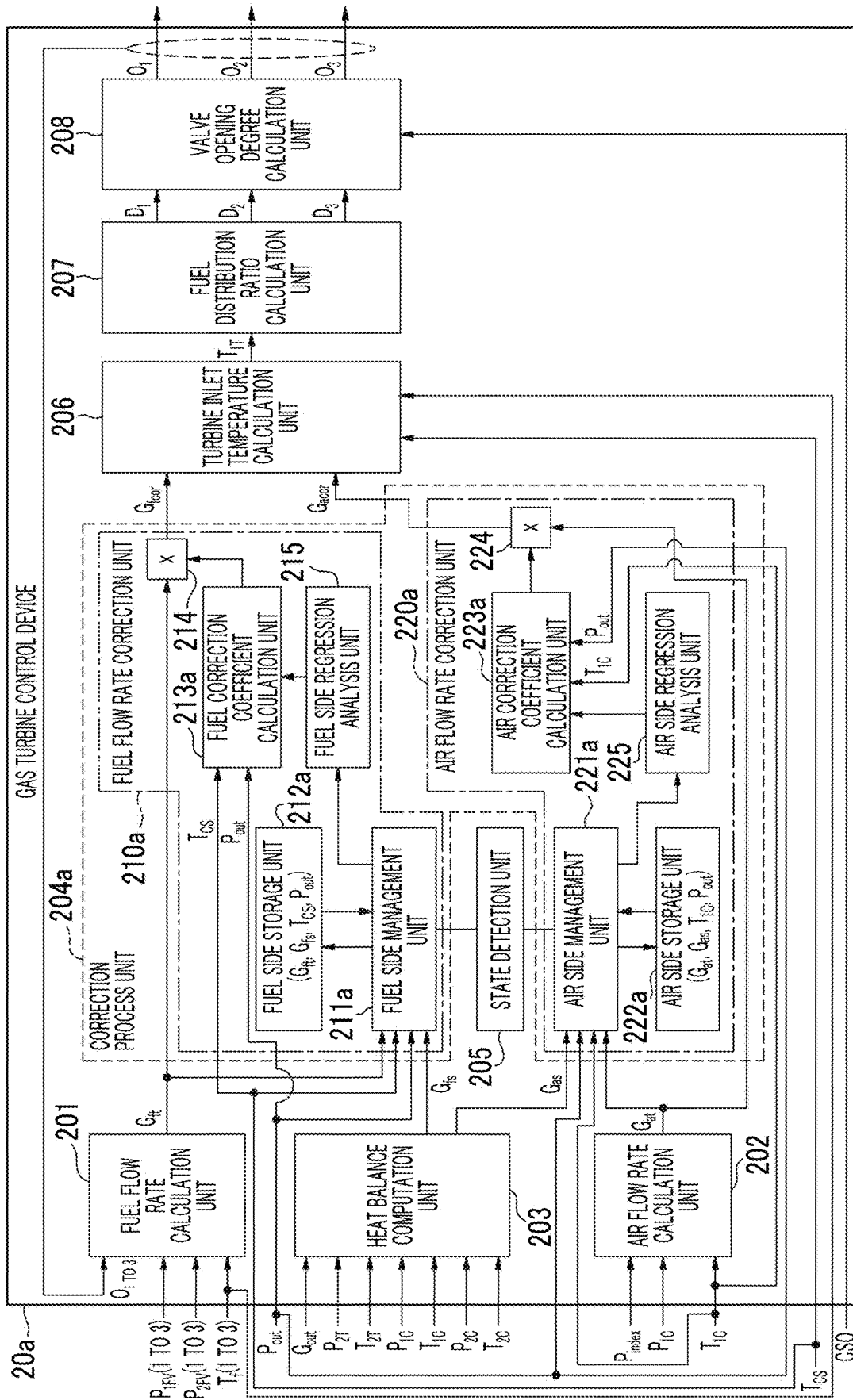


FIG. 10

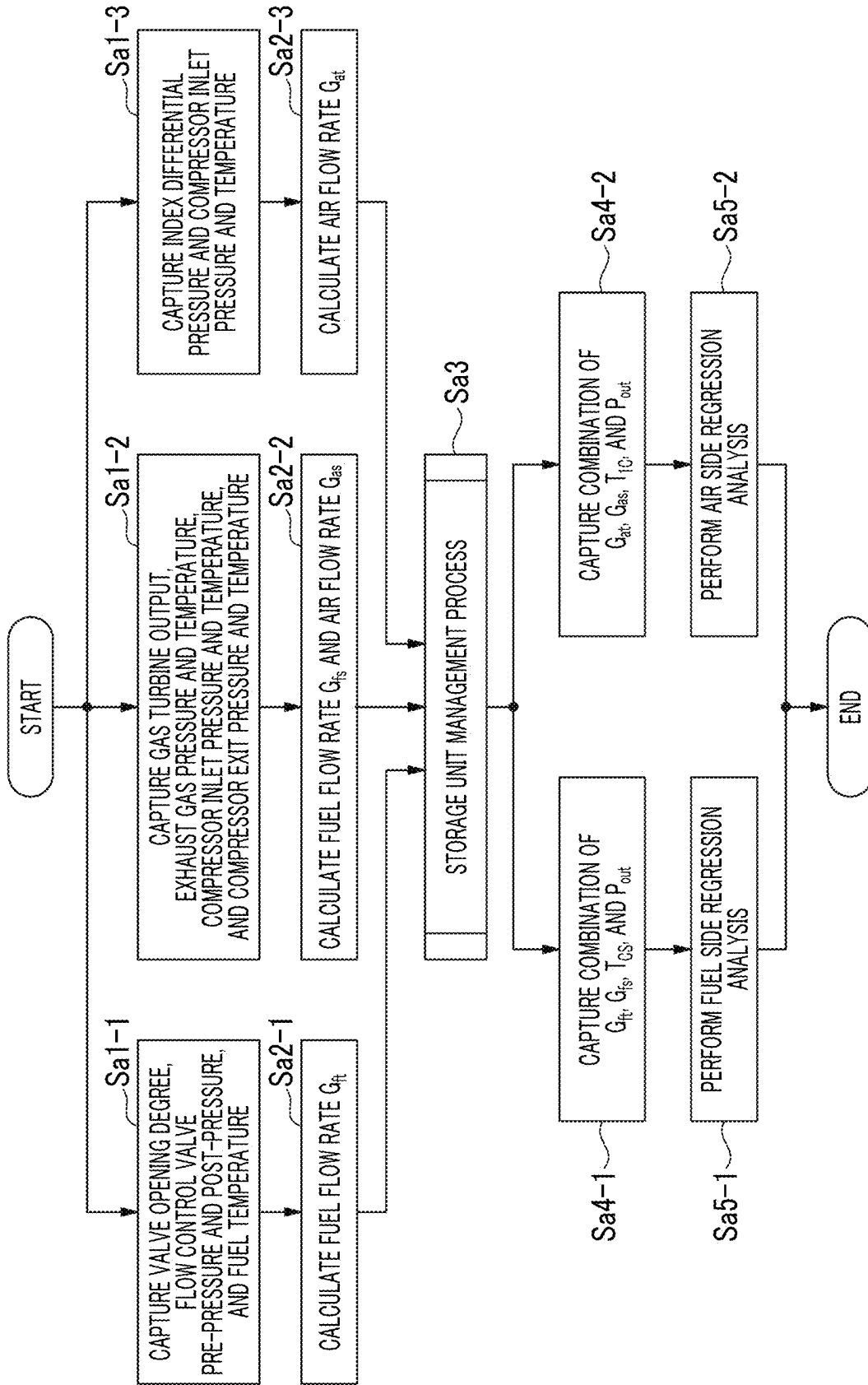


FIG. 11

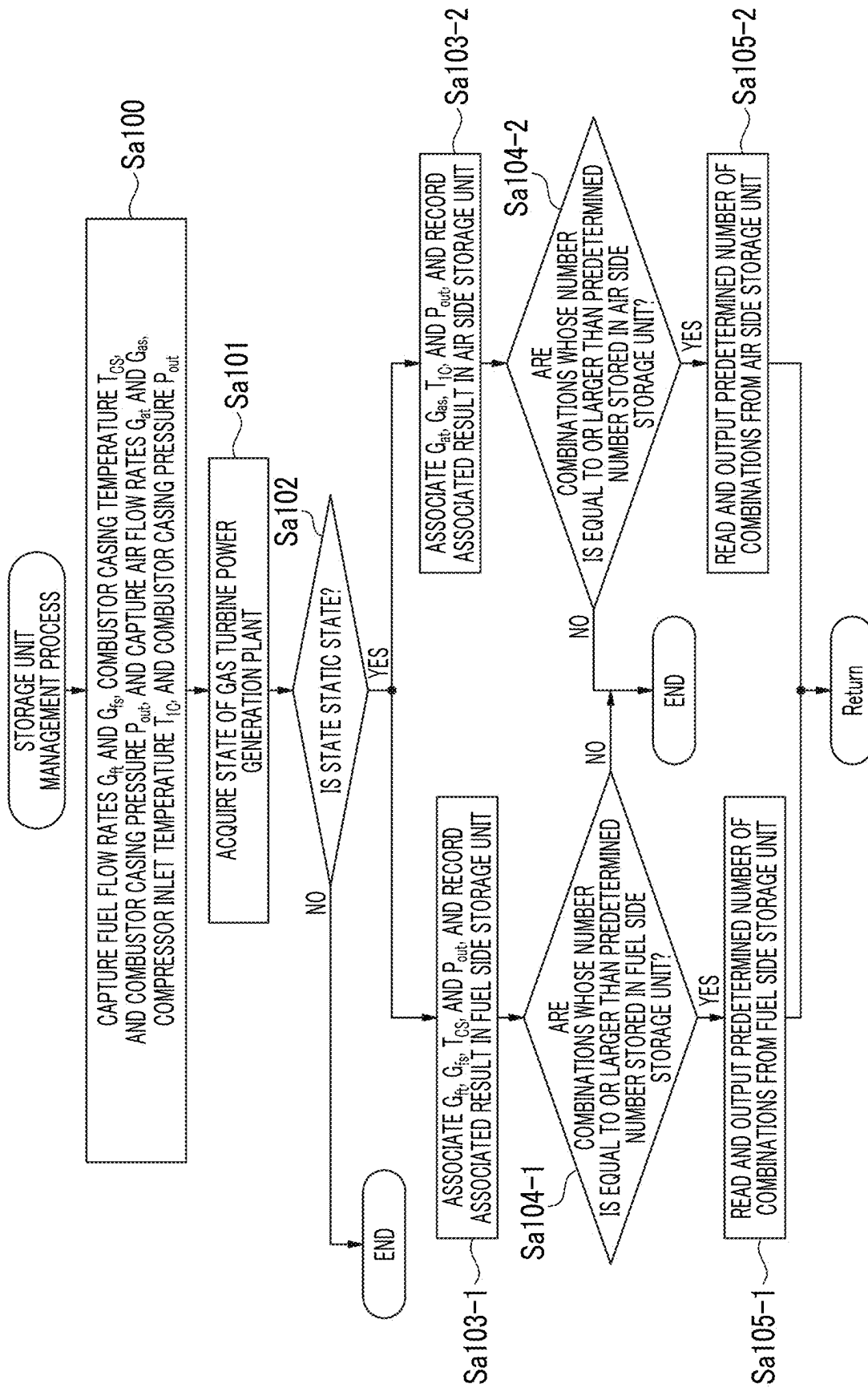


FIG. 12

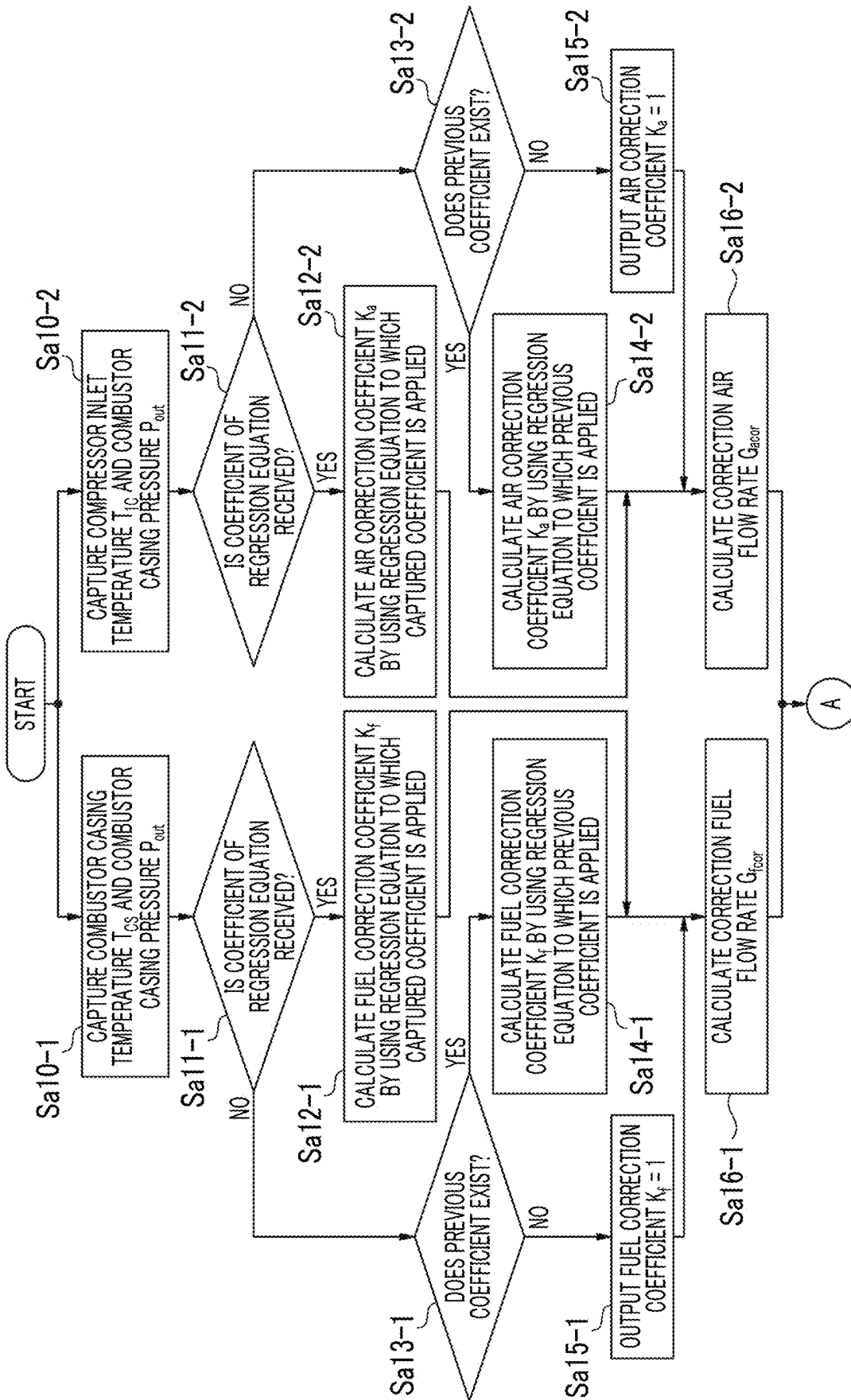


FIG. 13

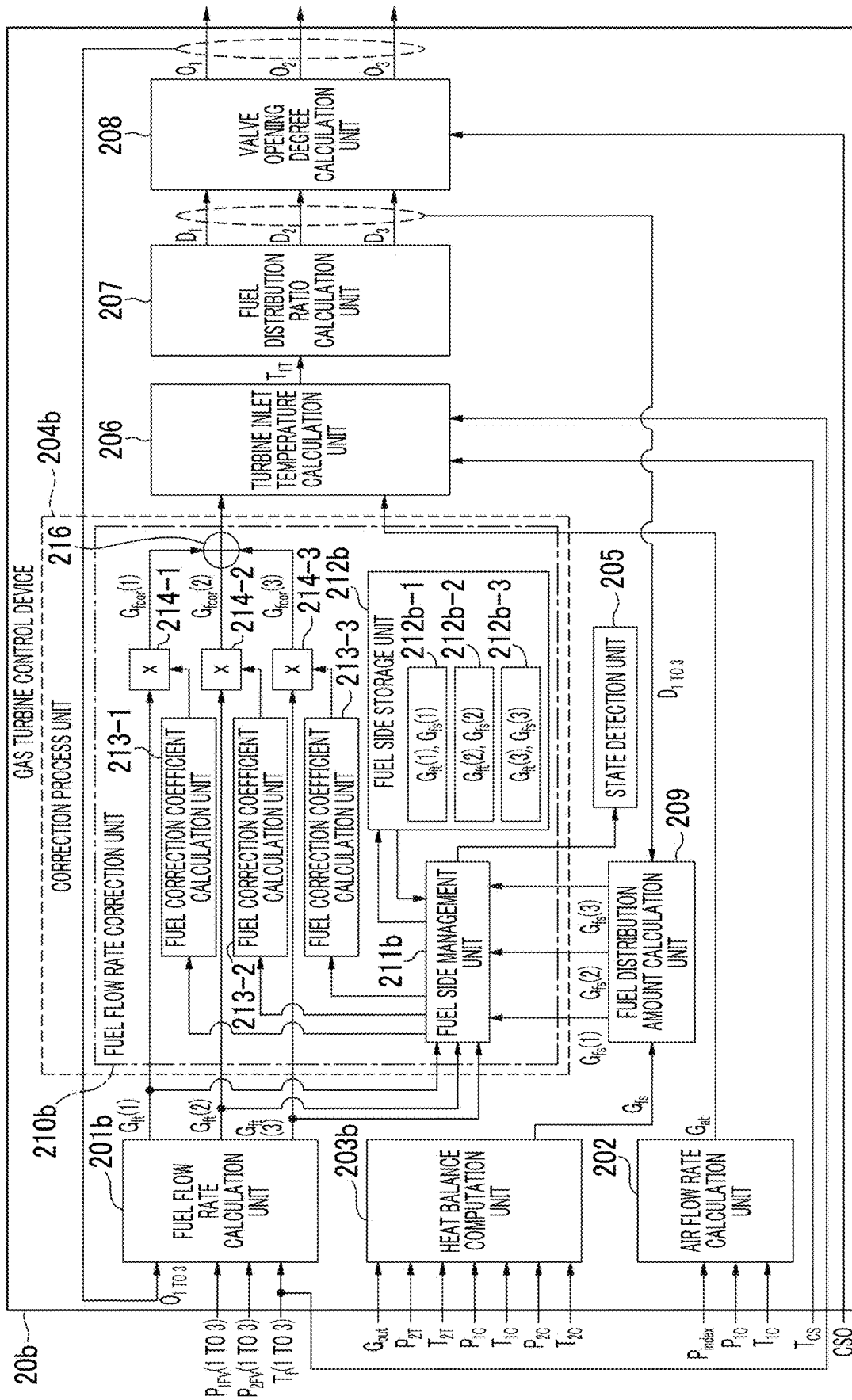


FIG. 14

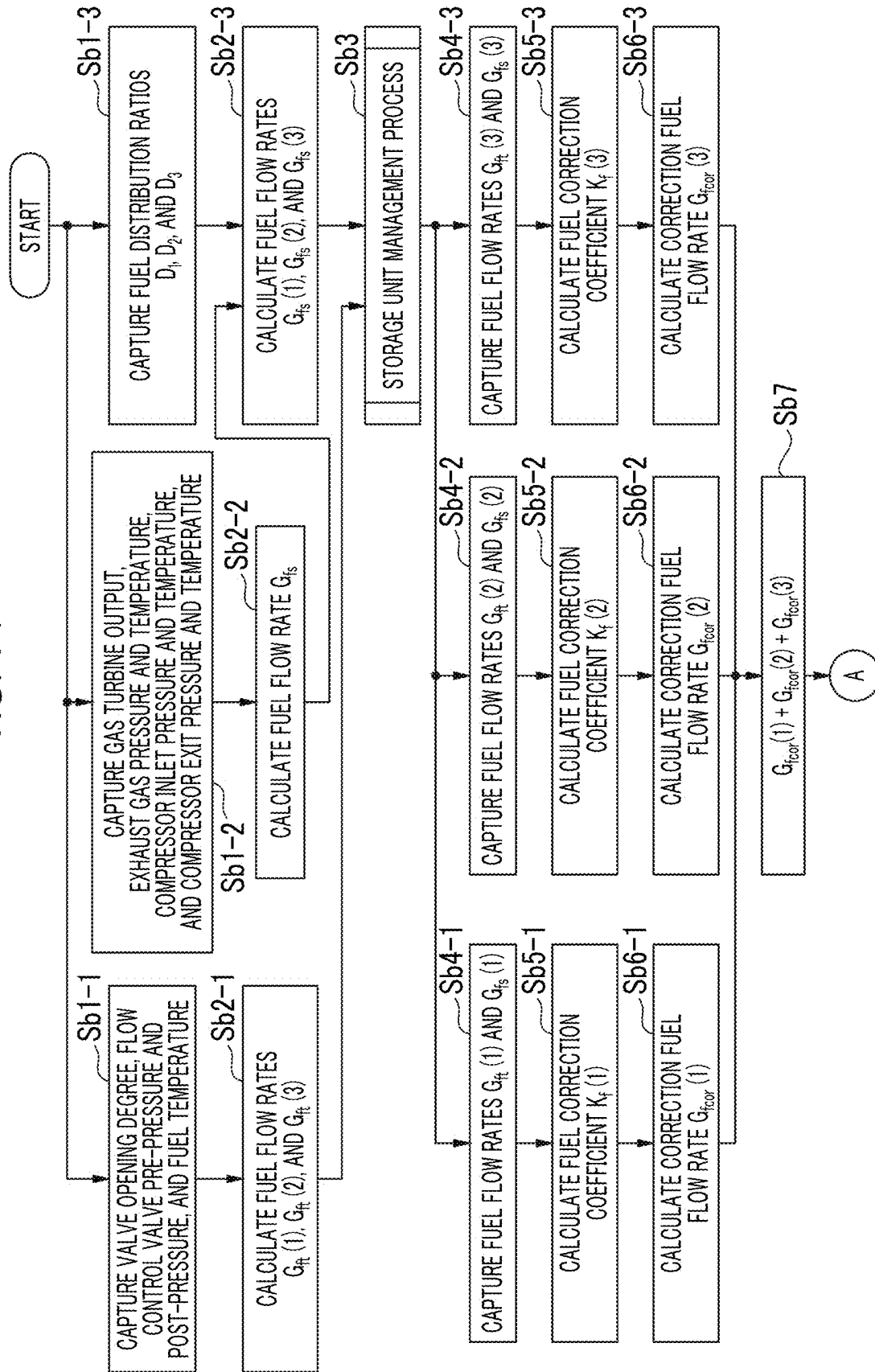


FIG. 15

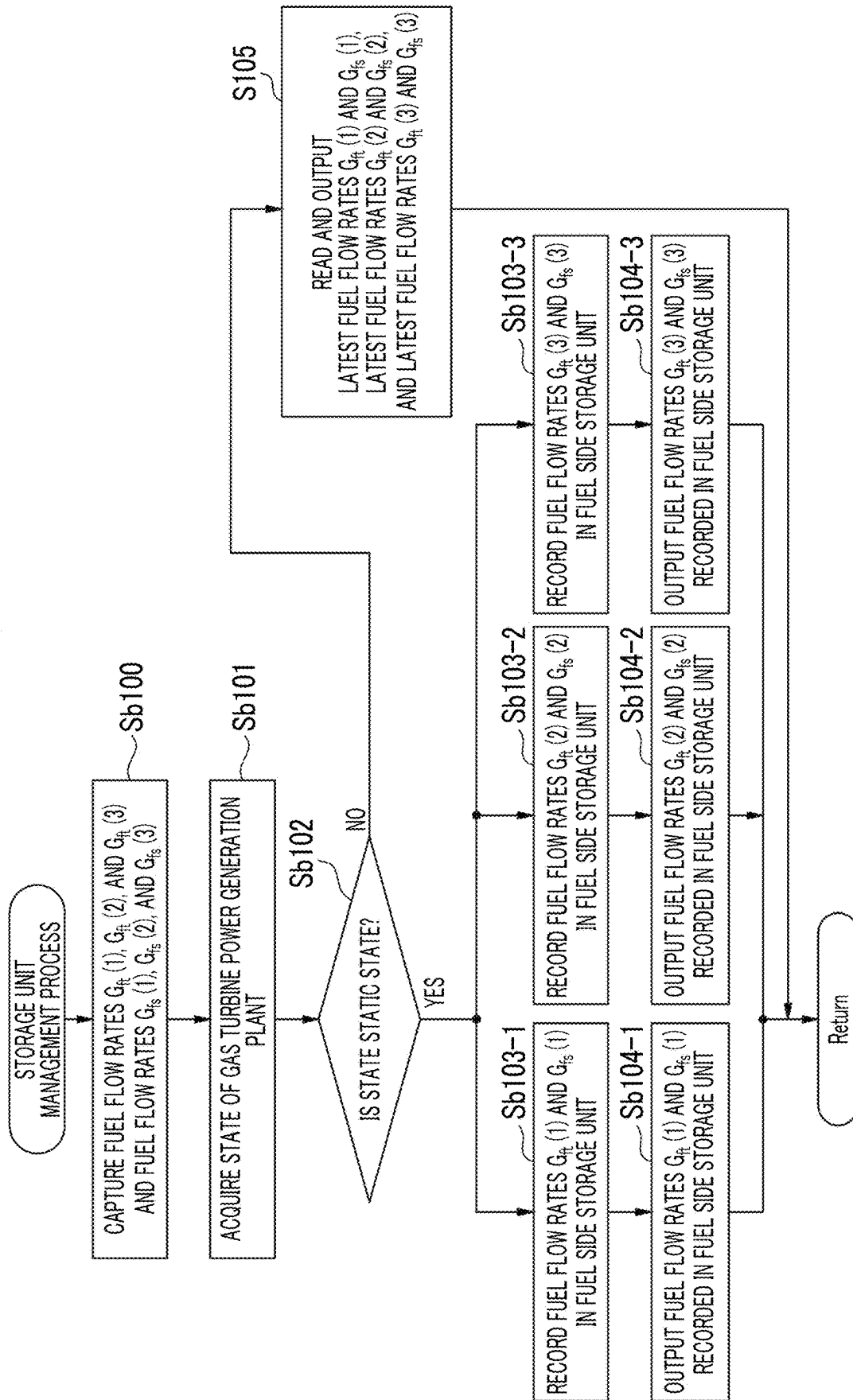
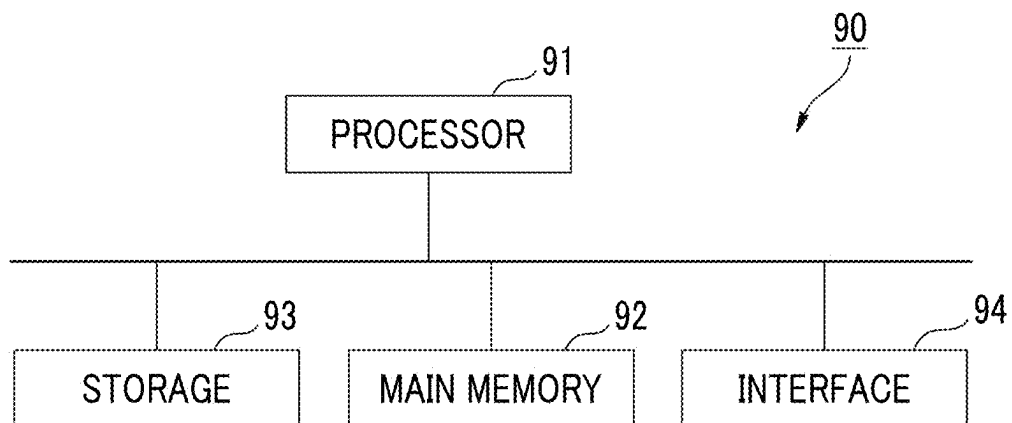


FIG. 16



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GAS TURBINE CONTROL DEVICE, GAS TURBINE CONTROL METHOD, AND PROGRAM

TECHNICAL FIELD

The present disclosure relates to a gas turbine control device, a gas turbine control method, and a program.

Priority is claimed on Japanese Patent Application No. 2022-076848, filed on May 9, 2022, the content of which is incorporated herein by reference.

BACKGROUND ART

In a gas turbine, for example, a fuel is injected to generate a combustion gas in a combustor that captures compressed air generated by a compressor. Power is generated in such a manner that the generated combustion gas drives the turbine. A fuel supply system that supplies the fuel to the combustor may be divided into a plurality of fuel supply systems from the viewpoint of combustion efficiency or combustion stability. A ratio for distributing the fuel to each of the plurality of fuel supply systems, that is, a fuel distribution ratio, can be calculated, based on a turbine inlet temperature.

However, there is no instrument can stably measure the turbine inlet temperature. Therefore, in general, the turbine inlet temperature is estimated, based on measurement data obtained by measurement of a sensor provided in the gas turbine, and the fuel distribution ratio is calculated by using the estimated turbine inlet temperature. Therefore, it is important to ensure accuracy of the estimated turbine inlet temperature. For example, PTL 1 discloses a technique as follows. According to the technique, the turbine inlet temperature after combustion is estimated from a fuel flow rate or an air flow rate flowing into the combustor. The estimated turbine inlet temperature is corrected by using a turbine exit temperature measured by a sensor, that is, by using an exhaust gas temperature. In this manner, accuracy of the turbine inlet temperature is improved.

CITATION LIST

Patent Literature

[PTL 1] Japanese Patent No. 6920456

SUMMARY OF INVENTION

Technical Problem

However, it is difficult to uniquely determine turbine efficiency which is a relationship between the turbine inlet temperature and the exhaust gas temperature. Therefore, the technique disclosed in PTL 1 has a problem as follows. Even when the turbine inlet temperature is corrected by using the exhaust gas temperature, in some cases, correction for improving the accuracy of the turbine inlet temperature may not be available. In addition, even when the turbine efficiency can be uniquely determined, since a gas turbine 10 is operated for a long period of time, performance of the gas turbine 10 is changed due to aged deterioration. Therefore, the uniquely determined turbine efficiency cannot be continuously used for a long period of time. Consequently, there is a problem in that the accuracy of the turbine inlet temperature is less likely to be stably maintained.

The present disclosure is made to solve the above-described problems, and aims to provide a gas turbine control

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device, a gas turbine control method, and a program that can maintain accuracy of a turbine inlet temperature for a long period of time without using turbine efficiency.

Solution to Problem

According to the present disclosure, in order to solve the above-described problems, there is provided a gas turbine control device including a fuel flow rate calculation unit that calculates a fuel flow rate indicating the amount of a fuel flowing into a combustor of a gas turbine, based on measurement data relating to a fuel supply system for supplying the fuel to the combustor; an air flow rate calculation unit that calculates an air flow rate indicating the amount of air suctioned by a compressor of the gas turbine, based on measurement data relating to the compressor; a turbine inlet temperature calculation unit that calculates a turbine inlet temperature, based on the fuel flow rate, the air flow rate, and a physical model equation relating to heat energy balance relating to the combustor; a heat balance computation unit that performs a computation for satisfying heat balance of the gas turbine, based on measurement data relating to energy balance of the gas turbine, and outputs a computation result; a state detection unit that detects a state of the gas turbine; and a correction process unit that corrects any one or both of the fuel flow rate and the air flow rate, based on the computation result output by the heat balance computation unit when the state of the gas turbine which is detected by the state detection unit is a static state, and provides the corrected flow rate to the turbine inlet temperature calculation unit.

According to the present disclosure, there is provided a gas turbine control method including a step of calculating a fuel flow rate indicating the amount of a fuel flowing into a combustor of a gas turbine, based on measurement data relating to a fuel supply system for supplying the fuel to the combustor; a step of calculating an air flow rate indicating the amount of air suctioned by a compressor of the gas turbine, based on measurement data relating to the compressor; a step of calculating a turbine inlet temperature, based on the fuel flow rate, the air flow rate, and a physical model equation relating to heat energy balance relating to the combustor; a step of performing a computation for satisfying heat balance of the gas turbine, based on measurement data relating to energy balance of the gas turbine, and outputting a computation result; a step of detecting a state of the gas turbine; and a step of correcting any one or both of the fuel flow rate and the air flow rate, based on the computation result output in the step of performing the computation for satisfying the heat balance when the state of the gas turbine which is detected in the step of detecting the state is a static state, and providing the corrected flow rate to the step of calculating the turbine inlet temperature.

According to the present disclosure, there is provided a program that causes a computer to function as fuel flow rate calculation means for calculating a fuel flow rate indicating the amount of a fuel flowing into a combustor of a gas turbine, based on measurement data relating to a fuel supply system for supplying the fuel to the combustor; air flow rate calculation means for calculating an air flow rate indicating the amount of air suctioned by a compressor of the gas turbine, based on measurement data relating to the compressor; turbine inlet temperature calculation means for calculating a turbine inlet temperature, based on the fuel flow rate, the air flow rate, and a physical model equation relating to heat energy balance relating to the combustor; heat balance computation means for performing a computation for satis-

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fyng heat balance of the gas turbine, based on measurement data relating to energy balance of the gas turbine, and outputting a computation result; state detection means for detecting a state of the gas turbine; and correction process means for correcting any one or both of the fuel flow rate and the air flow rate, based on the computation result output by the heat balance computation means when the state of the gas turbine which is detected by the state detection means is a static state, and providing the corrected flow rate to the turbine inlet temperature calculation means.

Advantageous Effects of Invention

According to a gas turbine control device, a gas turbine control method, and a program of the present disclosure, accuracy of a turbine inlet temperature can be maintained for a long period of time without using turbine efficiency.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a schematic view illustrating a configuration example of a gas turbine power generation plant according to a first embodiment of the present disclosure.

FIG. 2 is a block diagram illustrating a configuration example of a gas turbine control device according to the first embodiment of the present disclosure.

FIG. 3 is a flowchart (No. 1) illustrating an operation example of the gas turbine control device according to the first embodiment of the present disclosure.

FIG. 4 is a flowchart (No. 2) illustrating an operation example of the gas turbine control device according to the first embodiment of the present disclosure.

FIG. 5 is a flowchart (No. 3) illustrating an operation example of the gas turbine control device according to the first embodiment of the present disclosure.

FIG. 6 is a flowchart (No. 1) illustrating an operation example of a gas turbine control device according to another configuration example of the first embodiment of the present disclosure.

FIG. 7 is a flowchart (No. 2) illustrating an operation example of a gas turbine control device according to another configuration example of the first embodiment of the present disclosure.

FIG. 8 is a schematic view illustrating configuration example of a gas turbine power generation plant according to a second embodiment of the present disclosure.

FIG. 9 is a block diagram illustrating a configuration example of a gas turbine control device according to the second embodiment of the present disclosure.

FIG. 10 is a flowchart (No. 1) illustrating an operation example of the gas turbine control device according to the second embodiment of the present disclosure.

FIG. 11 is a flowchart (No. 2) illustrating an operation example of the gas turbine control device according to the second embodiment of the present disclosure.

FIG. 12 is a flowchart (No. 3) illustrating an operation example of the gas turbine control device according to the second embodiment of the present disclosure.

FIG. 13 is a block diagram illustrating a configuration example of a gas turbine control device according to a third embodiment of the present disclosure.

FIG. 14 is a flowchart (No. 1) illustrating an operation example of the gas turbine control device according to the third embodiment of the present disclosure.

FIG. 15 is a flowchart (No. 2) illustrating an operation example of the gas turbine control device according to the third embodiment of the present disclosure.

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FIG. 16 is a schematic block diagram illustrating a configuration of a computer according to at least one embodiment.

DESCRIPTION OF EMBODIMENTS

Hereinafter, a gas turbine control device, a gas turbine control method, and a program according to embodiments of the present disclosure will be described with reference to FIGS. 1 to 16. FIG. 1 and FIG. 8 are schematic views illustrating configuration examples of gas turbine power generation plants 100 and 100a, respectively, according to first and second embodiments of the present disclosure. FIG. 2, FIG. 9, and FIG. 13 are block diagrams illustrating configuration examples of gas turbine control devices 20, 20a, and 20b, respectively, according to first to third embodiments of the present disclosure. FIGS. 3 to 5 are flowcharts illustrating an operation example of the gas turbine control device 20 according to the first embodiment of the present disclosure. FIGS. 6 and 7 are flowcharts illustrating an operation example of the gas turbine control device 20 according to another configuration example of the first embodiment of the present disclosure. FIGS. 10 to 12 are flowcharts illustrating an operation example of the gas turbine control device 20a according to the second embodiment of the present disclosure. FIGS. 14 and 15 are flowcharts illustrating an operation example of the gas turbine control device 20b according to the third embodiment of the present disclosure. FIG. 16 is a schematic block diagram illustrating a configuration of a computer according to at least one embodiment of the present disclosure. In each drawing, the same reference numerals will be assigned to the same or corresponding configurations, and description thereof will be omitted as appropriate.

In each embodiment according to the present disclosure, as illustrated in FIGS. 2 and 9, description will be made on an assumption that all computations are performed inside a single gas turbine control device. However, this assumption is merely an example of the configuration. That is, in another embodiment, for example, an aspect may be adopted as follows. A fuel flow rate and a turbine inlet temperature may be calculated by the gas turbine control device. Heat balance may be computed, and a correction coefficient may be calculated by a different computer. Alternatively, a combination of each computation process and hardware that executes the computation process can be changed as appropriate.

First Embodiment

(Configuration Example of Gas Turbine Power Generation Plant of First Embodiment)

FIG. 1 is an overall configuration diagram of a gas turbine power generation plant 100 of the first embodiment. The gas turbine power generation plant 100 includes a gas turbine 10, a gas turbine control device 20 that controls the gas turbine 10, a fuel supply device 30 that supplies a fuel, a fuel supply system 40 that guides the fuel supplied by the fuel supply device 30 to the gas turbine 10, a generator 50 driven by power of the gas turbine 10 to generate power, and various sensors 25a to 25f that measure various state amounts and that transmit measurement data detected by the measurement to the gas turbine control device 20.

The gas turbine 10 includes a compressor 11, a combustor 12, and a turbine 13. The compressor 11 suctions air, and compresses the suctioned air to generate high-pressure compressed air. The compressed air generated by the compressor

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11 is supplied to the combustor 12 and the turbine 13 on a downstream side. The compressor 11 includes a compressor rotor 111 that rotates around a main axis As serving as a central axis, a compressor casing 112 that covers the compressor rotor 111 from an outer peripheral side, and an inlet guide vane 14 (IGV) for adjusting an amount of air suctioned by the compressor 11. The inlet guide vane 14 is connected to the gas turbine control device 20 by a control line such as a communication line as indicated by a dotted line arrow, and the amount of the air suctioned by the compressor 11 is adjusted by the inlet guide vane 14 that receives a command value from the gas turbine control device 20.

The turbine 13 includes a turbine rotor 131 that rotates around the main axis As serving as the central axis, and a turbine casing 132 that covers the turbine rotor 131 from an outer peripheral side. The compressor rotor 111 and the turbine rotor 131 are connected to each other, and rotate integrally with the main axis As serving as the central axis. The combustor 12 generates a high-pressure and high-temperature combustion gas by injecting and combusting the fuel supplied from the fuel supply system 40 to the compressed air generated by the compressor 11. The combustor 12 includes a combustor inner cylinder 121 and a combustor casing 16 that covers the combustor inner cylinder 121. The combustor inner cylinder 121 injects and ignites the fuel supplied from the fuel supply system 40 to combust the fuel inside the combustor casing 16. The combustor casing 16 is connected to the compressor casing 112 and the turbine casing 132, and the combustion gas generated in the combustor casing 16 is supplied to the turbine casing 132 to rotate the turbine rotor 131. The compressor rotor 111 is integrally rotated by the rotation of the turbine rotor 131 to suction the air.

The fuel supply system 40 supplies the fuel supplied by the fuel supply device 30 to the combustor 12 at a predetermined pressure and flow rate through a fuel pipe. In FIG. 1, in addition to the fuel pipe connecting the fuel supply device 30 and the gas turbine 10 which is the fuel pipe indicated by a solid line arrow or a broken line arrow, the fuel supply system 40 includes a flow regulation valve (hereinafter, referred to as a flow control valve) 41 provided in the fuel pipe, a nozzle 42, and a manifold pipe 43.

The fuel supply system 40 includes three fuel supply systems of a fuel supply system 40-1, a fuel supply system 40-2, and a fuel supply system 40-3. The fuel supply device 30 branches the supplied fuel into three, and supplies each of the branched fuels to the fuel supply system 40-1, the fuel supply system 40-2, and the fuel supply system 40-3. For example, the fuel supply system 40-1 is a top hat fuel supply system for supplying the fuel to a top hat portion of the combustor 12. The fuel supply system 40-2 is a pilot fuel supply system for supplying the fuel to a central portion of the combustor inner cylinder 121. The fuel supply system 40-3 is a main fuel supply system for supplying the fuel to a portion surrounding the central portion of the combustor inner cylinder 121.

Since the fuel supply system 40 includes the three fuel supply systems 40-1 to 40-3, the flow control valve 41 has flow control valves 41-1, 41-2, and 41-3 corresponding to each of the fuel supply systems 40-1 to 40-3. Each of the flow control valves 41-1, 41-2, and 41-3 is connected to the gas turbine control device 20 by a control line such as a communication line as indicated by the dotted line arrow. When receiving a command value indicating a valve opening degree for each of the flow control valves 41-1, 41-2, and 41-3 from the gas turbine control device 20 through the

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control line, each of the flow control valves 41-1, 41-2, and 41-3 increases or decreases the valve opening degree at an opening degree according to the received command value.

As in the flow control valve 41, the nozzle 42 has nozzles 42-1, 42-2, and 42-3 corresponding to each of the fuel supply systems 40-1 to 40-3. With regard to the manifold pipe 43, only the manifold pipe 43 corresponding to the fuel supply system 40-1 is illustrated in FIG. 1 due to space limitations. However, in reality, the manifold pipes 43 are provided one by one for each of the fuel supply systems 40-1 to 40-3. Hereinafter, when it is necessary to individually describe the manifold pipes 43 corresponding to each of the fuel supply systems 40-1 to 40-3, all of these will be respectively referred to as the manifold pipes 43-1 to 43-3 by associating branch numbers of reference numerals. Each of the manifold pipes 43-1 to 43-3 is connected to the combustor 12. For example, the manifold pipe 43-1 supplies the fuel to the top hat portion of the combustor 12. The manifold pipe 43-2 supplies the fuel to the central portion of the combustor inner cylinder 121. The manifold pipe 43-3 supplies the fuel to the portion surrounding the central portion of the combustor inner cylinder 121.

A rotor 15 is connected to the compressor rotor 111, and integrally rotates around the main axis As serving as the central axis, together with the compressor rotor 111 and the turbine rotor 131. The generator 50 is connected to one end of the compressor rotor 111 via the rotor 15. The generator 50 is driven by the rotation of the rotor 15 to generate power.

Next, the various sensors 25a to 25j connected to the gas turbine control device 20 will be described. Each of the various sensors 25a to 25j is connected to the gas turbine control device 20 by a control line such as a communication line as indicated by a dotted line arrow, and measurement data detected through the measurement is transmitted to the gas turbine control device 20 through the control line.

A flow control valve pre-pressure sensor 25a, a flow control valve post-pressure sensor 25b, and a fuel temperature sensor 25c detect measurement data relating to the fuel supply system 40 as described below. The flow control valve pre-pressure sensor 25a has flow control valve pre-pressure sensors 25a-1, 25a-2, and 25a-3 provided for each of the fuel supply systems 40-1, 40-2, and 40-3. The flow control valve pre-pressure sensors 25a-1, 25a-2, and 25a-3 respectively detect flow control valve pre-pressures $P_{1FV}(1)$, $P_{1FV}(2)$, and $P_{1FV}(3)$ which are pressures of the fuel flowing into the respectively corresponding flow control valves 41-1, 41-2, and 41-3. The flow control valve post-pressure sensor 25b has flow control valve post-pressure sensors 25b-1, 25b-2, and 25b-3 provided for each of the fuel supply systems 40-1, 40-2, and 40-3. The flow control valve post-pressure sensors 25b-1, 25b-2, and 25b-3 respectively detect flow control valve post-pressures $P_{2FV}(1)$, $P_{2FV}(2)$, and $P_{2FV}(3)$ which are pressures of the fuel flowing out from each of the respectively corresponding flow control valves 41-1, 41-2, and 41-3.

The fuel temperature sensor 25c has fuel temperature sensors 25c-1, 25c-2, and 25c-3 provided for each of the fuel supply systems 40-1, 40-2, and 40-3. The fuel temperature sensors 25c-1, 25c-2, and 25c-3 respectively detect fuel temperatures $T_f(1)$, $T_f(2)$, and $T_f(3)$ which are temperatures of the fuel flowing through the fuel pipes of the respectively corresponding fuel supply systems 40-1, 40-2, and 40-3.

A power meter sensor 25d, an index differential pressure sensor 25e, a compressor inlet pressure sensor 25f, a compressor inlet temperature sensor 25g, a compressor exit pressure sensor 25h, a compressor exit temperature sensor 25i, a combustor casing temperature sensor 25j, an exhaust

gas pressure sensor **25k**, and an exhaust gas temperature sensor **25l** detect measurement data relating to the gas turbine as described below. The power meter sensor **25d** detects the power generated by the generator **50**, and detects the measured power as a gas turbine output G_{out} .

The index differential pressure sensor **25e** detects a compressor index differential pressure P_{index} . Here, the compressor index differential pressure P_{index} is a pressure difference between the pressure in a casing suction port of the compressor **11** and the pressure in the vicinity of a vane inside the compressor **11**, and is a value serving as an index of the air flow rate suctioned by the compressor **11**. The compressor inlet pressure sensor **25f** detects a compressor inlet pressure P_{1C} which is the pressure in the casing suction port of the compressor **11**. The compressor inlet temperature sensor detects a **25g** compressor inlet temperature T_{1C} which is the temperature in the casing suction port of the compressor **11**. The compressor exit pressure sensor **25h** detects a compressor exit pressure P_{2C} which is the pressure in an exit of the compressor **11**. The compressor exit temperature sensor **25i** detects a compressor exit temperature T_{2C} which is the temperature in the exit of the compressor **11**. The combustor casing temperature sensor **25j** detects a combustor casing temperature T_{CS} which is a combustion gas temperature, that is, the temperature of the air inside the combustor casing **16**. The exhaust gas pressure sensor **25k** detects an exhaust gas pressure P_{2T} which is the pressure in an exit of the turbine **13**. The exhaust gas temperature sensor **25l** detects an exhaust gas temperature T_{2T} which is the temperature in the exit of the turbine **13**.

(Configuration Example of Gas Turbine Control Device of First Embodiment)

As illustrated in FIG. 2, the gas turbine control device **20** includes a fuel flow rate calculation unit **201**, an air flow rate calculation unit **202**, a heat balance computation unit **203**, a correction process unit **204**, a state detection unit **205**, a turbine inlet temperature calculation unit **206**, a fuel distribution ratio calculation unit **207**, and a valve opening degree calculation unit **208**.

In the gas turbine control device **20**, the fuel flow rate calculation unit **201** calculates a fuel flow rate G_{ft} indicating the amount of the fuel flowing into the combustor **12** of the gas turbine **10** per unit time by substituting each of data indicating valve opening degrees O_1 , O_2 , and O_3 provided as command values to the flow control valve **41** and the measurement data relating to each of the fuel supply systems **40-1**, **40-2**, and **40-3** into a predetermined fuel flow rate calculation function $f_1(\bullet)$. Here, the measurement data relating to each of the fuel supply systems **40-1**, **40-2**, and **40-3** is data of the flow control valve pre-pressures $P_{1FV}(1)$, $P_{1FV}(2)$, and $P_{1FV}(3)$ detected by the flow control valve pre-pressure sensor **25a**, the flow control valve post-pressures $P_{2FV}(1)$, $P_{2FV}(2)$, $P_{2FV}(3)$ detected by the flow control valve post-pressure sensor **25b**, and the fuel temperatures $T_f(1)$, $T_f(2)$, and $T_f(3)$ detected by the fuel temperature sensor **25c**. Based on the data, the fuel flow rate calculation unit **201** performs a computation using the fuel flow rate calculation function $f_1(\bullet)$ for each of the fuel supply systems **40-1**, **40-2**, and **40-3** as represented in Equations (1), (2), and (3) below, and calculates a fuel flow rate $G_{ft}(1)$, a fuel flow rate $G_{ft}(2)$, and a fuel flow rate $G_{ft}(3)$ flowing into the combustor **12** per unit time.

$$G_{ft}(1) = f_1(O_1, P_{1FV}(1), P_{2FV}(1), T_f(1)) \quad (1)$$

$$G_{ft}(2) = f_1(O_2, P_{1FV}(2), P_{2FV}(2), T_f(2)) \quad (2)$$

$$G_{ft}(3) = f_1(O_3, P_{1FV}(3), P_{2FV}(3), T_f(3)) \quad (3)$$

As represented in Equation (4) below, the fuel flow rate calculation unit **201** calculates a fuel flow rate G_{ft} flowing into the combustor **12** through the fuel supply system **40** per unit time by summing up a fuel flow rate $G_{ft}(1)$ corresponding to the fuel supply system **40-1**, a fuel flow rate $G_{ft}(2)$ corresponding to the fuel supply system **40-2**, and a fuel flow rate $G_{ft}(3)$ corresponding to the fuel supply system **40-3**.

$$G_{ft} = G_{ft}(1) + G_{ft}(2) + G_{ft}(3) \quad (4)$$

As represented in Equation (5) below, the air flow rate calculation unit **202** calculates an air flow rate G_{st} indicating the amount of the air suctioned by the compressor **11** of the gas turbine **10** per unit time by substituting the measurement data relating to the compressor **11** of the gas turbine **10**, that is, the compressor index differential pressure P_{index} detected by the index differential pressure sensor **25e**, the compressor inlet pressure P_{1C} detected by the compressor inlet pressure sensor **25f**, and the compressor inlet temperature T_{1C} detected by the compressor inlet temperature sensor **25g**, into a predetermined air flow rate calculation function $f_2(\bullet)$.

$$G_{at} = f_2(P_{index}, P_{1C}, T_{1C}) \quad (5)$$

The heat balance computation unit **203** calculates a fuel flow rate G_{fs} and an air flow rate G_{as} when the measurement data is obtained, by applying the measured measurement data to overall energy balance of the gas turbine **10**, that is, a computation equation indicating that work balance or heat balance matches thereafter, referred to as a gas turbine heat balance computation equation). For example, the measurement data applied to the gas turbine heat balance computation equation by the heat balance computation unit **203** is the gas turbine output G_{out} detected by the power meter sensor **25d**, the exhaust gas pressure P_{2T} detected by the exhaust gas pressure sensor **25k**, the exhaust gas temperature T_{2T} detected by the exhaust gas temperature sensor **25l**, the compressor inlet pressure P_{1C} detected by the compressor inlet pressure sensor **25f**, the compressor inlet temperature T_{1C} detected by the compressor inlet temperature sensor **25g**, the compressor exit pressure P_{2C} detected by the compressor exit pressure sensor **25h**, and the compressor exit temperature T_{2C} detected by the compressor exit temperature sensor **25i**.

The state detection unit **205** detects whether a state of the gas turbine power generation plant **100** is a transient state or a static state, based on a temporal change amount of a state amount of the gas turbine power generation plant **100** represented by various measurement data detected by the various sensors **25a** to **25l** connected to the gas turbine control device **20** or on the power generated by the generator **50**. For example, the state detection unit **205** detects the state of the gas turbine power generation plant **100** as the static state when the temporal change amount in the state amount is smaller than a predetermined threshold value. In contrast,

the state detection unit **205** detects the state of the gas turbine power generation plant **100** as the transient state when the temporal change amount is equal to or greater than the threshold value.

The correction process unit **204** includes a fuel flow rate correction unit **210** and an air flow rate correction unit **220**. The fuel flow rate correction unit **210** includes a fuel side management unit **211**, a fuel side storage unit **212**, a fuel correction coefficient calculation unit **213**, and a fuel correction coefficient application unit **214**. The fuel side management unit **211** associates the fuel flow rate G_{ft} calculated by the fuel flow rate calculation unit **201** and the fuel flow rate G_{fs} calculated by the heat balance computation unit **203** with each other in accordance with a state detected by the state detection unit **205**, and writes and stores a result in the fuel side storage unit **212**. The fuel side management unit **211** outputs a latest combination of the fuel flow rate G_{ft} and the fuel flow rate G_{fs} to the fuel correction coefficient calculation unit **213**.

The fuel side storage unit **212** stores the combination of the fuel flow rate G_{ft} and the fuel flow rate G_{fs} in a time-series order. The fuel correction coefficient calculation unit **213** calculates a fuel correction coefficient $K_f (=G_{fs}/G_{ft})$ by dividing the latest fuel flow rate G_{fe} by the latest fuel flow rate G_{ft} . For example, the fuel correction coefficient application unit **214** is a multiplier, corrects the fuel flow rate G_{ft} by multiplying the fuel flow rate G_{ft} calculated by the fuel flow rate calculation unit **201** by the fuel correction coefficient K_f calculated by the fuel correction coefficient calculation unit **213**, and outputs a correction fuel flow rate $G_{fcor} (=G_{ft} \times K_f)$ which is the corrected fuel flow rate G_{ft} .

The air flow rate correction unit **220** includes an air side management unit **221**, an air side storage unit **222**, an air correction coefficient calculation unit **223**, and an air correction coefficient application unit **224**. The air side management unit **221** associates the air flow rate G_{at} calculated by the air flow rate calculation unit **202** and the air flow rate G_{as} calculated by the heat balance computation unit **203** with each other in accordance with a state detected by the state detection unit **205**, and writes and stores the associated result in the air side storage unit **222**. The air side management unit **221** outputs the latest combination of the air flow rate G_{at} and the air flow rate G_{as} to the air correction coefficient calculation unit **223**.

The air side storage unit **222** stores the combination of the air flow rate G_{at} and the air flow rate G_{as} in a time-series order. The air correction coefficient calculation unit **223** divides the latest air flow rate G_{as} by the latest air flow rate G_{at} to calculate an air correction coefficient $K_a (=G_{at}/G_{as})$. For example, the air correction coefficient application unit **224** is a multiplier, corrects the air flow rate G_{at} by multiplying the air flow rate G_{at} calculated by the air flow rate calculation unit **202** by the air correction coefficient K_a calculated by the air correction coefficient calculation unit **223**, and outputs a correction air flow rate $G_{acor} (=G_{at} \times K_a)$ which is the corrected air flow rate G_{at} .

As represented in Equation (6) below, the turbine inlet temperature calculation unit **206** calculates the turbine inlet temperature T_{1T} by substituting the correction fuel flow rate G_{fcor} , the correction air flow rate G_{acor} , the combustor casing temperature T_{c3} detected by the combustor casing temperature sensor **25j**, and the fuel temperature T_f which is an average value of the fuel temperatures $T_f(1)$, $T_f(2)$, and $T_f(3)$ detected by the fuel temperature sensors **25c-1** to **25c-3** into an equation for solving the heat balance around the predetermined combustor **12**; that is, a function $f_3(\bullet)$

which is a physical model equation relating to the heat energy balance relating to the combustor **12**.

$$T_{1T} = f_3(G_{fcor}, G_{acor}, T_{c3}, T_f) \quad (6)$$

Here, for example, the physical model equation relating to the heat energy balance represented by the function $f_3(\bullet)$ of Equation (6) above is an equation obtained by modifying a non-stationary physical model. For example, the equation representing the non-stationary physical model is a model equation representing that the heat energy flowing into the combustor **12** and the heat energy flowing out from the combustor **12** are equal to each other. A physical model equation relating to the heat energy balance can be obtained by being transformed into an equation in which the turbine inlet temperature T_{1T} included in the model equation is set to a left side and other elements are set to a right side. The heat energy flowing into the combustor **12** in the non-stationary physical model is represented by a total of the heat energy of the fuel, the heat energy of the air, and the heat generation energy of the combustion gas. The heat energy flowing out from the combustor **12** is represented by the heat energy in the inlet of the turbine **13**.

Based on the turbine inlet temperature T_{1T} calculated by the turbine inlet temperature calculation unit **206**, the fuel distribution ratio calculation unit **207** calculates fuel distribution ratios D_1 , D_2 , and D_3 with respect to each of the fuel supply systems **40-1**, **40-2**, and **40-3** by using a relational equation between the predetermined turbine inlet temperature T_{1T} and the fuel distribution ratio. The total of the three fuel distribution ratios D_1 , D_2 , and D_3 calculated by the fuel distribution ratio calculation unit **207** is 100%.

The valve opening degree calculation unit **208** calculates the valve opening degrees O_1 , O_2 , and O_3 for each of the flow control valves **41-1**, **41-2**, and **41-3** by using a predetermined valve opening degree calculation equation, based on the fuel distribution ratios D_1 , D_2 , and D_3 calculated by the fuel distribution ratio calculation unit **207** and on a fuel control signal command value (CSO: control signal output) provided from the outside. The fuel control signal command value (CSO) is a control output signal for controlling the fuel flow rate to be supplied to the combustor **12**. The valve opening degree calculation unit **208** provides each of the calculated valve opening degrees O_1 , O_2 , and O_3 as command values for the respectively corresponding flow control valves **41-1**, **41-2**, and **41-3**. In this manner, the opening degree of the flow control valve **41-1** is adjusted in accordance with the valve opening degree O_1 . The opening degree of the flow control valve **41-2** is adjusted in accordance with the valve opening degree O_2 . The opening degree of the flow control valve **41-3** is adjusted in accordance with the valve opening degree O_3 . The valve opening degree calculation unit **208** feeds back the calculated valve opening degrees O_1 , O_2 , and O_3 to the fuel flow rate calculation unit **201**. (Operation Example of Gas Turbine Control Device of First Embodiment)

Processes performed by the gas turbine control device **20** will be described with reference to FIGS. **3** to **5**. The flow control valve pre-pressure sensors **25a-3** to **25a-3**, the flow control valve post-pressure sensors **25b-1** to **25b-3**, and the fuel temperature sensors **25c-1** to **25c-3** perform the measurement at each detection cycle, and transmit the measurement data detected by performing the measurement to the gas turbine control device **20** through the control line.

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The fuel flow rate calculation unit **201** receives and captures the flow control valve pre-pressures P_{1FV} (**1**), P_{1FV} (**2**), and P_{1FV} (**3**) transmitted by the flow control valve pre-pressure sensors **25a-1** to **25a-3**, the flow control valve post-pressures P_{2FV} (**1**), P_{2FV} (**2**), and P_{2FV} (**3**) transmitted by the flow control valve post-pressure sensors **25b-1** to **25b-3**, and the fuel temperatures T_f (**1**), T_f (**2**), and T_f (**3**) transmitted by the fuel temperature sensors **25c-1** to **25c-3**. The fuel flow rate calculation unit **201** captures the valve opening degrees O_1 , O_2 , and O_3 calculated and output by the valve opening degree calculation unit **208** in the previous detection cycle (**S1-1**).

The fuel flow rate calculation unit **201** substitutes the data captured in the process in Step **S1-1** into Equations (1), (2), and (3) as described above, and calculates the fuel flow rates G_{f1} (**1**), G_{f1} (**2**), and G_{f1} (**3**). As represented in Equation (4), the fuel flow rate calculation unit **201** sums up the fuel flow rates G_{f1} (**1**), G_{f1} (**2**), and G_{f1} (**3**), and calculates the fuel flow rate G_{f1} indicating the amount of the fuel flowing into the combustor **12** of the gas turbine **10** per unit time. The fuel flow rate calculation unit **201** outputs the calculated fuel flow rate G_{f1} to the fuel side management unit **211** and the fuel correction coefficient application unit **214** (**S2-1**).

The power meter sensor **25d**, the exhaust gas pressure sensor **25k**, the exhaust gas temperature sensor **25l**, the compressor inlet pressure sensor **25f**, the compressor inlet temperature sensor **25g**, the compressor exit pressure sensor **25h**, and the compressor exit temperature sensor **25i** transmit the measurement data detected by performing the measurement at each detection cycle to the gas turbine control device **20** through the control line.

The heat balance computation unit **203** receives and captures the gas turbine output G_{out} transmitted by the power meter sensor **25d**, the exhaust gas pressure P_{2T} transmitted by the exhaust gas pressure sensor **25k**, the exhaust gas temperature T_{2T} transmitted by the exhaust gas temperature sensor **25l**, the compressor inlet pressure P_{1C} transmitted by the compressor inlet pressure sensor **25f**, the compressor inlet temperature T_{1C} transmitted by the compressor inlet temperature sensor **25g**, the compressor exit pressure P_{2C} transmitted by the compressor exit pressure sensor **25h**, and the compressor exit temperature T_{2C} transmitted by the compressor exit temperature sensor **25i** (**S1-2**). The heat balance computation unit **203** calculates the fuel flow rate G_{fs} and the air flow rate G_{as} , based on the data captured in the process in Step **S1-2** and on the gas turbine heat balance computation equation. The heat balance computation unit **203** outputs the calculated fuel flow rate G_{fs} to the fuel side management unit **211**, and outputs the calculated air flow rate G_{as} to the air side management unit **221** (**S2-2**).

The index differential pressure sensor **25e** performs the measurement at each detection cycle, and transmits the data of the compressor index differential pressure P_{index} which is the measurement data detected by performing the measurement to the gas turbine control device **20** through the control line. The air flow rate calculation unit **202** receives and captures the compressor index differential pressure P_{index} transmitted by the index differential pressure sensor **25e**, the compressor inlet pressure P_{1C} transmitted by the compressor inlet pressure sensor **25f**, and the compressor inlet temperature T_{1C} transmitted by the compressor inlet temperature sensor **25g** (**S1-3**). As described above, the air flow rate calculation unit **202** substitutes the data captured in the process in Step **S1-3** into Equation (5), and calculates the air flow rate G_{at} indicating the amount of the air auctioned by the compressor **11** of the gas turbine **10** per unit time. The air flow rate calculation unit **202** outputs the calculated air

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flow rate G_{at} to the air side management unit **221** and the air correction coefficient application unit **224** (**S2-3**). In FIG. 3, a set of processes in Steps **S1-1** and **S2-1**, a set of processes in Steps **S1-2** and **S2-2**, and a set of processes in Steps **S1-3** and **S2-3** are performed in parallel.

When receiving the fuel flow rate G_{fs} , the fuel flow rate G_{fs} , the air flow rate G_{at} , and the air flow rate G_{as} , the correction process unit **204** starts a subroutine of a storage unit management process illustrated in FIG. 5 (**S3**).

In the fuel flow rate correction unit **210**, the fuel side management unit **211** captures the fuel flow rate G_{f1} output by the fuel flow rate calculation unit **201** and the fuel flow rate G_{fs} output by the heat balance computation unit **203**. In the air flow rate correction unit **220**, the air side management unit **221** captures the air flow rate G_{at} output by the air flow rate calculation unit **202** and the air flow rate G_{as} output by the heat balance computation unit **203** (**S100**). Each of the fuel side management unit **211** and the air side management unit **221** acquires the data indicating a state of the gas turbine power generation plant **100** detected by the state detection unit **205** from the state detection unit **205** at a time at which the process in Step **S100** performed by each of the fuel side management unit **211** and the air side management unit **221** is completed (**S101**). Each of the fuel side management unit **211** and the air side management unit **221** determines whether or not the acquired data indicating the state indicates a static state (**S102**).

When the fuel side management unit **211** determines in the process in Step **S102** that the acquired data indicating the state indicates the static state (**S102**, Yes), the fuel side management unit **211** associates the fuel flow rate G_{f1} and the fuel flow rate G_{fs} which are captured in the process in Step **S100** in the fuel side storage unit **212** with each other, and writes and stores the associated result in the fuel side storage unit **212**. The fuel side management unit **211** writes a combination of the fuel flow rate G_{f1} and the fuel flow rate G_{fs} which are sequentially written in the fuel side storage unit **212** to be stored in the fuel side storage unit **212** in a time-series order, for example, so that a newly written combination is stored in the fuel side storage unit **212** at the beginning of the time-series order (**S103-1**). The fuel side management unit **211** outputs the combination of the fuel flow rate G_{f1} and the fuel flow rate G_{fs} written in the fuel side storage unit **212** to the fuel correction coefficient calculation unit **213** (**S104-1**).

On the other hand, when the fuel side management unit **211** determines in the process in Step **S102** that the acquired data indicating the state does not indicate the static state (**S102**, No), the fuel side management unit **211** discards the fuel flow rate G_{f1} and the fuel flow rate G_{fs} which are captured in the process in Step **S100**, and reads the latest combination of the fuel flow rate G_{f1} and the fuel flow rate G_{fs} out of the combinations stored in the fuel side storage unit **212**. As described above, when the fuel side storage unit **212** stores the latest combination to be the beginning of the time-series order, the fuel side management unit **211** reads the combination stored at the beginning of the fuel side storage unit **212** as the latest combination of the fuel flow rate G_{f1} and the fuel flow rate G_{fs} . The fuel side management unit **211** outputs the read latest combination of the fuel flow rate G_{f1} and the fuel flow rate G_{fs} to the fuel correction coefficient calculation unit **213** (**S105-1**).

When the air side management unit **221** determines in the process in Step **S102** that the acquired data indicating the state indicates the static state (**S302**, Yes), the processes in Steps **S103-2** and **S104-2** are performed. On the other hand, when the air side management unit **221** determines in the

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process in Step S102 that the acquired data indicating the state does not indicate the static state (S102, No), the process in Step S105-2 is performed. Here, each of the processes in Steps S103-2, S104-2, and S105-2 is the following process. In the processes in Steps S103-1, S104-1, and S105-1 in which the branch number of each step number is replaced with “-1”, the fuel flow rate G_{fi} is replaced with the air flow rate G_{ar} , the fuel flow rate G_{fs} is replaced with the air flow rate G_{as} , the fuel side management unit 211 is replaced with the air side management unit 221, the fuel side storage unit 212 is replaced with the air side storage unit 222, and the fuel correction coefficient calculation unit 213 is replaced with the air correction coefficient calculation unit 223.

In this manner, when the state detected by the state detection unit 205 is the static state, the fuel side management unit 211 outputs the recently captured combination of the fuel flow rate G_{fi} and the fuel flow rate G_{fs} to the fuel correction coefficient calculation unit 213. When the state detected by the state detection unit 205 is not the static state, the fuel side management unit 211 outputs the latest combination of the fuel flow rate G_{fi} and the fuel flow rate G_{fs} stored in the fuel side storage unit 212 to the fuel correction coefficient calculation unit 213. In other words, regardless of the state detected by the state detection unit 205, the fuel side management unit 211 outputs the latest combination of the fuel flow rate G_{fi} and the fuel flow rate G_{fs} in the static state to the fuel correction coefficient calculation unit 213. Similarly, regardless of the state detected by the state detection unit 205, the air side management unit 221 outputs the latest combination of the air flow rate G_{ar} and the air flow rate G_{as} in the static state to the air correction coefficient calculation unit 223. When the processes in Steps S104-1, S104-2, S105-1, and S105-2 are completed, a subroutine of the storage unit management process in FIG. 5 is completed, and the process returns to the process in FIG. 3. In FIG. 5, the process performed by the fuel side management unit 211 and the process performed by the air side management unit 221 are performed in parallel.

As illustrated in Step S4-1 in FIG. 3, the fuel correction coefficient calculation unit 213 captures the combination of the fuel flow rate G_{fi} and the fuel flow rate G_{fs} output by the fuel side management unit 211 (S4-1). The fuel correction coefficient calculation unit 213 calculates the fuel correction coefficient K_f by dividing the fuel flow rate G_{fs} by the fuel flow rate G_{fi} . The fuel correction coefficient calculation unit 213 outputs the calculated fuel correction coefficient K_f to the fuel correction coefficient application unit 214 (S5-1). The fuel correction coefficient application unit 214 captures the fuel flow rate G_{fi} output by the fuel flow rate calculation unit 201 and the fuel correction coefficient K_f output by the fuel correction coefficient calculation unit 213. The fuel correction coefficient application unit 214 calculates the correction fuel flow rate G_{fcor} by multiplying the fuel flow rate G_{fi} by the fuel correction coefficient K_f . The fuel correction coefficient application unit 214 outputs the calculated correction fuel flow rate G_{fcor} to the turbine inlet temperature calculation unit 206 (S6-1).

Processes in Steps S4-2, S5-2, and S6-2 in FIG. 3 are performed by the air flow rate correction unit 220. Here, each of the processes in Steps S4-2, S5-2, and S6-2 is the following process. In the processes in S4-1, S5-1, and S6-1 in which the branch number of each step number is replaced with “-1”, the fuel flow rate G_{fi} is replaced with the air flow rate G_{ar} , the fuel flow rate G_{fs} is replaced with the air flow rate G_{as} , the fuel correction coefficient calculation unit 213 is replaced with the air correction coefficient calculation unit 223, the fuel correction coefficient application unit 214 is

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replaced with the air correction coefficient application unit 224, the fuel flow rate calculation unit 201 is replaced with the air flow rate calculation unit 202, the fuel correction coefficient K_f is replaced with the air correction coefficient K_s , and the correction fuel flow rate G_{fcor} is replaced with the correction air flow rate G_{scor} .

In FIG. 3, a set of processes in Steps S4-1, S5-1, and S6-1 and a set of processes in Steps S4-2, S5-2, and S6-2 are performed in parallel. A mark “A” illustrated in FIG. 3 is a mark indicating that the process is continuously performed, and as indicated by the mark “A”, the process in FIG. 4 is performed thereafter.

As illustrated in FIG. 4, the turbine inlet temperature calculation unit 206 captures the correction fuel flow rate G_{fcor} output by the fuel correction coefficient application unit 214 and the correction air flow rate G_{scor} output by the air correction coefficient application unit 224. The turbine inlet temperature calculation unit 206 receives and captures the combustor casing temperature T_{CB} transmitted by the combustor casing temperature sensor 25j and the fuel temperatures $T_f(1)$, $T_f(2)$, and $T_f(3)$ transmitted by the fuel temperature sensors 25c-3 to 25c-3. The combustor casing temperature sensor 253 performs the measurement at each detection cycle, and transmits the data of the combustor casing temperature T_{CS} which is the measurement data detected by performing the measurement to the gas turbine control device 20 through the control line. The turbine inlet temperature calculation unit 206 calculates the fuel temperature T_f which is an average value of the captured fuel temperatures $T_f(1)$, $T_f(2)$, and $T_f(3)$. As described above, the turbine inlet temperature calculation unit 206 substitutes the correction fuel flow rate G_{fcor} , the correction air flow rate G_{scor} , the combustor casing temperature T_{CS} , and the fuel temperature T_f into equation (6), and calculates the turbine inlet temperature T_{1T} . The turbine inlet temperature calculation unit 206 outputs the calculated turbine inlet temperature T_{1T} to the fuel distribution ratio calculation unit 207 (S20).

The fuel distribution ratio calculation unit 207 captures the turbine inlet temperature T_{1T} output by the turbine inlet temperature calculation unit 206. Based on the captured turbine inlet temperature T_{1T} , the fuel distribution ratio calculation unit 207 calculates the fuel distribution ratios D_1 , D_2 , and D_3 for each of the fuel supply systems 40-1, 40-2, and 40-3 by using a relational equation of the turbine inlet temperature T_{1T} and the fuel distribution ratio. The fuel distribution ratio calculation unit 207 outputs the calculated fuel distribution ratios D_1 , D_2 , and D_3 to the valve opening degree calculation unit 208 (S21).

The valve opening degree calculation unit 208 captures the fuel distribution ratios D_1 , D_2 , and D_3 output by the fuel distribution ratio calculation unit 207, and the fuel control signal command value (CSO) provided from the outside. Based on the captured fuel distribution ratios D_1 , D_2 , and D_3 and the fuel control signal command value (CSO), the valve opening degree calculation unit 209 calculates the valve opening degrees O_1 , O_2 , and O_3 for each of the flow control valves 41-1, 41-2, and 41-3 by using the valve opening degree calculation equation. The valve opening degree calculation unit 208 outputs each of the calculated valve opening degrees O_1 , O_2 , and O_3 to the respectively corresponding flow control valves 41-1, 41-2, and 41-3. The valve opening degree calculation unit 208 outputs the calculated valve opening degrees O_1 , O_2 , and O_3 to the fuel flow rate calculation unit 201 (S22), and completes the process.

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A series of the above-described processes illustrated in FIGS. 3 and 4 are performed as a process in one detection cycle, and the series of processes illustrated in FIGS. 3 and 4 are performed for each cycle.

(Operational Effect in Configuration of First Embodiment)

As described above, the gas turbine control device 20 of the first embodiment calculates the fuel flow rates G_{fi} and G_{fs} in two ways. One fuel flow rate G_{fi} is calculated by the fuel flow rate calculation unit 201 using parameters such as the valve opening degrees O_1 , O_2 , and O_3 , the flow control valve pre-pressures P_{1FV} (1), P_{1FV} (2), and P_{1FV} (3), the flow control valve post-pressures P_{2FV} (1), P_{2FV} (2), and P_{2FV} (3), and the fuel temperatures T_f (1), T_f (2), and T_f (3). Since the parameter used by the fuel flow rate calculation unit 201 has a quick response, even when the state of the gas turbine power generation plant 100 is the transient state, the fuel flow rate G_{fi} can be calculated with constant accuracy, following drastic changes in the transient state. However, in the fuel flow rate calculation function $f_1(\bullet)$ used to calculate the fuel flow rate G_{fi} by the fuel flow rate calculation unit 201, there is a possibility that a Cv characteristic of the flow control valve 41 which is predetermined when constructing the function includes an absolute error such as an error caused by a deviation in specifications like a difference from the actual Cv characteristic of the flow control valve 41 and an error when modeling is performed to calculate the fuel flow rate. The Cv characteristic is a characteristic indicating how much the fuel can flow in the flow control valve 41. Since there is a possibility of including this absolute error, even when the accuracy of the parameter substituted into the fuel flow rate calculation function $f_1(\bullet)$ is improved, there is a possibility that the fuel flow rate G_{fi} obtained by using the function has low absolute accuracy.

In contrast, the other fuel flow rate G_{fs} is calculated by the heat balance computation unit 203 using the parameters such as the gas turbine output G_{out} , the exhaust gas pressure P_{2T} , the exhaust gas temperature T_{2T} , the compressor inlet pressure P_{1C} , the compressor inlet temperature T_{1C} , the compressor exit pressure P_{2C} , and the compressor exit temperature T_{2C} . The gas turbine heat balance computation equation used to calculate the fuel flow rate G_{fs} by the heat balance computation unit 203 is a computation equation indicating that the overall energy balance of the gas turbines 10 matches. Therefore, compared to the fuel flow rate calculation function $f_1(\bullet)$ used by the fuel flow rate calculation unit 201, it is possible to calculate the fuel flow rate G_{fs} having high absolute accuracy. However, in some cases of the parameters used by the heat balance computation unit 203, the measurement data of the temperature obtained from the gas turbine 10 such as the exhaust gas temperature T_{2T} has a considerable response delay. In addition, in some cases, the gas turbine output G_{out} indicating the power generated by the generator 50 may be affected by a disturbance such as an imbalance between supply and demand in a power system connected to the generator 50. Therefore, whereas the fuel flow rate G_{fs} calculated by the heat balance computation unit 203 has higher accuracy than the fuel flow rate G_{fi} when the state of the gas turbine power generation plant 100 is the static state, there is a possibility that the fuel flow rate G_{fs} has lower accuracy than the fuel flow rate G_{fi} when the state of the gas turbine power generation plant 100 is the transient state.

The fuel flow rate correction unit 210 of the correction process unit 204 uses the above-described two fuel flow rates G_{fi} and G_{fs} to perform the following process so that the accuracy of the fuel flow rate provided for the turbine inlet temperature calculation unit 206 is not lowered in both the

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static state and the transient state, due to the process described with reference to FIGS. 3 to 5. For example, at time t1, it is assumed that the heat balance computation unit 203 calculates the fuel flow rate G_{fs} , and the fuel flow rate calculation unit 201 calculates the fuel flow rate G_{fi} . Furthermore, when the fuel side management unit 211 captures the fuel flow rate G_{fs} at time t1 output by the heat balance computation unit 203, it is assumed that the state of the gas turbine power generation plant 100 detected by the state detection unit 205 is the static state. In this case, the fuel side management unit 211 records the combination of the fuel flow rates G_{fs} and G_{fi} at time t1 in the fuel side storage unit 212, and outputs the combination of the fuel flow rates G_{fi} and G_{fs} at time t1 to the fuel correction coefficient calculation unit 213. Here, the fuel correction coefficient K_f calculated by the fuel correction coefficient calculation unit 213 is $K_f = G_{fs}/G_{fi}$. Therefore, the fuel correction coefficient application unit 214 performs the computation of $G_{fi} \times K_f = G_{fs}$, and the correction fuel flow rate G_{fcor} calculated by the fuel correction coefficient application unit 214 becomes equal to the fuel flow rate G_{fs} at time t1. Since the fuel flow rate G_{fs} in this case is in the static state, the correction fuel flow rate G_{fcor} has higher accuracy than the fuel flow rate G_{fi} .

On the other hand, when the fuel side management unit 211 captures the fuel flow rate G_{fs} at time t1 output by the heat balance computation unit 203, in a case where the state of the gas turbine power generation plant 100 detected by the state detection unit 205 is not the static state, that is, in a case of the transient state, the combination of the fuel flow rates G_{fs} and G_{fi} at time t1 is not recorded in the fuel side storage unit 212. Instead of the combination of the fuel flow rates G_{fi} and G_{fs} at time t1, the fuel side management unit 211 outputs the latest combination of the fuel flow rates G_{fi} and G_{fs} written in the fuel side storage unit 212 to the fuel correction coefficient calculation unit 213. A time at which the latest combination of the fuel flow rates G_{fi} and G_{fs} is obtained is a time before time t1, and is a time of the recent static state at time t1. This time is defined as time t2. In this case, the fuel correction coefficient K_f calculated by the fuel correction coefficient calculation unit 213 is $K_f = (G_{fs} \text{ at time t2}) / (G_{fi} \text{ at time t2})$. Therefore, the fuel correction coefficient application unit 214 performs the computation of $(G_{fi} \text{ at time t1}) \times K_f = (G_{fi} \text{ at time t1}) \times (G_{fs} \text{ at time t2}) / (G_{fi} \text{ at time t2})$. Here, the computation of $(G_{fs} \text{ at time t2}) / (G_{fi} \text{ at time t2})$ for calculating the fuel correction coefficient K_f is the computation using the combination of the fuel flow rates G_{fi} and G_{fs} at time t2 which is a time in the recently static state at time t1. Since the accuracy of the fuel flow rate G_{fi} in the static state is high, the accuracy of the fuel correction coefficient K_f is also high. Therefore, the fuel flow rate G_{fi} at time t1 is multiplied by the highly accurate fuel correction coefficient K_f . In this manner, the correction fuel flow rate G_{fcor} having the higher accuracy than the fuel flow rate G_{fi} at time t1 can be obtained.

As described above, only in a case of the static state, the fuel side management unit 211 writes the fuel flow rates G_{fi} and G_{fs} in the fuel side storage unit 212 in association with each other. In this manner, even when shifted to the transient state, the fuel correction coefficient calculation unit 213 can calculate the highly accurate fuel correction coefficient K_f by using the fuel flow rate G_{fi} when the highly accurate fuel flow rate G_{fs} obtained in the recent static state is obtained. In addition, as described above, the fuel correction coefficient calculation unit 213 can calculate the highly accurate fuel correction coefficient K_f even in the static state. Therefore, the accuracy of the correction fuel flow rate G_{fcor} calculated by the fuel correction coefficient application unit 214 is

higher than the fuel flow rate G_{ft} calculated by the fuel flow rate calculation unit **201** in both the static state and the transient state.

In the gas turbine control device **20**, two types of air flow rates G_{st} and G_{ss} are calculated for the air flow rate. Since a relationship between the air flow rates G_{st} and G_{ss} can be the same as a relationship between the fuel flow rates G_{ft} and G_{fs} , the accuracy of the correction air flow rate G_{acor} calculated by the air correction coefficient application unit **224** is higher than the air flow rate G_{st} calculated by the air flow rate calculation unit **202** in both the static state and the transient state. Therefore, in both the static state and the transient state, the turbine inlet temperature calculation unit **206** can calculate the highly accurate turbine inlet temperature T_{1T} without using the turbine efficiency which is less likely to be uniquely determined.

Since the heat balance computation unit **203** uses the gas turbine heat balance computation equation, the heat balance computation unit **203** calculates the fuel flow rate G_{fs} and the air flow rate G_{as} which reflect the recent performance of the gas turbine **10**. The fuel flow rate correction unit **210** corrects the fuel flow rate G_{ft} calculated by the fuel flow rate calculation unit **201** by using the fuel flow rate G_{fs} which reflects the recent performance of the gas turbine **10**. Similarly, the air flow rate correction unit **220** corrects the air flow rate G_{st} calculated by the air flow rate calculation unit **202** by using the air flow rate G_{as} which reflects the recent performance of the gas turbine **10**. Therefore, the turbine inlet temperature calculation unit **206** can calculate the turbine inlet temperature T_{1T} which reflects the recent performance of the gas turbine **10**. In this case, even when the performance of the gas turbine **10** is changed due to aged deterioration, the correction process unit **204** can perform the correction following the change in the performance on the fuel flow rate G_{ft} and the air flow rate G_{st} . That is, through the correction performed by the correction process unit **204**, it is possible to estimate the turbine inlet temperature T_{1T} with high accuracy over a long period of time and with high robustness against aged deterioration.

(Method for Calculating Each Correction Coefficient from Each Average Value of Fuel Flow Rate and Air Flow Rate)

In the gas turbine control device **20** of the first embodiment described above, in the fuel flow rate correction unit **210**, in the process in Step S104-1 in FIG. 5, the fuel side management unit **211** outputs a set of the fuel flow rate G_{ft} and the fuel flow rate G_{fs} captured in the preceding process in Step S100 to the fuel correction coefficient calculation unit **213**, and in the process in Step S105-1, the fuel side management unit **211** outputs the latest set of fuel flow rate G_{ft} and the fuel flow rate G_{fs} stored in the fuel side storage unit **212** to the fuel correction coefficient calculation unit **213**. Similarly, in the process in Steps S104-2 and S105-2, the air flow rate correction unit **220** outputs a set of the air flow rate G_{st} and the air flow rate G_{ss} to the air correction coefficient calculation unit **223**. In contrast, for example, the following processes as illustrated in FIGS. 6 and 7 may be performed.

FIG. 6 is a flowchart of a subroutine of a storage unit management process performed instead of the subroutine of the storage unit management process illustrated in FIG. 5. FIG. 7 is a flowchart illustrating a process subsequent to Step S3 of the flowchart illustrated in FIG. 3, and illustrates a process performed instead of Steps S4-1 to S6-1 and Steps S4-2 to S6-2. In FIG. 3, after the processes up to Steps S2-1, S2-2, and S2-3 are performed, the subroutine of the storage unit management process illustrated in FIG. 6 is started as

the process in Step S3, and the same process as that in FIG. 5 is performed up to the process in Step S102 in FIG. 6.

When the fuel side management unit **211** determines in the process in Step S102 that the acquired data indicating the state indicates the static state (S102, Yes), the fuel side management unit **211** performs the same process as that in Step S103-1 in FIG. 5, that is, a process of associating the fuel flow rate G_{ft} and the fuel flow rate G_{fs} which are captured in the process in Step S100 with each other and writing and storing the associated result in the fuel side storage unit **212** (S103-1). The fuel side management unit **211** determines whether or not the number of combinations of the fuel flow rate G_{ft} and the fuel flow rate G_{fs} stored in the fuel side storage unit **212** is equal to or larger than a predetermined number set in advance (S105-1). When the fuel side management unit **211** determines that the number of combinations of the fuel flow rate G_{ft} and the fuel flow rate G_{fs} stored in the fuel side storage unit **212** is equal to or larger than the predetermined number set in advance (S105-1, Yes), the fuel side management unit **211** reads all combinations of the fuel flow rate G_{ft} and the fuel flow rate G_{fs} stored in the fuel side storage unit **212**, and outputs an update instruction signal including the read combinations of the fuel flow rate G_{ft} and the fuel flow rate G_{fs} to the fuel correction coefficient calculation unit **213** (S106-1). On the other hand, when the fuel side management unit **211** determines that the number of combinations of the fuel flow rate G_{ft} and the fuel flow rate G_{fs} stored in the fuel side storage unit **212** is not equal to or larger than the predetermined number set in advance (S105-1, No), the process proceeds to Step S107.

The air flow rate correction unit **220** performs the processes in Steps S103-2, S105-2, and S106-2 in FIG. 6. Here, each of the processes in Steps S103-2, S105-2, and S106-2 is the following process. In the processes in S103-1, S105-1, and S106-1 in which the branch number of each step number is replaced with “-1”, the fuel flow rate G_{ft} is replaced with the air flow rate G_{st} , the fuel flow rate G_{fs} is replaced with the air flow rate G_{ss} , the fuel side management unit **211** is replaced with the air side management unit **221**, the fuel side storage unit **212** is replaced with the air side storage unit **222**, and the fuel correction coefficient calculation unit **213** is replaced with the air correction coefficient calculation unit **223**.

When each of the fuel side management unit **211** and the air side management unit **221** determines in the process in Step S102 that the acquired data indicating the state does not indicate the static state (S102, No), each of the fuel side management unit **211** and the air side management unit **221** performs a process in Step S107. Here, when performed by the fuel side management unit **211**, the process in Step S107 is a process in which the fuel side management unit **211** outputs a non-update instruction signal to the fuel correction coefficient calculation unit **213**. When performed by the air side management unit **221**, the process in Step S107 is a process in which the air side management unit **221** outputs the non-update instruction signal to the air correction coefficient calculation unit **223** (S107).

When the processes in Steps S106-1, S106-2, and S107 are completed, the subroutine of the storage unit management process is completed, and a process subsequent to Steps S10-1 and S10-2 illustrated in FIG. 7 is started. In FIG. 6, the process performed by the fuel side management unit **211** and the process performed by the air side management unit **221** are performed in parallel.

The fuel correction coefficient calculation unit **213** determines a type of the instruction signal output by the fuel side management unit **211** (S10-1). When the fuel correction

coefficient calculation unit 213 determines that the instruction signal output by the fuel side management unit 211 is the update instruction signal (S10-1, update instruction signal), the fuel correction coefficient calculation unit 213 reads and captures a combination of a plurality of the fuel flow rates G_{fi} and the fuel flow rates G_{fs} included in the update instruction signal (S11-1). The fuel correction coefficient calculation unit 213 calculates an average value of the plurality of fuel flow rates G_{fi} and an average value of the plurality of fuel flow rates G_{fs} (S12-1). The fuel correction coefficient calculation unit 213 calculates the fuel correction coefficient K_f by dividing the calculated average value of the fuel flow rates G_{fs} by the calculated average value of the fuel flow rates G_{fi} . The fuel correction coefficient calculation unit 213 writes and stores the calculated fuel correction coefficient K_f in an internal storage area, and outputs the calculated fuel correction coefficient K_f to the fuel correction coefficient application unit 214 (S13-1). Thereafter, as the process in Step S17-1, the same process as the process in Step S6-1 in FIG. 3 is performed (S17-1).

On the other hand, when the fuel correction coefficient calculation unit 213 determines in the process in Step S10-1 that the instruction signal output by the fuel side management unit 211 is the non-update instruction signal (S10-1, non-update instruction signal), the fuel correction coefficient calculation unit 213 determines whether or not the previous fuel correction coefficient K_f is stored in the internal storage area (S14-1). When the fuel correction coefficient calculation unit 213 determines that the previous fuel correction coefficient K_f is stored in the internal storage area (S14-1, Yes), the fuel correction coefficient calculation unit 213 reads the previous fuel correction coefficient K_f from the internal storage area, and outputs the read fuel correction coefficient K_f to the fuel correction coefficient application unit 214 (S15-1). Thereafter, the process proceeds to Step S17-1. When the fuel correction coefficient calculation unit 213 determines that the previous fuel correction coefficient K_f is not stored in the internal storage area (S14-1, No), the fuel correction coefficient calculation unit 213 outputs the fuel correction coefficient K_f which is $K_f=1$ to the fuel correction coefficient application unit 214 (S16-1). Thereafter, the process proceeds to Step S17-1. In addition, when a value of the fuel correction coefficient K_f is "1", in the process in Step S17-1, since the fuel correction coefficient application unit 214 multiplies the fuel flow rate G_{fi} by "1", the correction is not substantially performed. The fuel correction coefficient application unit 214 outputs the fuel flow rate G_{fi} output by the fuel flow rate calculation unit 201 to the turbine inlet temperature calculation unit 206 as the correction fuel flow rate G_{fcor} .

Processes in Steps S10-2 to S17-2 in FIG. 7 are performed by the air flow rate correction unit 220. Here, each of the processes of Steps S10-2 to S17-2 is the following process. In the processes in S10-1 to S17-1 in which the branch number of each step number is replaced with "-1", the fuel flow rate G_{fi} is replaced with the air flow rate G_{si} , the fuel flow rate G_{fs} is replaced with the air flow rate G_{as} , the fuel side management unit 211 is replaced with the air side management unit 221, the fuel correction coefficient calculation unit 213 is replaced with the air correction coefficient calculation unit 223, the fuel correction coefficient application unit 214 is replaced with the air correction coefficient application unit 224, the fuel flow rate calculation unit 201 is replaced with the air flow rate calculation unit 202, the fuel correction coefficient K_f is replaced with the air correction coefficient K_s , and the correction fuel flow rate G_{fcor} is replaced with the correction air flow rate G_{acor} .

In FIG. 7, a set of processes in Steps S10-1 to S17-1 and a set of processes in Steps S10-2 to S17-2 are performed in parallel. The mark "A" illustrated in FIG. 7 is a mark indicating that the process is continuously performed, and as indicated by the mark "A", the process in FIG. 4 is performed thereafter.

The fuel correction coefficient K_f obtained by performing the above-described process in FIG. 7 is a value calculated by an average value of the plurality of fuel flow rates G_{fi} and an average value of the plurality of fuel flow rates G_{fs} . Therefore, the fuel correction coefficient K_f is a value reflecting a relationship between the fuel flow rates G_{fi} and G_{fs} in the past static state. The air correction coefficient K_a is also a value reflecting a relationship between the air flow rates G_{si} and G_{as} in the past static state. Therefore, compared to a case of using one set of the fuel flow rates G_{fi} and G_{fs} and one set of the air flow rates G_{si} and G_{as} as in the gas turbine control device 20 of the first embodiment described above, it is possible to obtain the fuel correction coefficient K_f and the air correction coefficient K_a which are highly accurate in terms of reliability. Since the correction is performed by using the highly accurate fuel correction coefficient K_f and the highly accurate air correction coefficient K_a , it is possible to improve the accuracy of the correction fuel flow rate G_{fcor} and the correction air flow rate G_{acor} which are obtained by the correction. Therefore, based on the highly accurate correction fuel flow rate G_{fcor} and the highly accurate correction air flow rate G_{acor} , the accuracy of the turbine inlet temperature T_{1T} calculated by the turbine inlet temperature calculation unit 206 is improved.

As a predetermined number used in the processes in Steps S105-3 and S105-2 in FIG. 6, a value indicating the number of combinations adopted to such an extent that the reliability is improved is set in advance. The processes in Steps S16-1 and S16-2 in FIG. 7, that is, an outputting process as the fuel correction coefficient $K_f=1$ and an outputting process as the air correction coefficient $K_s=1$, are processes performed while the number of combinations stored in the fuel side storage unit 212 and the air side storage unit 222 is smaller than a predetermined value. The processes are not intended to correct each of the fuel flow rate G_{fi} and the air flow rate G_{si} while each average value of the fuel flow rates G_{fi} and G_{fs} and the air flow rates G_{si} and G_{as} cannot be calculated. The processes in Steps S15-1 and S15-2 in FIG. 7, that is, a process of using the previous fuel correction coefficient K_f for the correction and a process of using the previous air correction coefficient K_a for the correction, are processes performed when the state detected by the state detection unit 205 is not the static state, that is, when the state is the transient state. The processes are not intended to update the fuel correction coefficient K_f and the air correction coefficient K_a when the state is the transient state.

As the processes in Steps S106-1 and S106-2 in FIG. 6, the following processes may be performed. In the process in Step S106-1 in FIG. 6, instead of reading all combinations of the fuel flow rates G_{fi} and G_{fs} stored in the fuel side storage unit 212 as described above, the fuel side management unit 211 may read a predetermined number of combinations so that a new combination is sequentially included in a time-series order in which the latest combination is set as the beginning. Similarly, in the process in Step S106-2 in FIG. 6, instead of reading all combinations of the air flow rates G_{si} and G_{as} stored in the air side storage unit 222, the air side management unit 221 may read a predetermined number of combinations so that a new combination is sequentially included in a time-series order in which the latest combination is set as the beginning. In this way, the

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fuel correction coefficient K_f and the air correction coefficient K_a based on a movement average value can be obtained. Therefore, a computation amount can be reduced, compared to a case of using all combinations stored in the fuel side storage unit 212 and the air side storage unit 222. In addition, it is possible to obtain the highly accurate turbine inlet temperature T_{1T} reflecting the most recent performance of the gas turbine 10.

In addition, at a stage where a predetermined number or more of combinations are stored in the fuel side storage unit 212 and the air side storage unit 222, it is assumed that a change in values included in individual combinations stored in a time-series order is reduced. On a premise of this assumption, in the process in Step S106-1 in FIG. 6, the fuel side management unit 211 may read the latest combination of one set of the fuel flow rates G_{ft} and G_{fs} , and may skip the process in Step S12-1 in FIG. 7, and in the process in Step S13-1, the fuel correction coefficient calculation unit 213 may calculate the fuel correction coefficient $K_f (=G_{fs}/G_{ft})$ by dividing the latest fuel flow rate G_{fs} by the latest fuel flow rate G_{ft} . Similarly, in the process in Step S106-2 in FIG. 6, the air side management unit 221 may read the latest combination of one set of the air flow rates G_{at} and G_{as} , and may skip the process in Step S12-2 in FIG. 7. In the process in Step S13-2, the air correction coefficient calculation unit 223 may calculate the air correction coefficient $K_s (=G_{as}/G_{at})$ by dividing the latest air flow rate G_{as} by the latest air flow rate G_{at} . (Other Configuration Example in First Embodiment)

In the first embodiment described above, the correction process unit 204 includes the fuel flow rate correction unit 210 and the air flow rate correction unit 220. In contrast, the correction process unit 204 may include only the fuel flow rate correction unit 210. In this case, the heat balance computation unit 203 calculates only the fuel flow rate G_{fs} , and outputs the fuel flow rate G_{fs} to the fuel side management unit 211. The air flow rate G_{at} calculated by the air flow rate calculation unit 202 is directly provided for the turbine inlet temperature calculation unit 206. In addition, the correction process unit 204 may include only the air flow rate correction unit 220. In this case, the heat balance computation unit 203 calculates only the air flow rate G_{as} , and outputs the air flow rate G_{as} to the air side management unit 221. The fuel flow rate G_{ft} calculated by the fuel flow rate calculation unit 201 is directly provided for the turbine inlet temperature calculation unit 206.

Second Embodiment

In the first embodiment, the fuel flow rate G_{ft} calculated by the fuel flow rate calculation unit 201 and the air flow rate G_{at} calculated by the air flow rate calculation unit 202 are corrected by using the fuel flow rate G_{fs} and the air flow rate G_{st} which are calculated by the heat balance computation unit 203. It is conceivable that the fuel correction coefficient K_f and the air correction coefficient K_s do not depend only on the measurement data referred to in the heat balance computation unit 203, and that both of these vary depending on various state amounts in the gas turbine power generation plant 100. As described below, a second embodiment includes a configuration in which the fuel correction coefficient K_f and the air correction coefficient K_s are variable in accordance with a change in this state amount. (Configuration Example of Gas Turbine Power Generation Plant of Second Embodiment)

FIG. 8 is an overall configuration diagram of a gas turbine power generation plant 100a of the second embodiment. A

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difference between the gas turbine power generation plant 100 according to the first embodiment and the gas turbine power generation plant 100a according to the second embodiment is as follows. The gas turbine power generation plant 100a according to the second embodiment includes a gas turbine control device 20a instead of the gas turbine control device 20. Furthermore, a combustor casing pressure sensor 25m is provided. The combustor casing pressure sensor 25m is connected to the gas turbine control device 20a by a control line such as a communication line as indicated by a dotted line arrow, detects a combustor casing pressure P_{out} which is a pressure inside the combustor casing 16 by performing measurement in each detection cycle, and transmits the detected combustor casing pressure P_{out} to the gas turbine control device 20a through the control line. (Configuration Example of Gas Turbine Control Device of Second Embodiment)

As illustrated in FIG. 9, the gas turbine control device 20a includes the fuel flow rate calculation unit 201, the air flow rate calculation unit 202, the heat balance computation unit 203, a correction process unit 204a, the state detection unit 205, the turbine inlet temperature calculation unit 206, the fuel distribution ratio calculation unit 207, and the valve opening degree calculation unit 208. The correction process unit 204a includes a fuel flow rate correction unit 210a and an air flow rate correction unit 220a. The fuel flow rate correction unit 210a includes a fuel side management unit 211a, a fuel side storage unit 212a, a fuel correction coefficient calculation unit 213a, a fuel correction coefficient application unit 214, and a fuel side regression analysis unit 215.

In the fuel flow rate correction unit 210a, in accordance with a state detected by the state detection unit 205, the fuel side management unit 211a associates the fuel flow rate G_{ft} calculated by the fuel flow rate calculation unit 201, the fuel flow rate G_{fs} calculated by the heat balance computation unit 203, the combustor casing temperature T_{CS} detected by the combustor casing temperature sensor 25j, and the combustor casing pressure P_{out} detected by the combustor casing pressure sensor 25m with each other, and writes and stores the associated result in the fuel side storage unit 212a. The fuel side management unit 211a outputs a predetermined number of combinations of the fuel flow rate G_{ft} , the fuel flow rate G_{fs} , the combustor casing temperature T_{CS} , and the combustor casing pressure P_{out} which are set in advance to the fuel side regression analysis unit 215.

The fuel side storage unit 212a stores the combination of the fuel flow rate G_{ft} , the fuel flow rate G_{fs} , the combustor casing temperature T_{CS} , and the combustor casing pressure P_{out} in a time-series order. Based on the predetermined number of combinations of the fuel flow rate G_{ft} , the fuel flow rate G_{fs} , the combustor casing temperature T_{CS} , and the combustor casing pressure P_{out} , the fuel side regression analysis unit 215 performs a predetermined regression analysis in which an explanatory variable is set to the combustor casing temperature T_{CS} and the combustor casing pressure P_{out} and an objective variable is set to the fuel correction coefficient $K_f (= \text{fuel flow rate } G_{fs} / \text{fuel flow rate } G_{ft})$, and calculates a coefficient of a regression equation corresponding to the predetermined regression analysis. The fuel correction coefficient calculation unit 213a calculates the fuel correction coefficient K_f by substituting the combustor casing temperature T_{CS} and the combustor casing pressure P_{out} into the regression equation to which the coefficient of the regression equation calculated by the fuel side regression analysis unit 215 is applied.

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The air flow rate correction unit **220a** includes an air side management unit **221a**, an air side storage unit **222a**, an air correction coefficient calculation unit **223a**, the air correction coefficient application unit **224**, and an air side regression analysis unit **225**. In accordance with a state detected by the state detection unit **205**, the air side management unit **221a** associates the air flow rate G_{at} calculated by the air flow rate calculation unit **202**, the air flow rate G_{as} calculated by the heat balance computation unit **203**, the compressor inlet temperature T_{1C} detected by the compressor inlet temperature sensor **25g**, and the combustor casing pressure P_{out} detected by the combustor casing pressure sensor **25m** with each other, and writes and stores the associated result in the air side storage unit **222a**. The air side management unit **221a** outputs a predetermined number of combinations of the air flow rate G_{at} , the air flow rate G_{ss} , the compressor inlet temperature T_{1C} , and the combustor casing pressure P_{out} which are set in advance to the air side regression analysis unit **225**.

The air side storage unit **222a** stores the combination of the air flow rate G_{at} , the air flow rate G_{ss} , the compressor inlet temperature T_{1C} , and the combustor casing pressure P_{out} in a time-series order. Based on the predetermined number of combinations of the air flow rate G_{at} , the air flow rate G_{ss} , the compressor inlet temperature T_{1C} , and the combustor casing pressure P_{out} , the air side regression analysis unit **225** performs a predetermined regression analysis in which the explanatory variable is set to the compressor inlet temperature T_{1C} and the combustor casing pressure P_{out} , and the objective variable is set to the air correction coefficient K_a (=air flow rate G_{as} /air flow rate G_{at}), and calculates a coefficient of a regression equation corresponding to the predetermined regression analysis. The air correction coefficient calculation unit **223a** calculates the air correction coefficient K_s by substituting the compressor inlet temperature T_{1C} and the combustor casing pressure P_{out} into the regression equation to which the coefficient of the regression equation calculated by the air side regression analysis unit **225** is applied.

Here, for example, the predetermined regression analysis performed by the fuel side regression analysis unit **215** and the air side regression analysis unit **225** is a linear multiple regression analysis. In addition, as the predetermined number, for example, a value indicating the number of combinations required for calculating an accurate coefficient in the predetermined regression analysis is set in advance. (Operation Example of Gas Turbine Control Device of Second Embodiment)

A process performed by the gas turbine control device **20a** will be described with reference to FIGS. **10** to **12**. A process illustrated in FIG. **10** and a process illustrated in FIG. **11** and performed as the subroutine of the storage unit management process in the process illustrated in FIG. **10** are processes performed by the fuel side management unit **211a**, the fuel side regression analysis unit **215**, the air side management unit **221a**, and the air side regression analysis unit **225**, and will be referred to as a "regression analysis process" hereinafter. A process illustrated in FIG. **12** is a process performed by the fuel correction coefficient calculation unit **213a**, the fuel correction coefficient application unit **214**, the air correction coefficient calculation unit **223a**, and the air correction coefficient application unit **224**, and will be referred to as a "correction process" hereinafter. (Regarding Regression Analysis Process)

In FIG. **10**, in Steps Sa1-1 and Sa2-1, the same processes as those in Steps S1-1 and S2-1 in FIG. **3** are performed. In Steps Sa1-2 and Sa2-2, the same processes as those in Steps

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S1-2 and S2-2 in FIG. **3** are performed. In Steps Sa1-3 and Sa2-3, the same processes as those in Steps S1-3 and S2-3 in FIG. **3** are performed. A set of processes in Steps Sa1-1 and Sa2-1, a set of processes in Steps Sa1-2 and Sa2-2, and a set of processes in Steps Sa1-3 and Sa2-3 are performed in parallel. After the processes in Steps Sa2-1, Sa2-2, and Sa2-3, the subroutine of the storage unit management process illustrated in FIG. **11** is performed as the process in Step Sa3.

The fuel side management unit **211a** receives and captures the combustor casing temperature T_{CS} transmitted by the combustor casing temperature sensor **25j** and the combustor casing pressure P_{out} transmitted by the combustor casing pressure sensor **25m**. The fuel side management unit **211a** captures the fuel flow rate G_{ft} output by the fuel flow rate calculation unit **201** and the fuel flow rate G_{fs} output by the heat balance computation unit **203**. The air side management unit **221a** receives and captures the compressor inlet temperature T_{1C} transmitted by the compressor inlet temperature sensor **25g** and the combustor casing pressure P_{out} transmitted by the combustor casing pressure sensor **25m**. The air flow rate G_{ss} output by the air flow rate calculation unit **202** and the air flow rate G_{as} output by the heat balance computation unit **203** are captured (Sa100).

Each of the fuel side management unit **211a** and the air side management unit **221a** acquires data indicating the state of the gas turbine power generation plant **100a** detected by the state detection unit **205** at a time at which the process in Step Sa100 performed by each of the fuel side management unit **211a** and the air side management unit **221a** is completed, from the state detection unit **205** (Sa101). The fuel side management unit **211a** and the air side management unit **221a** determine whether or not the acquired data indicating the state indicates the static state (Sa102).

When the fuel side management unit **211a** determines that the acquired data indicating the state indicates the static state (Sa102, Yes), the fuel side management unit **211a** associates the fuel flow rate G_{ft} , the fuel flow rate G_{fs} , the combustor casing temperature T_{CS} , and the combustor casing pressure P_{out} which are captured in the process in Step Sa100 with each other, and writes and stores the associated result in the fuel side storage unit **212a**. The fuel side management unit **211a** writes a combination of the fuel flow rate G_{ft} , the fuel flow rate G_{fs} , the combustor casing temperature T_{CS} , and the combustor casing pressure P_{out} which are sequentially written in the fuel side storage unit **212a** to be stored in the fuel side storage unit **212a** in a time-series order, for example, so that a newly written combination is stored in the fuel side storage unit **212a** at the beginning of the time-series order (Sa103-1).

The fuel side management unit **211a** determines whether or not the number of combinations stored in the fuel side storage unit **212a** is equal to or larger than a predetermined number (Sa104-1). It is assumed that the fuel side management unit **211a** determines that the number of combinations stored in the fuel side storage unit **212a** is not equal to or larger than the predetermined number (Sa104-1, No). In this case, the fuel side regression analysis unit **215** does not have sufficient data to perform the predetermined regression analysis, and cannot proceed to the process in Step Sa4-1 in FIG. **10**. Therefore, the process is completed.

On the other hand, when the fuel side management unit **211a** determines that the number of combinations stored in the fuel side storage unit **212a** is equal to or larger than a predetermined number (Sa104-1, Yee), the fuel side management unit **211a** reads a predetermined number of combinations of the fuel flow rate G_{ft} , the fuel flow rate G_{fs} , the

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combustor casing temperature T_{C3} , and the combustor casing pressure P_{out} in a time-series order in which the latest combination is set as the beginning, outputs the read predetermined number of combinations to the fuel side regression analysis unit **215** (Sa105-1), and completes the process of the subroutine of the storage unit management process.

When the air side management unit **221** determines in the process in Step Sa102 that the acquired data indicating the state indicates the static state (Sa102, Yes), processes in Steps Sa103-2, Sa104-2, and Sa105-2 are performed. Here, each of the processes in Steps Sa103-2, Sa104-2, and Sa105-2 is the following process. In the process in Steps Sa103-1, Sa104-1, and Sa105-1 in which the branch number of each step number is replaced with “-1”, the fuel flow rate G_{ft} is replaced with the air flow rate G_{st} , the fuel flow rate G_{fs} is replaced with the air flow rate G_{as} , the combustor casing temperature T_{CS} is replaced with the compressor inlet temperature T_{1C} , the fuel side management unit **211a** is replaced with the air side management unit **221a**, the fuel side storage unit **212a** is replaced with the air side storage unit **222a**, and the fuel side regression analysis unit **215** is replaced with the air side regression analysis unit **225**.

On the other hand, in the process in Step Sa102, it is assumed that the fuel side management unit **211a** and the air side management unit **221a** determine that the acquired data indicating the state does not indicate the static state (Sa102, No). In this case, in each of the fuel side management unit **211a** and the air side management unit **221a**, the data captured in the process in Step Sa100 is data obtained in the transient state. Therefore, the data does not need to be recorded in the respectively corresponding fuel side storage unit **212a** and air side storage unit **222a**. In addition, the data stored in the respectively corresponding fuel side storage unit **212a** and air side storage unit **222a** is not updated. Therefore, the coefficient of the regression equation does not need to be updated. Therefore, the process is completed.

When the processes in Steps Sa105-1 and Sa105-2 are completed, the subroutine of the storage unit management process is completed, and the process returns to the process in FIG. 10. In FIG. 11, the process performed by the fuel side management unit **211a** and the process performed by the air side management unit **221a** are performed in parallel.

With reference back to FIG. 10, the fuel side regression analysis unit **215** captures a predetermined number of combinations of the fuel flow rate G_{ft} , the fuel flow rate G_{fs} , the combustor casing temperature T_{C3} , and the combustor casing pressure P_{out} which are output by the fuel side management unit **211a** (Sa4-1). Based on the captured predetermined number of combinations of the fuel flow rate G_{ft} , the fuel flow rate G_{fs} , the combustor casing temperature T_{CS} , and the combustor casing pressure P_{out} , the fuel side regression analysis unit **215** performs the predetermined regression analysis, and calculates the coefficient of the regression equation corresponding to the predetermined regression analysis. The fuel side regression analysis unit **215** outputs the calculated coefficient to the fuel correction coefficient calculation unit **213a** (Sa5-1), and completes the process.

Similarly, the air side regression analysis unit **225** performs the process in Steps Sa4-2 and Sa5-2. Here, each of the processes in Steps Sa4-2 and Sa5-2 is the following process. In the processes in Steps Sa4-1 and Sa5-1 in which the branch number of each step number is replaced with “-1”, the fuel flow rate G_{fs} is replaced with the air flow rate G_{as} , the fuel flow rate G_{ft} is replaced with the air flow rate G_{st} , the combustor casing temperature T_{CS} is replaced with the compressor inlet temperature T_{1C} , the fuel side management unit **211a** is replaced with the air side management unit

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221a, the fuel side regression analysis unit **215** is replaced with the air side regression analysis unit **225**, and the fuel correction coefficient calculation unit **213a** is replaced with the air correction coefficient calculation unit **223a**.

In FIG. 10, a set of processes in Steps Sa4-1 and Sa5-1 and a set of processes in Steps Sa4-2 and Sa5-2 are performed in parallel. The above-described process illustrated in FIG. 10 is performed as one detection cycle process, and the process illustrated in FIG. 10 is performed for each cycle.

(Regarding Correction Process)

As illustrated in FIG. 12, the fuel correction coefficient calculation unit **213a** receives and captures the combustor casing temperature T_{C3} a transmitted by the combustor casing temperature sensor **25j**, and the combustor casing pressure P_{out} transmitted by the combustor casing pressure sensor **25m** (Sa10-1). The fuel correction coefficient calculation unit **213a** determines whether or not the coefficient of the regression equation is received from the fuel side regression analysis unit **215** (Sa11-1). When the fuel correction coefficient calculation unit **213a** determines that the coefficient of the regression equation is received (Sa11-1, Yes), the fuel correction coefficient calculation unit **213a** captures the received coefficient, and writes and stores the received coefficient in an internal storage area. The fuel correction coefficient calculation unit **213a** calculates the fuel correction coefficient K_f by substituting the combustor casing temperature T_{CS} and the combustor casing pressure P_{out} into the regression equation to which the captured coefficient is applied (Sa12-1).

On the other hand, when the fuel correction coefficient calculation unit **213a** determines in the process in Step Sa11-1 that the coefficient of the regression equation is not received (Sa11-1, No), the fuel correction coefficient calculation unit **213a** determines whether or not the previous coefficient is stored in the internal storage area (Sa13-1). When the fuel correction coefficient calculation unit **213** determines that the previous coefficient is stored in the internal storage area (Sa13-1, Yes), the fuel correction coefficient calculation unit **213** reads the previous coefficient from the internal storage area, and calculates the fuel correction coefficient K_f by substituting the combustor casing temperature T_{CS} and the combustor casing pressure P_{out} into the regression equation to which the read coefficient is applied (Sa14-1). When the fuel correction coefficient calculation unit **213** determines that the previous coefficient is not stored in the internal storage area (Sa13-1, No), the fuel correction coefficient calculation unit **213** outputs the fuel correction coefficient K_f which is $K_f=1$ to the fuel correction coefficient application unit **214** (Sa15-1). After the processes in Steps Sa12-1, Sa14-1, and Sa15-1, the same process as the process in Step S6-1 in FIG. 3 is performed as the process in Step Sa16-1 (Sa16-1).

The air correction coefficient calculation unit **223a** receives and captures the compressor inlet temperature T_{1C} transmitted by the compressor inlet temperature sensor **25g**, and the combustor casing pressure P_{out} transmitted by the combustor casing pressure sensor **25m** (Sa10-2). Thereafter, the processes in Steps Sa11-2 to Sa16-2 are performed by the air correction coefficient calculation unit **223a** and the air correction coefficient application unit **224**. Here, each of the processes in Steps Sa11-2 to Sa14-2 is the following process. In the processes in Steps Sa11-1 to Sa16-1 in which the branch number of each step number is replaced with “-1”, the fuel flow rate G_{ft} is replaced with the air flow rate G_{st} , the fuel flow rate G_{fs} is replaced with the air flow rate G_{as} , the combustor casing temperature T_{C3} is replaced with the

compressor inlet temperature T_{1C} , the fuel correction coefficient calculation unit **213a** is replaced with the air correction coefficient calculation unit **223a**, the fuel correction coefficient application unit **214** is replaced with the air correction coefficient application unit **224**, the fuel correction coefficient K_f is replaced with the air correction coefficient K_a , the fuel flow rate calculation unit **201** is replaced with the air flow rate calculation unit **202**, and the correction fuel flow rate G_{fcor} is replaced with the correction air flow rate G_{acor} .

A set of processes in Steps Sa10-1 to Sa16-1 and a set of processes in Steps Sa10-2 to Sa16-2 are performed in parallel. The mark "A" illustrated in FIG. 12 is a mark indicating that the process is continuously performed, and as indicated by the mark "A", the same process as the process in FIG. 4 of the first embodiment is performed. A series of the above-described processes illustrated in FIGS. 12 and 4 are performed as the process of one detection cycle, and the series of processes illustrated in FIGS. 12 and 4 are performed for each cycle.

(Operational Effect in Configuration of Second Embodiment)

As described above, when the state detected by the state detection unit **205** is the static state, the fuel flow rate correction unit **210a** of the second embodiment records the combination of the fuel flow rate G_{fs} , the fuel flow rate G_{fs} , the combustor casing temperature T_{C3} , and the combustor casing pressure P_{out} in the fuel side storage unit **212a**, and updates the fuel correction coefficient K_f by using a predetermined number of combinations recorded in the fuel side storage unit **212a**. On the other hand, when the state detected by the state detection unit **205** is not the static state, the fuel flow rate correction unit **210a** does not record the combination in the fuel side storage unit **212a**, and does not update the fuel correction coefficient K_f . Similarly, when the state detected by the state detection unit **205** is the static state, the air flow rate correction unit **220a** records the combination of the air flow rate G_{as} , the air flow rate G_{as} , the compressor inlet temperature T_{1C} , and the combustor casing pressure P_{out} in the air side storage unit **222a**, and updates the air correction coefficient K_a by using a predetermined number of combinations recorded in the air side storage unit **222a**. On the other hand, when the state detected by the state detection unit **205** is not the static state, the air flow rate correction unit **220a** does not record the combination in the air side storage unit **222a**, and does not update the air correction coefficient K_a . Therefore, in the configuration of the second embodiment, as in the configuration of the first embodiment, the accuracy of the fuel flow rate and the air flow rate provided for the turbine inlet temperature calculation unit **206** is not lowered in both the static state and the transient state. In this manner, the turbine inlet temperature calculation unit **206** can maintain the accuracy of the turbine inlet temperature T_{1T} for a long period of time without using the turbine efficiency which is less likely to be uniquely determined.

Furthermore, in the second embodiment, in the fuel flow rate correction unit **210a**, the predetermined number of combinations of the fuel flow rate G_{fs} , the fuel flow rate G_{fs} , the combustor casing temperature T_{C3} , and the combustor casing pressure P_{out} are used to perform a predetermined regression analysis in which the explanatory variable is set to the combustor casing temperature T_{C3} and the combustor casing pressure P_{out} , and the objective variable is set to the fuel correction coefficient K_f . Therefore, the fuel correction coefficient K_f can be variable in accordance with the state amount indicating the state of the gas turbine power generation plant **100a**, such as the combustor casing tempera-

ture T_{C3} and the combustor casing pressure P_{out} . Similarly, in the air flow rate correction unit **220a**, a predetermined number of combinations of the air flow rate G_{as} , the air flow rate G_{as} , the compressor inlet temperature T_{1C} , and the combustor casing pressure P_{out} are used to perform a predetermined regression analysis in which the explanatory variable is set to the compressor inlet temperature T_{1C} and the combustor casing pressure P_{out} , and the objective variable is set to the air correction coefficient K_a . Therefore, the air correction coefficient K_a can be variable in accordance with the state amount indicating the state of the gas turbine power generation plant **100a**, such as the compressor inlet temperature T_{1C} and the combustor casing pressure P_{out} . In this manner, the fuel correction coefficient K_f and the air correction coefficient K_a which are highly accurate in terms of reliability in accordance with the state of the gas turbine power generation plant **100a** can be obtained. Since the correction is performed by using the highly accurate fuel correction coefficient K_f and the highly accurate air correction coefficient K_a , it is possible to improve the accuracy of the correction fuel flow rate G_{fcor} and the correction air flow rate G_{acor} which are obtained by the correction. Therefore, based on the highly accurate correction fuel flow rate G_{fcor} and the highly accurate correction air flow rate G_{acor} , the accuracy of the turbine inlet temperature T_{1T} calculated by the turbine inlet temperature calculation unit **206** is improved.

The processes in Steps Sa15-1 and Sa15-2 in FIG. 12, that is, the outputting process as the fuel correction coefficient $K_f=1$ and the outputting process as the air correction coefficient $K_a=1$, are processes performed while the number of combinations stored in the fuel side storage unit **212a** and the air side storage unit **222a** is smaller than a predetermined value. The processes are not intended to correct each of the fuel flow rate G_{fs} and the air flow rate G_{as} while the coefficient of the regression equation cannot be calculated. In addition, the processes in Steps Sa14-1 and Sa14-2, that is, the processes of applying the previous coefficient to the regression equation, are processes performed when the state detected by the state detection unit **205** is not the static state, that is, when the state is the transient state. When the state is the transient state, the processes are not intended to update the coefficient of the regression equation.

(Other Configuration Examples in Second Embodiment)

In the second embodiment described above, the linear multiple regression analysis is applied as the predetermined regression analysis. However, any desired regression analysis method other than the linear multiple regression analysis may be applied as the predetermined regression analysis. For example, a method for the regression analysis performed by machine learning such as Support Vector Regression (SVR) and random forest may be applied.

In the second embodiment described above, the fuel side regression analysis unit **215** performs a predetermined regression analysis by setting the explanatory variable to the combustor casing temperature T_{C3} and the combustor casing pressure P_{out} . In addition, the air side regression analysis unit **225** performs a predetermined regression analysis by setting the explanatory variable to the compressor inlet temperature T_{1C} and the combustor casing pressure P_{out} . However, the explanatory variables are described as examples. The fuel side regression analysis unit **215** may use any one of the combustor casing temperature T_{C3} and the combustor casing pressure P_{out} as the explanatory variable. The air side regression analysis unit **225** may use any one of the compressor inlet temperature T_{1C} and the combustor casing pressure P_{out} as the explanatory variable. Any one or

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a combination of a plurality of the measurement data measured by any desired sensors **25a** to **25m** provided in the gas turbine power generation plant **100a** may be used as the explanatory variable. In addition, any desired command value output for controlling the gas turbine **10** by the gas turbine control device **20a**, for example, the above-described fuel control signal command value (CSO), may be used as the explanatory variable, or a combination of the measurement data and the command value may be used as the explanatory variable.

In the second embodiment described above, in the process in Step Sa105-1 in FIG. 11, the fuel side management unit **211a** reads a predetermined number of combinations of the fuel flow rate G_{ft} , the fuel flow rate G_{fs} , the combustor casing temperature T_{C3} , and the combustor casing pressure P_{out} from the fuel side storage unit **212a**, and outputs the combinations to the fuel side regression analysis unit **235**. In contrast, in the process in Step Sa105-1, the fuel side management unit **211a** may read all combinations stored in the fuel side storage unit **212a**, and may output the combinations to the fuel side regression analysis unit **215**. Similarly, in the process in Step Sa105-2 in FIG. 11, the air side management unit **221a** reads a predetermined number of combinations of the air flow rates G_{at} , the air flow rate G_{as} , the compressor inlet temperature T_{1C} , and the combustor casing pressure P_{out} from the air side storage unit **222a**, and outputs the combinations to the air side regression analysis unit **225**. In contrast, in the process in Step Sa105-2, the air side management unit **221a** may read all combinations stored in the air side storage unit **222a**, and may output the combinations to the air side regression analysis unit **225**.

In the second embodiment described above, the correction process unit **204a** includes the fuel flow rate correction unit **210a** and the air flow rate correction unit **220a**. In contrast, the correction process unit **204a** may include only the fuel flow rate correction unit **210a**. In this case, the heat balance computation unit **203** calculates and outputs only the fuel flow rate G_{fs} , and the air flow rate G_{at} calculated by the air flow rate calculation unit **202** is provided for the turbine inlet temperature calculation unit **206**. In addition, the correction process unit **204a** may include only the air flow rate correction unit **220a**. In this case, the heat balance computation unit **203** calculates and outputs only the air flow rate G_{as} , and the fuel flow rate G_{ft} calculated by the fuel flow rate calculation unit **201** is provided for the turbine inlet temperature calculation unit **206**. In addition, the correction process unit **204a** may include the fuel flow rate correction unit **210a** and the air flow rate correction unit **220** of the first embodiment, or may include the air flow rate correction unit **220a** and the fuel flow rate correction unit **210** of the first embodiment.

Third Embodiment

In the first and second embodiments, the fuel flow rate G_{ft} calculated by the fuel flow rate calculation unit **201** and the fuel flow rate G_{fs} calculated by the heat balance computation unit **203** become a total value of the fuel flow rates supplied by the respective three fuel supply systems **40-1**, **40-2**, and **40-3**. In contrast, the gas turbine power generation plants **100** and **100a** of the first and second embodiments include the three fuel supply systems **40-1**, **40-2**, and **40-3**. Therefore, it is conceivable that the more highly accurate turbine inlet temperature T_{1T} can be obtained when each of the fuel flow rates G_{ft} is individually corrected by calculating the fuel flow rate G_{ft} and the fuel flow rate G_{fs} for each of the fuel supply systems **40-1**, **40-2**, and **40-3**. A gas turbine control

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device **20b** of a third embodiment includes a configuration for performing this individual correction as described below. Hereinafter, for convenience of description, in the gas turbine power generation plant **100** of the first embodiment, a configuration in which the gas turbine control device **20** is replaced with the gas turbine control device **20b** of the third embodiment will be referred to as a gas turbine power generation plant **100b**.

(Configuration Example of Gas Turbine Control Device of Third Embodiment)

FIG. 13 is a block diagram illustrating a configuration of the gas turbine control device **20b** according to the third embodiment. The gas turbine control device **20b** includes a fuel flow rate calculation unit **201b**, the air flow rate calculation unit **202**, a heat balance computation unit **203b**, a correction process unit **204b**, the state detection unit **205**, the turbine inlet temperature calculation unit **206**, the fuel distribution ratio calculation unit **207**, the valve opening degree calculation unit **208**, and a fuel distribution amount calculation unit **209**. The fuel flow rate calculation unit **201b** has the same configuration as that of the fuel flow rate calculation unit **201** of the first embodiment except for the following point. The fuel flow rate calculation unit **201** of the first embodiment calculates the fuel flow rates $G_{ft}(1)$, $G_{ft}(2)$, and $G_{ft}(3)$, and calculates the fuel flow rate G_{ft} by summing up the calculated fuel flow rates $G_{ft}(1)$, $G_{ft}(2)$, and $G_{ft}(3)$. In contrast, the fuel flow rate calculation unit **201b** of the third embodiment outputs the calculated fuel flow rates $G_{ft}(1)$, $G_{ft}(2)$, and $G_{ft}(3)$ without summing up the fuel flow rates. The heat balance computation unit **203b** has the same configuration as that of the heat balance computation unit **203** of the first embodiment except for the following point. The heat balance computation unit **203** of the first embodiment calculates and outputs the fuel flow rate G_{fs} and the air flow rate G_{ss} . In contrast, the heat balance computation unit **203b** of the third embodiment calculates and outputs only the fuel flow rate G_{fs} .

The fuel distribution amount calculation unit **209** calculates three fuel flow rates $G_{fs}(1)$, $G_{fs}(2)$, and $G_{fs}(3)$ so that the fuel flow rate G_{fs} calculated by the heat balance computation unit **203b** becomes each ratio of the fuel distribution ratios D_1 , D_2 , and D_3 calculated by the fuel distribution ratio calculation unit **207**. The fuel distribution ratio D_1 corresponds to the fuel supply system **40-1**. Therefore, the fuel flow rate $G_{fs}(1)$ calculated corresponding to the fuel distribution ratio D_1 corresponds to the fuel supply system **40-1**. Similarly, the fuel supply system **40-2**, the fuel distribution ratio D_2 , and the fuel flow rate $G_{fs}(2)$ correspond to each other, and the fuel supply system **40-3**, the fuel distribution ratio D_3 , and the fuel flow rate $G_{fs}(3)$ correspond to each other.

The correction process unit **204b** includes a fuel flow rate correction unit **210b**. The fuel flow rate correction unit **210b** includes a fuel side management unit **211b**, a fuel side storage unit **212b**, fuel correction coefficient calculation units **213-1**, **213-2**, and **213-3**, fuel correction coefficient application units **214-1**, **214-2**, **214-3**, and an adder **216**. The fuel side storage unit **212b** includes fuel side storage areas **212b-1**, **212b-2**, and **212b-3** which are storage areas provided in advance in the fuel side storage unit **212b**.

The fuel side management unit **211b** associates the fuel flow rate $G_{ft}(1)$ corresponding to the fuel supply system **40-1** with the fuel flow rate $G_{fs}(1)$. The fuel side management unit **211b** associates the fuel flow rate $G_{ft}(2)$ corresponding to the fuel supply system **40-2** with the fuel flow rate $G_{fs}(2)$. The fuel side management unit **211b** associates the fuel flow rate $G_{ft}(3)$ corresponding to the fuel supply

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system **40-3** with the fuel flow rate G_{fs} (3). In accordance with the state detected by the state detection unit **205**, the fuel side management unit **211b** writes and stores the associated fuel flow rates G_{ft} (1) and G_{fs} (1), the associated fuel flow rates G_{ft} (2) and G_{fs} (2), and the associated fuel flow rates G_{ft} (3) and G_{fs} (3) in the respective fuel side storage areas **212b-1**, **212b-2**, and **212b-3**. The fuel side management unit **211b** outputs a latest combination of the fuel flow rates G_{ft} (3) and G_{fs} (3), a latest combination of the fuel flow rates G_{ft} (2) and G_{fs} (2), and a latest combination of the fuel flow rates G_{ft} (3) and G_{fs} (3) to the respective fuel correction coefficient calculation units **213-1**, **213-2**, and **213-3**.

In the fuel side storage unit **212b**, the fuel side storage area **212b-1** stores the combination of the fuel flow rates G_{ft} (1) and G_{fs} (1) in a time-series order. The fuel side storage area **212b-2** stores the combination of the fuel flow rates G_{ft} (2) and G_{fs} (2) in a time-series order. The fuel side storage area **212b-3** stores the combination of the fuel flow rates G_{ft} (3) and G_{fs} (3) in a time-series order. Each of the fuel correction coefficient calculation units **213-1**, **213-2**, and **213-3** has the same configuration as that of the fuel correction coefficient calculation unit **213** of the first embodiment. Each of the fuel correction coefficient application units **214-1**, **214-2**, and **214-3** has the same configuration as that of the fuel correction coefficient application unit **214** of the first embodiment. The adder **216** calculates the correction fuel flow rate G_{fcor} by summing up the correction fuel flow rates G_{fcor} (1), G_{fcor} (2), and G_{fcor} (3) calculated by each of the fuel correction coefficient application units **214-1**, **214-2**, and **214-3**.

(Operation Example of Gas Turbine Control Device of Third Embodiment)

A process performed by the gas turbine control device **20b** will be described with reference to FIGS. **14** and **15**. In FIG. **14**, as the process in Step Sb1-1, the same process as that in Step S1-1 in FIG. **3** is performed by the fuel flow rate calculation unit **201b**. As the process in Step Sb1-2, the same process as that in Step S1-2 in FIG. **3** is performed by the heat balance computation unit **203b**. The fuel distribution amount calculation unit **209** captures the fuel distribution ratios D_1 , D_2 , and D_3 calculated and output by the fuel distribution ratio calculation unit **207** in the previous detection cycle (Step Sb1-3).

The fuel flow rate calculation unit **201b** calculates the fuel flow rates G_{ft} (1), G_{ft} (2), and G_{ft} (3) by substituting the data captured in the process in Step Sb1-1 into Equations (1), (2), and (3). The fuel flow rate calculation unit **201b** outputs the calculated fuel flow rates G_{ft} (1), G_{ft} (2), and G_{ft} (3) to the fuel side management unit **211b**. The fuel flow rate calculation unit **201b** outputs the calculated fuel flow rate G_{ft} (1) to the fuel correction coefficient application unit **214-1**, outputs the fuel flow rate G_{ft} (2) to the fuel correction coefficient application unit **214-2**, and outputs the fuel flow rate G_{ft} (3) to the fuel correction coefficient application unit **214-3** (Sb2-1).

The heat balance computation unit **203b** calculates the fuel flow rate G_{fs} , based on the data captured in the process in Step Sb1-2 and on the gas turbine heat balance computation equation. The heat balance computation unit **203b** outputs the calculated fuel flow rate G_{fs} to the fuel distribution amount calculation unit **209** (Sb2-2).

The fuel distribution amount calculation unit **209** calculates three fuel flow rates G_{fs} (1), G_{fs} (2), and G_{fs} (3) so that the fuel flow rate G_{fs} calculated by the heat balance computation unit **203b** becomes each ratio of the fuel distribution ratios D_1 , D_2 , and D_3 captured in the process in Step

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Sb1-3. For example, the fuel distribution amount calculation unit **209** calculates the fuel flow rates as G_{fs} (1) = $G_{fs} \times D_1 / (D_1 + D_2 + D_3)$, G_{fs} (2) = $G_{fs} \times D_2 / (D_1 + D_2 + D_3)$, and G_{fs} (3) = $G_{fs} \times D_3 / (D_1 + D_2 + D_3)$. The fuel distribution amount calculation unit **209** outputs the calculated fuel flow rates G_{fs} (1), G_{fs} (2), and G_{fs} (3) to the fuel side management unit **211b** (Sb2-3).

A set of processes in Steps Sb1-1 and Sb1-2, a set of processes in Steps Sb1-2 and Sb2-2, and a process in Step Sb1-3 are performed in parallel.

When receiving the fuel flow rates G_{ft} (1), G_{ft} (2), and G_{ft} (3) and the fuel flow rates G_{fs} (1), G_{fs} (2), and G_{fs} (3), the fuel flow rate correction unit **210b** of the correction process unit **204b** starts the subroutine of the storage unit management process illustrated in FIG. **15** (Sb3).

In the fuel flow rate correction unit **210b**, the fuel side management unit **211b** captures the fuel flow rates G_{ft} (1), G_{ft} (2), and G_{ft} (3) output by the fuel flow rate calculation unit **201b**, and the fuel flow rates G_{fs} (1), G_{fs} (2), and G_{fs} (3) output by the fuel distribution amount calculation unit **209**. The fuel flow rate correction unit **210b** associates the captured fuel flow rate G_{ft} (1) with the fuel flow rate G_{fs} (1), associates the captured fuel flow rate G_{ft} (2) with the fuel flow rate G_{fs} (2), and associates the captured fuel flow rate G_{ft} (3) with the fuel flow rate G_{fs} (3) (Sb100). The fuel side management unit **211b** acquires the data indicating the state of the gas turbine power generation plant **100b** which is detected by the state detection unit **205** from the state detection unit **205**, at a time when the process in Step Sb100 is completed (Sb101).

The fuel side management unit **211b** determines whether or not the acquired data indicating the state indicates the static state (Sb102). When the fuel side management unit **211b** determines that the acquired data indicating the state indicates the static state (Sb102, Yes), the fuel side management unit **211b** performs the processes in Steps Sb103-1, Sb103-2, and Sb103-3, and thereafter, performs the processes in Steps Sb104-1, Sb104-2, and Sb104-3.

That is, the fuel side management unit **211b** writes and stores the associated combination of the fuel flow rate G_{ft} (1) and the fuel flow rate G_{fs} (1) in the fuel side storage area **212b-1**. The fuel side management unit **211b** writes the combination of the fuel flow rates G_{ft} (1) and G_{fs} (1) to be sequentially written in the fuel side storage area **212b-1** to be stored in the fuel side storage area **212b-1** in a time-series order, for example, so that a newly written combination is stored in the fuel side storage area **212b-1** at the beginning of the time-series order (Sb103-1). The fuel side management unit **211b** outputs the combination of the fuel flow rate G_{ft} (1) and the fuel flow rate G_{fs} (1) written in the fuel side storage area **212b-1** to the fuel correction coefficient calculation unit **213-1** (Sb104-1).

As the processes in Steps Sb103-2 and Sb104-2, the fuel side management unit **211b** performs the following processes. In the processes in Steps Sb103-1 and Sb104-1, the fuel flow rate G_{ft} (1) is replaced with the fuel flow rate G_{ft} (2), the fuel flow rate G_{fs} (1) is replaced with the fuel flow rate G_{fs} (2), the fuel side storage area **212b-1** is replaced with the fuel side storage area **212b-2**, and the fuel correction coefficient calculation unit **213-1** is replaced with the fuel correction coefficient calculation unit **213-2**. Similarly, as the processes in Steps Sb103-3 and Sb104-3, the fuel side management unit **211b** performs the following processes. In the processes in Steps Sb103-1 and Sb104-1, the fuel flow rate G_{ft} (1) is replaced with the fuel flow rate G_{ft} (3), the fuel flow rate G_{fs} (1) is replaced with the fuel flow rate G_{fs} (3), the fuel side storage area **212b-1** is replaced with the fuel side storage area **212b-3**, and the fuel correction coefficient

calculation unit **213-1** is replaced with the fuel correction coefficient calculation unit **213-3**.

When the fuel side management unit **211b** determines in the process in Step **Sb102** that the acquired data indicating the state does not indicate the static state (**Sb102**, No), the fuel side management unit **211b** discards the fuel flow rates $G_{ft}(1)$, $G_{ft}(2)$, and $G_{ft}(3)$, and the fuel flow rates $G_{fs}(3)$, $G_{fs}(2)$, and $G_{fs}(1)$ which are captured in the process in Step **Sb100**, and reads the latest combination out of the combinations stored in each of the fuel side storage areas **212b-1**, **212b-2**, and **212b-3**. As described above, when the fuel side storage areas **212b-1**, **212b-2**, and **212b-3** store the latest combination to be the beginning in a time-series order, the fuel side management unit **211b** reads the combination stored at the beginning of each of the fuel side storage areas **212b-1**, **212b-2**, and **212b-3** as the latest combination. The fuel side management unit **211b** outputs the latest combination of the fuel flow rates $G_{ft}(1)$ and $G_{fs}(1)$ read from the fuel side storage area **212b-1** to the fuel correction coefficient calculation unit **213-1**, outputs the latest combination of the fuel flow rates $G_{ft}(2)$ and $G_{fs}(2)$ read from the fuel side storage area **212b-2** to the fuel correction coefficient calculation unit **213-2**, and outputs the latest combination of the fuel flow rates $G_{ft}(3)$ and $G_{fs}(3)$ read from the fuel side storage area **212b-3** to the fuel correction coefficient calculation unit **213-3** (**Sb105**).

In the process in FIG. 15, in the fuel side management unit **211b**, a set of the processes in Steps **Sb103-1** and **Sb104-1**, a set of the processes in Steps **Sb103-2** and **Sb104-2**, and a set of the processes in Steps **Sb103-3** and **Sb104-3** may be performed in any order, or may be performed in parallel. In addition, in the process in Step **Sb105**, the fuel side management unit **211b** may read the latest combination from each of the fuel side storage areas **212b-1**, **212b-2**, and **212b-3**. The processes of outputting the read latest combination to the fuel correction coefficient calculation units **213-1**, **213-2**, and **213-3** may be performed in any order, or may be performed in parallel.

With reference back to FIG. 14, the fuel correction coefficient calculation unit **213-1** captures the combination of the fuel flow rates $G_{ft}(1)$ and $G_{fs}(1)$ output by the fuel side management unit **211b** (**Sb4-1**). The fuel correction coefficient calculation unit **213-1** calculates the fuel correction coefficient $K_f(1)$ by dividing the fuel flow rate $G_{fs}(1)$ by the fuel flow rate $G_{ft}(1)$. The fuel correction coefficient calculation unit **213-1** outputs the calculated fuel correction coefficient $K_f(1)$ to the fuel correction coefficient application unit **214-1** (**Sb5-1**). The fuel correction coefficient application unit **214-1** captures the fuel flow rate $G_{ft}(1)$ output by the fuel flow rate calculation unit **201b** and the fuel correction coefficient $K_f(1)$ output by the fuel correction coefficient calculation unit **213-1**. The fuel correction coefficient application unit **214-1** calculates the correction fuel flow rate $G_{fcor}(1)$ by multiplying the fuel flow rate $G_{ft}(1)$ by the fuel correction coefficient $K_f(1)$. The fuel correction coefficient application unit **214-1** outputs the calculated correction fuel flow rate $G_{fcor}(1)$ to the adder **216** (**Sb6-1**).

Similarly, the processes in Steps **Sb4-2**, **Sb5-2**, and **Sb6-2** and the processes in Steps **Sb4-3**, **Sb5-3**, and **Sb6-3** are performed. Here, each of the processes of Steps **Sb4-2**, **Sb5-2**, and **Sb6-2** is the following process. In the processes in Steps **Sb4-1**, **Sb5-1**, and **Sb6-1** in which the branch number of each step number is replaced with “-1”, the fuel flow rate $G_{ft}(1)$ is replaced with the fuel flow rate $G_{ft}(2)$, the fuel flow rate $G_{fs}(1)$ is replaced with the fuel flow rate $G_{fs}(2)$, the fuel correction coefficient calculation unit **213-1** is replaced with the fuel correction coefficient calculation unit

213-2, the fuel correction coefficient $K_f(1)$ is replaced with the fuel correction coefficient $K_f(2)$, the fuel correction coefficient application unit **214-1** is replaced with the fuel correction coefficient application unit **214-2**, and the correction fuel flow rate $G_{fcor}(1)$ is replaced with the correction fuel flow rate $G_{fcor}(2)$. In addition, each of the processes in Steps **Sb4-3**, **Sb5-3**, and **Sb6-3** is the following process. In the processes in Steps **Sb4-1**, **Sb5-1**, and **Sb6-1** in which the branch number of each step number is replaced with “-1”, the fuel flow rate $G_{ft}(1)$ is replaced with the fuel flow rate $G_{ft}(3)$, the fuel flow rate $G_{fs}(1)$ is replaced with the fuel flow rate $G_{fs}(3)$, the fuel correction coefficient calculation unit **213-1** is replaced with the fuel correction coefficient calculation unit **213-3**, the fuel correction coefficient $K_f(1)$ is replaced with the fuel correction coefficient $K_f(3)$, the fuel correction coefficient application unit **214-1** is replaced with the fuel correction coefficient application unit **214-3**, and the correction fuel flow rate $G_{fcor}(1)$ is replaced with the correction fuel flow rate $G_{fcor}(3)$.

The adder **216** captures the correction fuel flow rates $G_{fcor}(1)$, $G_{fcor}(2)$, and $G_{fcor}(3)$ output by each of the fuel correction coefficient application units **214-1**, **214-2**, and **214-3**, and calculates the correction fuel flow rate G_{fcor} by summing up the captured correction fuel flow rates $G_{fcor}(1)$, $G_{fcor}(2)$, and $G_{fcor}(3)$. The adder **216** outputs the calculated correction fuel flow rate G_{fcor} to the turbine inlet temperature calculation unit **206** (**Sb7**). The mark “A” illustrated in FIG. 14 is a mark indicating that the process is continuously performed, and as indicated by the mark “A”, the same process as the process in FIG. 4 of the first embodiment is performed thereafter. A series of the above-described processes illustrated in FIGS. 14 and 4 are performed as the process of one detection cycle, and the series of processes illustrated in FIGS. 14 and 4 are performed for each cycle. (Operational Effect in Third Embodiment)

As described above, regardless of whether the state detected by the state detection unit **205** is the static state or the transient state, the fuel flow rate correction unit **210b** of the third embodiment calculates the fuel correction coefficient K_f by using the fuel flow rates $G_{ft}(1)$ and $G_{fs}(1)$, the fuel flow rates $G_{ft}(2)$ and $G_{fs}(2)$, and the fuel flow rates $G_{ft}(3)$ and $G_{fs}(3)$ in the recent static state. Therefore, in the configuration of the third embodiment, as in the configuration of the first embodiment, the accuracy of the fuel flow rate provided for the turbine inlet temperature calculation unit **206** is not lowered in both the static state and the transient state. In this manner, the turbine inlet temperature calculation unit **206** can maintain the accuracy of the turbine inlet temperature T_{17} for a long period of time without using the turbine efficiency which is less likely to be uniquely determined.

Furthermore, in the third embodiment, the fuel distribution amount calculation unit **209** calculates the fuel flow rates $G_{fs}(1)$, $G_{fs}(2)$, and $G_{fs}(3)$ in which the fuel flow rate G_{fs} calculated by the heat balance computation unit **203b** is distributed in accordance with the fuel distribution ratios D_1 , D_2 , and D_3 . Each of the fuel correction coefficient calculation units **213-1**, **213-2**, and **213-3** calculates the fuel correction coefficients $K_f(1)$, $K_f(2)$, and $K_f(3)$, based on the respectively corresponding fuel flow rates $G_{fs}(1)$, $G_{fs}(2)$, and $G_{fs}(3)$. Each of the fuel correction coefficient application units **214-1** to **214-3** corrects each of the fuel flow rates $G_{ft}(1)$, $G_{ft}(2)$, and $G_{ft}(3)$ for each of the fuel supply systems **40-1**, **40-2**, and **40-3** by using the respectively corresponding fuel correction coefficients $K_f(1)$, $K_f(2)$, and $K_f(3)$. Therefore, in the third embodiment, an inherent error existing in each of the fuel supply systems **40-1**, **40-2**, and **40-3**, that is, an

error of the fuel flow rate G_{fs} (1) and the fuel flow rate G_{ft} (1), an error of the fuel flow rate G_{fs} (2) and the fuel flow rate G_{ft} (2), and an error of the fuel flow rate G_{fs} (3) and the fuel flow rate G_{ft} (3) which are recorded in the static state can be individually corrected. Therefore, in the third embodiment, the fuel flow rate G_{ft} can be highly accurately corrected, compared to the configuration in which the correction process unit 204 includes only the fuel flow rate correction unit 210 in the configuration of the first embodiment. Accordingly, it is possible to further improve the accuracy of the turbine inlet temperature T_{1T} calculated by the turbine inlet temperature calculation unit 206.

(Other Configuration Examples in Third Embodiment)

In the third embodiment described above, the fuel distribution amount calculation unit 209 uses the fuel distribution ratios D_1 , D_2 , and D_3 calculated by the fuel distribution ratio calculation unit 207 to calculate the fuel flow rates G_{fs} (1), G_{fs} (2), and G_{fs} (3) from the fuel flow rate G_{fs} calculated by the heat balance computation unit 203b. In contrast, the fuel distribution amount calculation unit 209 may use the valve opening degrees O_1 , O_2 , and O_3 calculated by the valve opening degree calculation unit 208 to calculate the fuel flow rates G_{fs} (1), G_{fs} (2), and G_{fs} (3) from the fuel flow rate G_{fs} .

In addition, each of the fuel supply systems 40-1, 40-2, and 40-3 may include a sensor for measuring the flow rate of the fuel flowing through each of the fuel supply systems 40-1, 40-2, and 40-3, and may use each fuel flow rate of the fuel supply systems 40-1, 40-2, and 40-3 which is obtained from the sensor, and may calculate the fuel flow rates G_{fs} (1), G_{fs} (2), and G_{fs} (3) from the fuel flow rate G_{fs} . However, the data of the fuel flow rate obtained by the sensor measuring the flow rate of the fuel flowing through each of the fuel supply systems 40-1, 40-2, and 40-3 is the data which is sufficiently accurate to determine an approximate ratio of the fuel flowing through each of the fuel supply systems 40-1, 40-2, and 40-3. However, the data includes an absolute error in the measurement. Therefore, the data has a value different from the fuel flow rates G_{ft} (1), G_{ft} (2), and G_{ft} (3) calculated by the fuel flow rate calculation unit 201b. In addition, even when a total value of the data is provided for the turbine inlet temperature calculation unit 206, the data cannot obtain the accurate turbine inlet temperature T_{1T} .

In the third embodiment described above, the “method for calculating each correction coefficient from the average value of each of the fuel flow rate and the air flow rate” described in the first embodiment may be applied to the fuel flow rate correction unit 210b. In addition, the configuration of the second embodiment may be applied to the fuel flow rate correction unit 210b. In addition, in the third embodiment, the heat balance computation unit 203b may be replaced with the heat balance computation unit 203 of the first embodiment, and thereafter, the correction process unit 204b may include the air flow rate correction unit 220 of the first embodiment. The correction process unit 204b may include the air flow rate correction unit 220a of the second embodiment.

(Other Configuration Examples in First to Third Embodiments)

In the first to third embodiments described above, the gas turbine power generation plants 100, 100a, and 100b include the fuel temperature sensors 25c-1, 25c-2, and 25c-3 respectively corresponding to the fuel supply systems 40-1, 40-2, and 40-3. For example, when there is no significant difference in the temperatures of the fuels respectively flowing through the fuel supply systems 40-1, 40-2, and 40-3, any one of the fuel temperature sensors 25c-1, 25c-2, and 25c-3

may be provided. A sensor for measuring the temperature of the fuel before the fuel is branched into three may be provided in the fuel supply device 30. For example, when only the fuel temperature sensor 25c-1 is provided, each of the fuel temperatures T_f (2) and T_f (3) is regarded to have the same value as the fuel temperature T_f (1), and the processes according to the first to third embodiments described above are performed. As an example, only the fuel temperature T_f (1) is provided for the fuel flow rate calculation units 201 and 201b and the turbine inlet temperature calculation unit 205. When the computation of Equation (2) and Equation (3) is performed, the fuel flow rate calculation units 201 and 201b substitute the fuel temperature T_f (1) instead of the fuel temperatures T_f (2) and T_f (3). In addition, when the turbine inlet temperature calculation unit 206 substitutes the fuel temperature T_f into Equation (6), the turbine inlet temperature calculation unit 206 does not need to calculate the average value of the fuel temperatures T_f (1), T_f (2), and T_f (3), and substitutes the fuel temperature T_f (1) into Equation (6) as the fuel temperature T_f .

In the first to third embodiments described above, the fuel flow rate calculation units 201 and 201b adopt a method for calculating the fuel flow rates G_{ft} (1), G_{ft} (2), and G_{ft} (3), based on the flow control valve pre-pressures P_{1FV} (1), P_{1FV} (2), and P_{1FV} (3) detected by each of the flow control valve pre-pressure sensors 25a-1 to 25a-3, and on the flow control valve post-pressures P_{2FV} (1), P_{2FV} (2), and P_{2FV} (3) detected by each of the flow control valve post-pressure sensors 25b-1 to 25b-3. However, the method is an example, and the fuel flow rate calculation units 201 and 201b can calculate the fuel flow rates G_{ft} (1), G_{ft} (2), and G_{ft} (3) by using the fuel flow rate calculation function f_1 (•) modified so that the measurement data can be applied, as long as the measurement data relating to the fuel supply systems 40-1, 40-2, and 40-3, more specifically, the measurement data relating to the pressure which can be measured at two locations of each of the fuel supply systems 40-1, 40-2, and 40-3, can be applied. For example, nozzle pre-pressure sensors for measuring the pressure of the fuel flowing into the nozzles 42-1, 42-2, and 42-3 may be provided instead of the flow control valve pre-pressure sensors 25a-1 to 25a-3, and nozzle post-pressure sensors for measuring the pressure of the fuel flowing out from the nozzles 42-1, 42-2, and 42-3 may be provided instead of the flow control valve post-pressure sensors 25b-1 to 25b-3. In this case, the fuel flow rate calculation units 201 and 201b capture the pressure measured by the nozzle pre-pressure sensor and the pressure measured by the nozzle post-pressure sensor, and calculate the fuel flow rates G_{ft} (1), G_{ft} (2), and G_{ft} (3) by using the fuel flow rate calculation function f_1 (•) modified so that the two pressures can be applied, the valve opening degrees O_1 to O_3 , the two pressures, and the fuel temperatures T_f (1) to T_f (3). In addition, for example, a first sensor for measuring the pressure of the fuel inside the manifold pipes 43-1, 43-2, and 43-3 corresponding to each of the fuel supply systems 40-1, 40-2, and 40-3 may be provided instead of the flow control valve pre-pressure sensors 25a-1 to 25a-3, and a second sensor for measuring the pressure of the fuel flowing out from each of the manifold pipes 43-1, 43-2, and 43-3, that is, an exit pressure of a combustion nozzle, may be provided instead of in the flow control valve post-pressure sensors 25b-1 to 25b-3. An example of the second sensor is the combustor casing pressure sensor 25m for measuring the combustor casing pressure P_{out} described in the second embodiment.

In the first to third embodiments described above, the valve opening degree calculation unit 208 included in the

gas turbine control devices **20**, **20a**, and **20b** calculates the valve opening degrees O_1 , O_2 , and O_3 corresponding to each of the flow control valves **41-1**, **41-2**, and **41-3**. In contrast, the valve opening degree calculation unit **208** may calculate three Cv values corresponding to each of the flow control valves **41-1**, **41-2**, and **41-3**. In this case, the fuel flow rate calculation units **201** and **201b** capture the three Cv values calculated by the valve opening degree calculation unit **208**, and calculate fuel flow rates G_{f1} (1), G_{f1} (2), and G_{f1} (3) by substituting the three Cv values instead of the valve opening degrees O_1 to O_3 into the fuel flow rate calculation function f_1 (•) modified so that the Cv values can be applied.

In the first to third embodiments described above, the heat balance computation units **203** and **203b** capture the gas turbine output G_{out} , the exhaust gas pressure P_{2T} , the exhaust gas temperature T_{2T} , the compressor inlet pressure P_{1C} , the compressor inlet temperature T_{1C} , the compressor exit pressure P_{2C} , and the compressor exit temperature T_{2C} , but may capture the measurement data other than these.

In the first to third embodiments described above, as an example, a case where the fuel supply system **40** includes the three fuel supply systems **40-1** to **40-3** has been described. However, a configuration including two fuel supply systems **40-1** and **40-2**, or a configuration including four or more fuel supply systems **40-1**, **40-2**, **40-3**, and so forth may be adopted. In addition, in FIGS. **1** and **8**, as an example, a case where the gas turbine **10** includes one combustor **12** has been described. However, a configuration may be adopted as follows. The gas turbine **10** includes a plurality of the combustors **12**. Each of the combustors **12** is connected to the compressor **11** and the turbine **13**, and includes a plurality of the combustor inner cylinders **121** corresponding to each of the plurality of combustors **12**, and the fuel supply systems **40-1**, **40-2**, **40-3**, and so forth. In addition, a configuration may be adopted as follows. A plurality of combustor inner cylinders **121** are provided in one combustor **12**, and the fuel supply systems **40-1**, **40-2**, **40-3**, and so forth corresponding to each of the plurality of combustor inner cylinders **121** are provided. In this way, when the configuration is changed to change the number of the fuel supply systems **40-1**, **40-2**, and **40-3**, the number of the combustors **12**, and the number of the combustor inner cylinders **121** described in the first to third embodiments, the number of components such as the flow control valve **41** provided in the gas turbine power generation plants **100**, **100a**, and **100b** is changed in accordance with the configuration change, and a configuration of a functional unit included in the gas turbine control devices **20**, **20a**, and **20b** is also changed.

In the first to third embodiments described above, the gas turbine control devices **20**, **20a**, and **20b** include a configuration for calculating the command value for adjusting the amount of the air provided for the inlet guide vane **14** and suctioned by the inlet guide vane **14**, in addition to the configurations illustrated in FIGS. **2**, **9**, and **13**.

In the first to third embodiments described above, the gas turbine power generation plants **100**, **100a**, and **100b** include the flow control valves **41-1**, **41-2**, and **41-3**. However, a pressure regulation valve may be applied instead of the flow regulation valve.

In the first to third embodiments described above, the gas turbine power generation plants **100**, **100a**, and **100b** may further include a steam turbine, and may be configured as a gas turbine combined cycle power generation plant (GTCC) in which the steam turbine is connected to the rotor **15**.

In the first to third embodiments described above, the function f_3 (•) used by the turbine inlet temperature calcu-

lation unit **206** is a function constructed based on the non-stationary physical model. As described above, the heat energy flowing into the combustor **12** in the non-stationary physical model is represented by the total of the heat energy of the fuel, the heat energy of the air, and the heat generation energy of the combustion gas. The heat energy of the air varies depending on a specific enthalpy set in view of influence of humidity generated by steam vapor contained in the air flowing into the combustor casing **16**. Therefore, instead of calculating the heat energy of the air by using the parameter substituted into the function f_3 (•), that is, any one of the correction fuel flow rate G_{fcor} , the correction air flow rate G_{acor} , the combustor casing temperature T_{CS} , and the fuel temperature T_f which is the average value of the fuel temperatures T_f (1), T_f (2), and T_f (3), the turbine inlet temperature calculation unit **206** may calculate the heat energy of the air as follows. That is, a sensor for measuring the humidity of the air flowing into the combustor casing **16** may be newly provided. The turbine inlet temperature calculation unit **206** may calculate the specific enthalpy, based on the humidity of the air flowing into the combustor casing **16** which is measured by the sensor, and may calculate the heat energy of the air by using the calculated specific enthalpy.

In addition, in the configuration of the first to third embodiments described above, the state detection unit **205** performs a determination process using an unequal sign provided with an equal sign in the process of detecting the state of the gas turbine power generation plant **100**, the processes in Steps **S105-1** and **8105-2** in FIG. **6**, and the processes in Steps **Sa104-1** and **Sa104-2** in FIG. **11**. However, the present disclosure is not limited to the embodiment, and the determination process of “whether an object is equal to or greater than a threshold value” is merely an example. Depending on a setting method for the threshold value, each process may be replaced with the determination process of “whether or not the object exceeds the threshold value”.

Hitherto, the embodiments of the present disclosure have been described in detail with reference to the drawings. However, the specific configuration is not limited to the embodiments, and the present disclosure includes design within the scope not departing from the concept of the present disclosure.

<Computer Configuration>

In the first to third embodiments described above, for example, the gas turbine control devices **20**, **20a**, and **20b** are configured to include a central processing unit (CPU), a random access memory (RAM), a read only memory (ROM), and a computer-readable storage medium. As an example, a series of processes for realizing various functions included in the gas turbine control devices **20**, **20a**, and **20b** are stored in a storage medium in a form of a program, and various functions are realized in such a manner that the CPU reads the program from the RAM and executes information processing and computation processing. The program may adopt a form in which the program is installed in advance in the ROM or another storage medium, a form in which the program is provided in a stored state in the computer-readable storage medium, or a form in which the program is distributed via wired or wireless communication means. The computer-readable storage medium is a magnetic disc, a magneto-optical disc, a CD-ROM, a DVD-ROM, or a semiconductor memory.

More specifically, each of the gas turbine control devices **20**, **20a**, and **20b** according to the first to third embodiments described above is realized by a configuration of a computer illustrated in FIG. **16**. A computer **90** includes a processor

91, a main memory 92, a storage 93, and an interface 94. Functional units of the fuel flow rate calculation units 201 and 201b, the heat balance computation units 203 and 203b, the air flow rate calculation unit 202, the correction process units 204, 204a, and 204b, the state detection unit 205, the turbine inlet temperature calculation unit 206, the fuel distribution ratio calculation unit 207, the valve opening degree calculation unit 208, and the fuel distribution amount calculation unit 209 which are included in the gas turbine control devices 20, 20a, and 20b are installed in the computer 90. Operations of the functional units are stored in the storage 93 in a form of a program. The processor 91 reads the program from the storage 93, deploys the program into the main memory 92, and performs the processes described in the first to third embodiments in accordance with the program deployed in the main memory 92. In addition, for example, in accordance with the program, the processor 91 secures areas of the fuel side storage units 212, 212a, and 212b and the air side storage units 222 and 222a in the main memory 92. In addition, in accordance with the program, the processor 91 allocates the interface 94 for transmitting and receiving data among various sensors 25a to 20m, the inlet guide vane 14, and the flow control valve 41 to the functional units included in the gas turbine control devices 20, 20a, and 20b.

The program may partially realize functions fulfilled by the computer 90. For example, the program may cause the functions to be fulfilled in combination with another program previously stored in the storage 93 or in combination with another program installed in another device. In another embodiment, in addition to the above-described configuration or instead of the above-described configuration, the computer may include a custom large scale integrated circuit (LSI) such as a programmable logic device (PLD). Examples of the PLD include a programmable array logic (PAL), a generic array logic (GAL), a complex programmable logic device (CPLD), and a field programmable gate array (FPGA). In this case, the functions realized by the processor may be partially or entirely realized by the integrated circuit.

Examples of the storage 93 include a hard disk drive (HDD), a solid state drive (SSD), a magnetic disc, a magneto-optical disc, a compact disc read only memory (CD-ROM), a digital versatile disc read only memory (DVD-ROM), and a semiconductor memory. The storage 93 may be internal media directly connected to a bus of the computer 90, or external media connected to the computer 90 via the interface 94 or a communication line. In addition, when this program is distributed to the computer 90 via the communication line, the computer 90 receiving the distribution may deploy the program in the main memory 92, and may perform the above-described process. In addition, the storage 93 is a tangible storage medium that is not temporary.

Each of the gas turbine control devices 20, 20a, and 20b according to the first to third embodiments described above is not limited to being realized as an integrated computer. For example, in the first and third embodiments, configurations of the fuel flow rate calculation units 201 and 201b, the air flow rate calculation unit 202, the fuel correction coefficient application units 214 and 214-1 to 234-3, the air correction coefficient application unit 224, the turbine inlet temperature calculation unit 206, the fuel distribution ratio calculation unit 207, and the valve opening degree calculation unit 208 and a configuration including the functional unit which is not included in the configuration may be provided in respectively different computers, and two computers may transmit and receive the data through a commu-

nication line. In addition, in the second embodiment, configurations of the fuel flow rate calculation unit 201, the air flow rate calculation unit 202, the fuel correction coefficient calculation unit 213a, the fuel correction coefficient application unit 214, the air correction coefficient calculation unit 223a, the air correction coefficient application unit 224, the turbine inlet temperature calculation unit 206, the fuel distribution ratio calculation unit 207, and the valve opening degree calculation unit 208 and a configuration including the functional unit which is not included in the configuration may be provided in respectively different computers, and two computers may transmit and receive the data through the communication line.

Additional Notes

The gas turbine control device (for example, the gas turbine control devices 20, 20a, 20b) described in each embodiment is understood as follows, for example.

(1) According to a first aspect, there is provided the gas turbine control device including the fuel flow rate calculation unit (for example, fuel flow rate calculation units 201 and 201b) that calculates the fuel flow rate G_{fi} indicating the amount of the fuel flowing into the combustor 12 of the gas turbine 10, based on the measurement data (for example, valve opening degrees O_1 , O_2 , and O_3 , flow control valve pre-pressures P_{1FV} (1), P_{1FV} (2), and P_{1FV} (3), flow control valve post-pressures P_{2FV} (1), P_{2FV} (2), and P_{2FV} (3), and fuel temperatures T_f (1), T_f (2), and T_f (3)) relating to a fuel supply system 40 for supplying the fuel to the combustor; the air flow rate calculation unit 202 that calculates the air flow rate G_{st} indicating the amount of the air suctioned by the compressor 11 of the gas turbine, based on the measurement data (for example, compressor index differential pressure P_{index} , compressor inlet pressure P_{1C} , and compressor inlet temperature T_{1C}) relating to the compressor; the turbine inlet temperature calculation unit 206 that calculates the turbine inlet temperature T_{1T} , based on the fuel flow rate G_{fi} , the air flow rate G_{st} , and the physical model equation (for example, fuel flow rate calculation function f_1 (•)) relating to the heat energy balance relating to the combustor; the heat balance computation unit (for example, heat balance computation units 203 and 203b) that performs the computation for satisfying the heat balance of the gas turbine, based on the measurement data (for example, gas turbine output G_{out} , exhaust gas pressure P_{2T} , exhaust gas temperature T_{2T} , compressor inlet pressure P_{1C} , compressor inlet temperature T_{1C} , compressor exit pressure P_{2C} , and compressor exit temperature T_{2C}) relating to the energy balance of the gas turbine, and outputs the computation result; the state detection unit 205 that detects the state of the gas turbine; and the correction process unit (for example, correction process units 204, 204a, and 204b) that corrects any one or both of the fuel flow rate and the air flow rate, based on the computation result output by the heat balance computation unit when the state of the gas turbine which is detected by the state detection unit is the static state, and provides the corrected flow rate to the turbine inlet temperature calculation unit. According to the first aspect and each of the following aspects, the fuel flow rate G_{fi} and the air flow rate G_{st} can be corrected, based on the computation result of the heat balance computation unit. Therefore, correction can be performed without using the turbine efficiency which is less likely to be uniquely determined. Furthermore, the heat balance computation unit performs the computation, based on the measurement data obtained through repeated mea-

surement. Therefore, the accuracy of the turbine inlet temperature T_{1T} can be maintained for a long period of time.

(2) As the gas turbine control device according to a second aspect, in the gas turbine control device of (1), the fuel flow rate calculated by the fuel flow rate calculation unit is defined as the first fuel flow rate G_{f1} , and the air flow rate calculated by the air flow rate calculation unit is defined as the first air flow rate G_{a1} . The correction process unit includes any one correction unit or both correction units of the fuel flow rate correction unit (for example, fuel flow rate correction units **210**, **210a**, and **210b**) that acquires the second fuel flow rate G_{f2} included in the computation result output by the heat balance computation unit when the state of the gas turbine which is detected by the state detection unit is the static state, that corrects the first fuel flow rate calculated by the fuel flow rate calculation unit, based on the acquired second fuel flow rate and the first fuel flow rate when the second fuel flow rate is acquired, and that provides the corrected first fuel flow rate to the turbine inlet temperature calculation unit, and the air flow rate correction unit (for example, air flow rate correction units **220** and **220a**) that acquires the second air flow rate G_{a2} included in the computation result output by the heat balance computation unit when the state of the gas turbine which is detected by the state detection unit is the static state, that corrects the first air flow rate calculated by the air flow rate calculation unit, based on the acquired second air flow rate and the first air flow rate when the second air flow rate is acquired, and that provides the corrected first air flow rate to the turbine inlet temperature calculation unit. According to the second aspect, the first fuel flow rate G_{f1} can be corrected by the highly accurate second fuel flow rate G_{f2} included in the computation result of the heat balance computation unit, and the first air flow rate G_{a1} can be corrected by the highly accurate second air flow rate G_{a2} included in the computation result. Therefore, the highly accurate turbine inlet temperature T_{1T} can be calculated.

(3) As the gas turbine control device according to a third aspect, in the gas turbine control device of (2), the fuel flow rate correction unit includes the fuel correction coefficient calculation unit (for example, fuel correction coefficient calculation units **213** and **213-1** to **213-3**) that calculates the fuel correction coefficient, based on a latest second fuel flow rate in the second fuel flow rates acquired when the state of the gas turbine which is detected by the state detection unit is the static state, and on the first fuel flow rate when the latest second fuel flow rate is acquired, and the fuel correction coefficient application unit (for example, fuel correction coefficient application units **214** and **214-1** to **214-3**) that corrects the first fuel flow rate calculated by the fuel flow rate calculation unit by using the fuel correction coefficient calculated by the fuel correction coefficient calculation unit. According to the third aspect, the fuel correction coefficient K_f is calculated by using the latest second fuel flow rate G_{f2} and the first fuel flow rate G_{f1} when the latest second fuel flow rate G_{f2} is acquired, and the first fuel flow rate G_{f1} is corrected by using the calculated fuel correction coefficient K_f . Therefore, the highly accurate turbine inlet temperature T_{1T} can be calculated, and the accuracy of the turbine inlet temperature T_{1T} can be maintained for a long period of time.

(4) As the gas turbine control device according to a fourth aspect, in the gas turbine control device of (2), the fuel flow rate correction unit includes the fuel correction coefficient calculation unit (for example, fuel correction coefficient calculation units **213** and **213-1** to **213-3**) that calculates the fuel correction coefficient, based on the plurality of second fuel flow rates acquired when the state of the gas turbine

which is detected by the state detection unit is the static state, and on the first fuel flow rate when each of the plurality of second fuel flow rates is acquired, and the fuel correction coefficient application unit (for example, fuel correction coefficient application units **214** and **214-1** to **214-3**) that corrects the first fuel flow rate calculated by the fuel flow rate calculation unit by using the fuel correction coefficient calculated by the fuel correction coefficient calculation unit. According to the fourth aspect, the fuel correction coefficient K_f is calculated, based on the plurality of second fuel flow rates G_{f2} and the plurality of first fuel flow rates G_{f1} , and the first fuel flow rate G_{f1} is corrected by using the calculated fuel correction coefficient K_f . Therefore, the fuel correction coefficient K_f which is highly accurate in terms of reliability can be obtained.

(5) As the gas turbine control device according to a fifth aspect, in the gas turbine control device according to any one of (2) to (4), the air flow rate correction unit includes the air correction coefficient calculation unit **223** that calculates the air correction coefficient, based on a latest second air flow rate in the second air flow rates acquired when the state of the gas turbine which is detected by the state detection unit is the static state, and on the first air flow rate when the latest second air flow rate is acquired, and the air correction coefficient application unit **224** that corrects the first air flow rate calculated by the air flow rate calculation unit by using the air correction coefficient calculated by the air correction coefficient calculation unit. According to the fifth aspect, the air correction coefficient K_a is calculated by using the latest second air flow rate G_{a2} and the first air flow rate G_{a1} when the latest second air flow rate G_{a2} is acquired, and the first air flow rate G_{a1} is corrected by using the calculated air correction coefficient K_a . Therefore, the highly accurate turbine inlet temperature T_{1T} can be calculated, and the accuracy of the turbine inlet temperature T_{1T} can be maintained for a long period of time.

(6) As the gas turbine control device according to a sixth aspect, in the gas turbine control device according to any one of (2) to (4), the air flow rate correction unit includes the air correction coefficient calculation unit **223** that calculates the air correction coefficient, based on the plurality of second air flow rates acquired when the state of the gas turbine which is detected by the state detection unit is the static state, and on the first air flow rate when each of the plurality of second air flow rates is acquired, and the air correction coefficient application unit **224** that corrects the first air flow rate calculated by the air flow rate calculation unit by using the air correction coefficient calculated by the air correction coefficient calculation unit. According to the sixth aspect, the air correction coefficient K_a is calculated, based on the plurality of second air flow rates G_{a2} and the plurality of first air flow rates G_{a1} , and the first air flow rate G_{a1} is corrected by using the calculated air correction coefficient K_a . Therefore, the air correction coefficient K_a which is highly accurate in terms of reliability can be obtained.

(7) As the gas turbine control device according to a seventh aspect, in the gas turbine control device according to any one of (2), (5), and (6), the fuel flow rate correction unit includes the fuel side regression analysis unit **215** that performs the predetermined regression analysis (for example, linear multiple regression analysis), based on the second fuel flow rate acquired when the state of the gas turbine which is detected by the state detection unit is the static state, the first fuel flow rate when the second fuel flow rate is acquired, and data (for example, combustor casing temperature T_{CS} and the combustor casing pressure P_{out}) of the explanatory variable, and that calculates the coefficient

of the regression equation corresponding to the predetermined regression analysis, the fuel correction coefficient calculation unit (for example, fuel correction coefficient calculation units **213a** and **213-1** to **213-3**) that calculates the fuel correction coefficient by substituting the data of the explanatory variable when the fuel flow rate calculation unit calculates the first fuel flow rate, into the regression equation to which the coefficient calculated by the fuel side regression analysis unit is applied, and the fuel correction coefficient application unit (for example, fuel correction coefficient application units **214** and **214-1** to **214-3**) that corrects the first fuel flow rate corresponding to the data of the explanatory variable substituted into the regression equation by using the fuel correction coefficient calculated by the fuel correction coefficient calculation unit. According to the seventh aspect, the fuel correction coefficient K_f which is highly accurate in terms of reliability can be obtained by using the regression analysis.

(8) As the gas turbine control device according to an eighth aspect, in the gas turbine control device according to any one of (2), (3), (4), and (7), the air flow rate correction unit includes the air side regression analysis unit **225** that performs the predetermined regression analysis, based on the second air flow rate acquired when the state of the gas turbine which is detected by the state detection unit is the static state, the first air flow rate when the second air flow rate is acquired, and data of the explanatory variable, and that calculates the coefficient of the regression equation corresponding to the predetermined regression analysis, the air correction coefficient calculation unit **223a** that calculates the air correction coefficient by substituting the data of the explanatory variable when the air flow rate calculation unit calculates the first air flow rate, into the regression equation to which the coefficient calculated by the air side regression analysis unit is applied, and the air correction coefficient application unit **224** that corrects the first air flow rate corresponding to the data of the explanatory variable substituted into the regression equation by using the air correction coefficient calculated by the air correction coefficient calculation unit. According to the eighth aspect, the air correction coefficient K_a which is highly accurate in terms of reliability can be obtained by using the regression analysis.

(9) As the gas turbine control device according to a ninth aspect, in the gas turbine control device according to (7) or (8), the data of the explanatory variable is data in which the measurement data and data of the command value for controlling the gas turbine are combined in any desired way. According to the ninth aspect, the fuel correction coefficient K_f or the air correction coefficient K_a reflecting the measurement data acquired by the measurement and the data of the command value provided for the gas turbine **10** can be obtained.

(10) As the gas turbine control device according to a tenth aspect, in the gas turbine control device according to any one of (2) to (9), when the correction process unit includes the fuel flow rate correction unit, and there exist the plurality of fuel supply systems, the correction process unit includes the fuel distribution amount calculation unit **209** that calculates the second fuel flow rate for each of the plurality of fuel supply systems from the second fuel flow rates included in the computation result output by the heat balance computation unit, based on the fuel distribution ratio (for example, fuel distribution ratios D_1 , D_2 , and D_3) for each of the plurality of fuel supply systems. The fuel flow rate calculation unit **201b** calculates the first fuel flow rate (for example, fuel flow rates $G_{fs}(1)$, $G_{fs}(2)$, $G_{fs}(3)$) for each of

the fuel supply systems. Instead of the second fuel flow rate included in the computation result output by the heat balance computation unit **203b** when the state of the gas turbine which is detected by the state detection unit is the static state, the fuel flow rate correction unit **210b** acquires the second fuel flow rate (for example, fuel flow rates $G_{fs}(1)$, $G_{fs}(2)$, $G_{fs}(3)$) for each of the fuel supply systems calculated by the fuel distribution amount calculation unit, corrects the first fuel flow rate for each of the fuel supply systems calculated by the fuel flow rate calculation unit, based on each of the acquired second fuel flow rates and each of the first fuel flow rates for each of the fuel supply systems when each of the second fuel flow rates is acquired, and provides the total value of the corrected first fuel flow rate for each of the fuel supply systems to the turbine inlet temperature calculation unit. According to the tenth aspect, the fuel flow rates $G_{fs}(1)$, $G_{fs}(2)$, and $G_{fs}(3)$ for each of the fuel supply systems **40-1**, **40-2**, and **40-3** can be individually corrected.

INDUSTRIAL APPLICABILITY

According to the above-described aspects, the accuracy of the turbine inlet temperature is maintained for a long period of time without using the turbine efficiency.

REFERENCE SIGNS LIST

- 20:** Gas turbine control device
- 201:** Fuel flow rate calculation unit
- 202:** Air flow rate calculation unit
- 203:** Heat balance computation unit
- 204:** Correction process unit
- 205:** State detection unit
- 206:** Turbine inlet temperature calculation unit
- 207:** Fuel distribution ratio calculation unit
- 208:** Valve opening degree calculation unit
- 210:** Fuel flow rate correction unit
- 211:** Fuel side management unit
- 212:** Fuel side storage unit
- 213:** Fuel correction coefficient calculation unit
- 214:** Fuel correction coefficient application unit
- 220:** Air flow rate correction unit
- 221:** Air side management unit
- 222:** Air side storage unit
- 223:** Air correction coefficient calculation unit
- 224:** Air correction coefficient application unit

The invention claimed is:

1. A gas turbine control device comprising:
 - a fuel flow rate calculation unit that calculates a fuel flow rate indicating the amount of a fuel flowing into a combustor of a gas turbine, based on measurement data relating to a fuel supply system for supplying the fuel to the combustor;
 - an air flow rate calculation unit that calculates an air flow rate indicating the amount of air suctioned by a compressor of the gas turbine, based on measurement data relating to the compressor;
 - a turbine inlet temperature calculation unit that calculates a turbine inlet temperature, based on the fuel flow rate, the air flow rate, and a physical model equation relating to heat energy balance relating to the combustor;
 - a heat balance computation unit that performs a computation for satisfying heat balance of the gas turbine, based on measurement data relating to energy balance of the gas turbine, and outputs a computation result;
 - a state detection unit that detects a state of the gas turbine; and

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a correction process unit that corrects any one or both of the fuel flow rate and the air flow rate, based on the computation result output by the heat balance computation unit when the state of the gas turbine which is detected by the state detection unit is a static state, and provides the corrected flow rate to the turbine inlet temperature calculation unit,

wherein the gas turbine control device controls a flow control valve of the fuel supply system based on the turbine inlet temperature calculated by the turbine inlet temperature calculation unit.

2. The gas turbine control device according to claim 1, wherein the fuel flow rate calculated by the fuel flow rate calculation unit is defined as a first fuel flow rate, the air flow rate calculated by the air flow rate calculation unit is defined as a first air flow rate, and the correction process unit includes any one correction unit or both correction units of

a fuel flow rate correction unit that acquires a second fuel flow rate included in the computation result output by the heat balance computation unit when the state of the gas turbine which is detected by the state detection unit is the static state, that corrects the first fuel flow rate calculated by the fuel flow rate calculation unit, based on the acquired second fuel flow rate and the first fuel flow rate when the second fuel flow rate is acquired, and that provides the corrected first fuel flow rate to the turbine inlet temperature calculation unit, and

an air flow rate correction unit that acquires a second air flow rate included in the computation result output by the heat balance computation unit when the state of the gas turbine which is detected by the state detection unit is the static state, that corrects the first air flow rate calculated by the air flow rate calculation unit, based on the acquired second air flow rate and the first air flow rate when the second air flow rate is acquired, and that provides the corrected first air flow rate to the turbine inlet temperature calculation unit.

3. The gas turbine control device according to claim 2, wherein the fuel flow rate correction unit includes

a fuel correction coefficient calculation unit that calculates a fuel correction coefficient, based on a latest second fuel flow rate in the second fuel flow rates acquired when the state of the gas turbine which is detected by the state detection unit is the static state, and on the first fuel flow rate when the latest second fuel flow rate is acquired, and

a fuel correction coefficient application unit that corrects the first fuel flow rate calculated by the fuel flow rate calculation unit by using the fuel correction coefficient calculated by the fuel correction coefficient calculation unit.

4. The gas turbine control device according to claim 2, wherein the fuel flow rate correction unit includes

a fuel correction coefficient calculation unit that calculates a fuel correction coefficient, based on a plurality of the second fuel flow rates acquired when the state of the gas turbine which is detected by the state detection unit is the static state, and on the first fuel flow rate when each of the plurality of second fuel flow rates is acquired, and

a fuel correction coefficient application unit that corrects the first fuel flow rate calculated by the fuel

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flow rate calculation unit by using the fuel correction coefficient calculated by the fuel correction coefficient calculation unit.

5. The gas turbine control device according to claim 2, wherein the air flow rate correction unit includes

an air correction coefficient calculation unit that calculates an air correction coefficient, based on a latest second air flow rate in the second air flow rates acquired when the state of the gas turbine which is detected by the state detection unit is the static state, and on the first air flow rate when the latest second air flow rate is acquired, and

an air correction coefficient application unit that corrects the first air flow rate calculated by the air flow rate calculation unit by using the air correction coefficient calculated by the air correction coefficient calculation unit.

6. The gas turbine control device according to claim 2, wherein the air flow rate correction unit includes

an air correction coefficient calculation unit that calculates an air correction coefficient, based on a plurality of the second air flow rates acquired when the state of the gas turbine which is detected by the state detection unit is the static state, and on the first air flow rate when each of the plurality of second air flow rates is acquired, and

an air correction coefficient application unit that corrects the first air flow rate calculated by the air flow rate calculation unit by using the air correction coefficient calculated by the air correction coefficient calculation unit.

7. The gas turbine control device according to claim 2, wherein the fuel flow rate correction unit includes

a fuel side regression analysis unit that performs a predetermined regression analysis, based on the second fuel flow rate acquired when the state of the gas turbine which is detected by the state detection unit is the static state, the first fuel flow rate when the second fuel flow rate is acquired, and data of an explanatory variable, and that calculates a coefficient of a regression equation corresponding to the predetermined regression analysis,

a fuel correction coefficient calculation unit that calculates a fuel correction coefficient by substituting the data of the explanatory variable when the fuel flow rate calculation unit calculates the first fuel flow rate, into the regression equation to which the coefficient calculated by the fuel side regression analysis unit is applied, and

a fuel correction coefficient application unit that corrects the first fuel flow rate corresponding to the data of the explanatory variable substituted into the regression equation by using the fuel correction coefficient calculated by the fuel correction coefficient calculation unit.

8. The gas turbine control device according to claim 7, wherein the data of the explanatory variable is data in which the measurement data and data of a command value for controlling the gas turbine are combined in any desired way.

9. The gas turbine control device according to claim 2, wherein the air flow rate correction unit includes

an air side regression analysis unit that performs a predetermined regression analysis, based on the second air flow rate acquired when the state of the gas turbine which is detected by the state detection unit is the static state, the first air flow rate when the

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second air flow rate is acquired, and data of an explanatory variable, and that calculates a coefficient of a regression equation corresponding to the predetermined regression analysis,

an air correction coefficient calculation unit that calculates an air correction coefficient by substituting the data of the explanatory variable when the air flow rate calculation unit calculates the first air flow rate, into the regression equation to which the coefficient calculated by the air side regression analysis unit is applied, and

an air correction coefficient application unit that corrects the first air flow rate corresponding to the data of the explanatory variable substituted into the regression equation by using the air correction coefficient calculated by the air correction coefficient calculation unit.

10. The gas turbine control device according to claim 2, wherein when the correction process unit includes the fuel flow rate correction unit, and there exist a plurality of the fuel supply systems, the correction process unit includes a fuel distribution amount calculation unit that calculates the second fuel flow rate for each of the plurality of fuel supply systems from the second fuel flow rates included in the computation result output by the heat balance computation unit, based on a fuel distribution ratio for each of the fuel supply systems, the fuel flow rate calculation unit calculates the first fuel flow rate for each of the fuel supply systems, and instead of the second fuel flow rate included in the computation result output by the heat balance computation unit when the state of the gas turbine which is detected by the state detection unit is the static state, the fuel flow rate correction unit acquires the second fuel flow rate for each of the fuel supply systems calculated by the fuel distribution amount calculation unit, corrects the first fuel flow rate for each of the fuel supply systems calculated by the fuel flow rate calculation unit, based on each of the acquired second fuel flow rates and each of the first fuel flow rates for each of the fuel supply systems when each of the second fuel flow rates is acquired, and provides a total value of the corrected first fuel flow rate for each of the fuel supply systems to the turbine inlet temperature calculation unit.

11. A gas turbine control method comprising:

a step of calculating a fuel flow rate indicating the amount of a fuel flowing into a combustor of a gas turbine, based on measurement data relating to a fuel supply system for supplying the fuel to the combustor;

a step of calculating an air flow rate indicating the amount of air suctioned by a compressor of the gas turbine, based on measurement data relating to the compressor;

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a step of calculating a turbine inlet temperature, based on the fuel flow rate, the air flow rate, and a physical model equation relating to heat energy balance relating to the combustor;

a step of performing a computation for satisfying heat balance of the gas turbine, based on measurement data relating to energy balance of the gas turbine, and outputting a computation result;

a step of detecting a state of the gas turbine;

a step of correcting any one or both of the fuel flow rate and the air flow rate, based on the computation result output in the step of performing the computation for satisfying the heat balance when the state of the gas turbine which is detected in the step of detecting the state is a static state, and providing the corrected flow rate to the step of calculating the turbine inlet temperature; and

a step of controlling a flow control valve of the fuel supply system based on the turbine inlet temperature calculated in the step of calculating a turbine inlet temperature.

12. A non-transitory computer-readable recording medium storing a program that causes a computer to function as:

fuel flow rate calculation means for calculating a fuel flow rate indicating the amount of a fuel flowing into a combustor of a gas turbine, based on measurement data relating to a fuel supply system for supplying the fuel to the combustor;

air flow rate calculation means for calculating an air flow rate indicating the amount of air suctioned by a compressor of the gas turbine, based on measurement data relating to the compressor;

turbine inlet temperature calculation means for calculating a turbine inlet temperature, based on the fuel flow rate, the air flow rate, and a physical model equation relating to heat energy balance relating to the combustor;

heat balance computation means for performing a computation for satisfying heat balance of the gas turbine, based on measurement data relating to energy balance of the gas turbine, and outputting a computation result;

state detection means for detecting a state of the gas turbine;

correction process means for correcting any one or both of the fuel flow rate and the air flow rate, based on the computation result output by the heat balance computation means when the state of the gas turbine which is detected by the state detection means is a static state, and providing the corrected flow rate to the turbine inlet temperature calculation means; and

control means for controlling a flow control valve of the fuel supply system based on the turbine inlet temperature calculated by the turbine inlet temperature calculation means.

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